YEAR 12 MATHEMATICS EXTENSION 1

# Year 12 Integration Part 2 Exponential, Logarithmic and Trigonometric Integrals

Lesson and Homework Pack

| | | |
|---|---|---|
| Revision | Derivatives, log laws, radians and integration basics | 6B, 6G, 6K, 7B · Part 1 |
| Lesson 6D | Integration of exponential functions | Past papers: 5 marks · 10 min |
| Lesson 6E | Applications of integration | Past papers: 14 marks · 20 min |
| Lesson 6I | Integration of the reciprocal function | Past papers: 2 marks · 5 min |
| Lesson 6J | Applications of integration of $1/x$ | Past papers: timed practice |
| Lesson 6K | Calculus with other bases | Past papers: timed practice |
| Lesson 7D | Integrating the trigonometric functions | Past papers: 16 marks · 25 min |
| Lesson 7E | Applications of integration | Past papers: 17 marks · 25 min |
| Review | Chapter 6 and Chapter 7 reviews: integration questions | Mixed practice |

Each lesson runs in the same order: **concept** → **tutor demo** → **your attempt** → **worked example** to keep for homework. Then do the **textbook homework** and finish with the **past-paper set** under exam conditions. The Revision section reviews the derivatives that these integrals reverse, and the integration basics from Part 1. Answers and full past-paper solutions are in the separate Homework Answers booklet.

REVISION

# Derivatives, log laws, radians and integration basics

Every integral in Part 2 is found by reversing a derivative. This section reviews the standard derivatives of $e^x$, $\log_e x$, $a^x$, $\sin x$, $\cos x$ and $\tan x$ (Sections 6B, 6G, 6K and 7B), the log laws, radian measure and the exact values, and the integration basics from Part 1. Keep these pages open while you work through the lessons.

## SUMMARY · keep these pages open while you work

### THREE STANDARD DERIVATIVES FOR EXPONENTIAL FUNCTIONS (6B)

Two standard forms were established in Chapter 11 (Year 11). The second one needs the chain rule with $u = ax + b$. For other functions of $x$ in the index, use the chain rule, or learn it as a formula in one of its two forms.

$$\frac{d}{dx}e^x = e^x \qquad \frac{d}{dx}e^{ax+b} = a\,e^{ax+b}$$

$$\frac{d}{dx}e^u = e^u\frac{du}{dx} \quad \text{OR} \quad \frac{d}{dx}e^{f(x)} = e^{f(x)}f'(x)$$

A function such as $(e^{2x} - 3)^4$ also needs the formula for differentiating powers of functions of $x$ (Section 5I):

$$\frac{d}{dx}u^n = nu^{n-1}\frac{du}{dx} \qquad \text{OR} \qquad \frac{d}{dx}\big(f(x)\big)^n = n\big(f(x)\big)^{n-1}f'(x).$$

Products such as $x^3e^x$ use the product rule, and quotients such as $\dfrac{e^{5x}}{x}$ use the quotient rule, as usual.

WORKED EXAMPLE **Textbook Example 4 (6B)**

Differentiate:

a $y = e^x + e^{-x}$ b $y = 5e^{4x-3}$ c $y = e^{2-\frac{1}{2}x}$ d $y = \sqrt{e^x} + \dfrac{1}{\sqrt{e^x}}$

**SOLUTION**

**a** Given $y = e^x + e^{-x}$.
For $e^{-x}$, $a = -1$ and $b = 0$,
so $y' = e^x - e^{-x}$.

**b** Given $y = 5e^{4x-3}$.
Here $a = 4$ and $b = -3$,
so $y' = 20e^{4x-3}$.

**c** Given $y = e^{2-\frac{1}{2}x}$.
Here $a = -\frac{1}{2}$ and $b = 2$,
so $y' = -\frac{1}{2}e^{2-\frac{1}{2}x}$.

**d** Here $y = \sqrt{e^x} + \dfrac{1}{\sqrt{e^x}}$
$y = e^{\frac{1}{2}x} + e^{-\frac{1}{2}x}$,
so $y' = \frac{1}{2}e^{\frac{1}{2}x} - \frac{1}{2}e^{-\frac{1}{2}x}$.

WORKED EXAMPLE **Textbook Example 5 (6B)**

Use the chain rule to differentiate:

a $y = e^{1-x^2}$

b $y = (e^{2x} - 3)^4$

**SOLUTION**

**a** Here $y = e^{1-x^2}$.

Applying the chain rule,

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$
$$= -2x\, e^{1-x^2}.$$

Let $u = 1 - x^2$.

Then $y = e^u$.

Hence $\frac{du}{dx} = -2x$

and $\frac{dy}{du} = e^u$.

**b** Here $y = (e^{2x} - 3)^4$.

Applying the chain rule,

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$
$$= 4(e^{2x} - 3)^3 \times 2e^{2x}$$
$$= 8e^{2x}(e^{2x} - 3)^3.$$

Let $u = e^{2x} - 3$.

Then $y = u^4$.

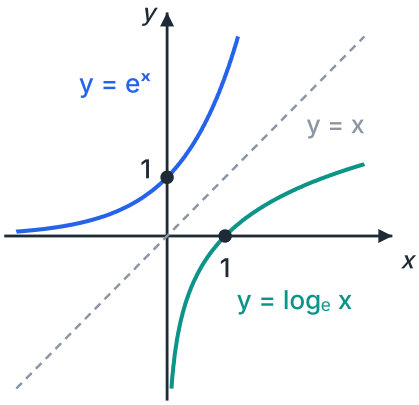
Hence $\frac{du}{dx} = 2e^{2x}$

and $\frac{dy}{du} = 4u^3$.

## THE DERIVATIVE OF THE LOGARITHMIC FUNCTION (6F, 6G)

The function $y = \log_e x$ (also written $\ln x$) is the inverse function of $y = e^x$, so the two graphs are reflections of each other in the line $y = x$. The important features of $y = \log_e x$ are:

- The domain is $x > 0$ and the range is all real numbers.
- The $x$-intercept is 1, and the gradient there is 1.
- The $y$-axis is a vertical asymptote.
- As $x \to \infty$, $y \to \infty$.
- Throughout its whole domain, $\log_e x$ is increasing at a decreasing rate.



**Proof that the derivative is the reciprocal function.**

Let $y = \log_e x.$

Then $x = e^y,$ by the definition of logarithms.

Differentiating, $\dfrac{dx}{dy} = e^y,$ because the exponential function is its own derivative,

$= x,$ because $e^y = x,$

and taking reciprocals, $\dfrac{dy}{dx} = \dfrac{1}{x}.$

Hence the derivative of the logarithmic function is the reciprocal function.

**Three standard forms for differentiating logarithmic functions** (the second and third come from the chain rule, as in Textbook Example 27):

$$\frac{d}{dx}\log_e x = \frac{1}{x} \qquad \frac{d}{dx}\log_e(ax+b) = \frac{a}{ax+b}$$

$$\frac{d}{dx}\log_e u = \frac{u'}{u} \quad \text{OR} \quad \frac{d}{dx}\log_e f(x) = \frac{f'(x)}{f(x)}$$

The third form will be needed later for integration (Lesson 6I).

WORKED EXAMPLE **Textbook Example 28 (6G)**

Using the standard forms developed above, differentiate:

a $y = \log_e(4x-9)$ b $y = \log_e(1-\frac{1}{2}x)$ c $y = \log_e(4+x^2)$

**SOLUTION**

**a** For $y = \log_e(4x-9)$, use the second standard form with $ax+b = 4x-9$.

Thus $y' = \dfrac{4}{4x-9}.$

**b** For $y = \log_e(1-\frac{1}{2}x)$, use the second standard form with $ax+b = -\frac{1}{2}x+1$.

Thus $y' = \dfrac{-\frac{1}{2}}{-\frac{1}{2}x+1}$

$= \dfrac{1}{x-2},$ after multiplying top and bottom by $-2$.

**c** For $y = \log_e(4+x^2)$,

$y' = \dfrac{2x}{4+x^2}.$

Alternatively, use the chain rule.

Let $u = 4+x^2.$

Then $u' = 2x.$

$\dfrac{d}{dx}\log_e u = \dfrac{u'}{u}$

OR let $f(x) = 4+x^2$, then $f'(x) = 2x$

and $\dfrac{d}{dx}\log_e f(x) = \dfrac{f'(x)}{f(x)}.$

## EXPONENTIAL AND LOGARITHMIC FUNCTIONS WITH OTHER BASES (6K)

Throughout, the base $a$ is positive and not equal to 1. Lesson 6K teaches this in full; here are the facts you will meet first.

- Every positive real number can be written as a power of $e$: $a = e^{\log_e a}$.
- Every exponential function can be written as an exponential function base $e$: $a^x = e^{x \log_e a}$.
- Every logarithmic function is a multiple of $\log_e x$: $\log_a x = \dfrac{\log_e x}{\log_e a}$ (the change-of-base formula).

**Differentiating $a^x$.** Write the function as a power of $e$, then differentiate:

First, $a^x = e^{x \log_e a}$.

Differentiating, $\dfrac{d}{dx} a^x = \dfrac{d}{dx} e^{x \log_e a}$

$= e^{x \log_e a} \times \log_e a,$ because $\dfrac{d}{dx} e^{kx} = k e^{kx}$,

$= a^x \log_e a,$ because $e^{x \log_e a} = a^x$.

$$\frac{d}{dx} a^x = a^x \log_e a \qquad \text{and} \qquad \frac{d}{dx} \log_a x = \frac{1}{x \log_e a}$$

The constant $\log_e a$ is 1 when $a = e$, so the formulae are simplest when the base is $e$.

## STANDARD DERIVATIVES OF THE TRIGONOMETRIC FUNCTIONS (7B)

In calculus, angles are always in radians. With radian measure, the tangent to $y = \sin x$ at the origin has gradient exactly 1, just as the tangent to $y = e^x$ at its $y$-intercept has gradient exactly 1.

**Standard derivatives of trigonometric functions**

$$\frac{d}{dx} \sin x = \cos x$$
$$\frac{d}{dx} \cos x = -\sin x$$
$$\frac{d}{dx} \tan x = \sec^2 x$$

**Standard derivatives of functions of $ax + b$**

(the chain rule with $u = ax + b$)

$$\frac{d}{dx} \sin(ax + b) = a \cos(ax + b)$$
$$\frac{d}{dx} \cos(ax + b) = -a \sin(ax + b)$$
$$\frac{d}{dx} \tan(ax + b) = a \sec^2(ax + b)$$

**Differentiation of trigonometric functions as phase shift**

- Differentiation of $y = \sin x$ and $y = \cos x$ shifts each curve left $\frac{\pi}{2}$, advancing the phase $\frac{\pi}{2}$:

$$\frac{d}{dx} \sin x = \cos x = \sin\left(x + \tfrac{\pi}{2}\right) \qquad \text{and} \qquad \frac{d}{dx} \cos x = -\sin x = \cos\left(x + \tfrac{\pi}{2}\right)$$

- The second derivatives of $\sin x$ and $\cos x$ reflect each curve in the $x$-axis:

$$\frac{d^2}{dx^2}\sin x = -\sin x \qquad \text{and} \qquad \frac{d^2}{dx^2}\cos x = -\cos x$$

- Differentiation of $\sin x$ and $\cos x$ has order 4:

$$\frac{d^4}{dx^4}\sin x = \sin x \qquad \text{and} \qquad \frac{d^4}{dx^4}\cos x = \cos x$$

Other compound functions, products and quotients use the chain, product and quotient rules as usual.

WORKED EXAMPLE **Textbook Example 4 (7B)**

Differentiate these functions.

a $y = \sin x + \cos x$ b $y = x - \tan x$

Hence find the gradient of each curve when $x = \frac{\pi}{4}$.

**SOLUTION**

**a**

The function is $y = \sin x + \cos x.$

Differentiating, $y' = \cos x - \sin x.$

When $x = \frac{\pi}{4}$, $y' = \cos\frac{\pi}{4} - \sin\frac{\pi}{4}$

$$= \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}$$

$$= 0.$$

**b**

The function is $y = x - \tan x.$

Differentiating, $y' = 1 - \sec^2 x.$

When $x = \frac{\pi}{4}$, $y' = 1 - \sec^2\frac{\pi}{4}$

$$= 1 - \left(\sqrt{2}\right)^2$$

$$= -1.$$

WORKED EXAMPLE **Textbook Example 7 (7B)**

Use the extended standard forms given in the box above to differentiate:

a $y = \cos 7x$ b $y = 4\sin\left(3x - \frac{\pi}{3}\right)$ c $y = \tan\frac{3}{2}x$

**SOLUTION**

**a** The function is $y = \cos 7x$, so $a = 7$ and $b = 0$,

and $\dfrac{dy}{dx} = -7\sin 7x.$

**b** The function is $y = 4\sin\left(3x - \frac{\pi}{3}\right)$, so $a = 3$ and $b = -\frac{\pi}{3}$,

and $\dfrac{dy}{dx} = 12\cos\left(3x - \frac{\pi}{3}\right).$

**c** The function is $y = \tan\frac{3}{2}x$, so $a = \frac{3}{2}$ and $b = 0$,

and $\dfrac{dy}{dx} = \frac{3}{2}\sec^2\frac{3}{2}x.$

## LOG LAWS, THE INVERSE IDENTITIES AND CHANGE OF BASE (YEAR 11 ADVANCED 7D, 6F)

$\log_a x = y$ means $x = a^y$. 'The log is the index, when the number is written as a power of the base.' In this course, $\ln x$ means $\log_e x$. On the calculator, the **ln** key means $\log_e x$ and the **log** key means $\log_{10} x$.

### Three laws for logarithms

**1** The log of a product is the sum of the logs:
$\log_a xy = \log_a x + \log_a y$

**2** The log of a quotient is the difference of the logs:
$\log_a \frac{x}{y} = \log_a x - \log_a y$

**3** The log of a power is the multiple of the log:
$\log_a x^n = n \log_a x$

### Some particular values and identities

$\log_a 1 = 0$, because $1 = a^0$.

$\log_a a = 1$, because $a = a^1$.

$\log_a \sqrt{a} = \frac{1}{2}$, because $\sqrt{a} = a^{\frac{1}{2}}$.

$\log_a \frac{1}{a} = -1$, because $\frac{1}{a} = a^{-1}$.

$\log_a \frac{1}{x} = -\log_a x$, because $\log_a \frac{1}{x} = \log_a x^{-1} = -\log_a x$.

**The inverse identities** (6F): the composition of $y = e^x$ and $y = \log_e x$, in either order, is the identity function,

$$\log_e e^x = x, \text{ for all real } x \qquad \text{and} \qquad e^{\log_e x} = x, \text{ for all } x > 0.$$

**The change-of-base formula** (6F): suppose that the new base $b$ is a positive number not equal to 1. Then

$$\log_b x = \frac{\log_e x}{\log_e b}.$$

'The log of the number over the log of the base.'

WORKED EXAMPLE **Textbook Example 21 (6F)**

Simplify:

a $\log_e e^6$ b $\log_e e$ c $\log_e \frac{1}{e}$ d $\log_e \frac{1}{\sqrt{e}}$ e $e^{\log_e 10}$ f $e^{\log_e 0.1}$

**SOLUTION**

**a** $\log_e e^6 = 6$

**b** $\log_e e = \log_e e^1 = 1$

**c** $\log_e \frac{1}{e} = \log_e e^{-1} = -1$

**d** $\log_e \frac{1}{\sqrt{e}} = \log_e e^{-\frac{1}{2}} = -\frac{1}{2}$

**e** $e^{\log_e 10} = 10$

**f** $e^{\log_e 0.1} = 0.1$

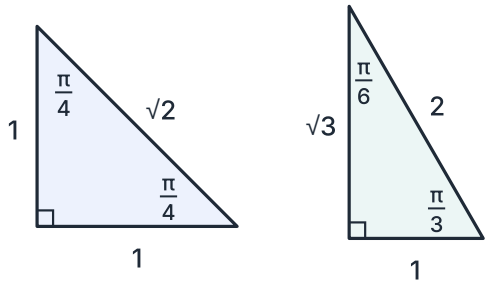
## RADIAN MEASURE (YEAR 11 ADVANCED 9G)

The size of $\angle AOB$ in radians is $\frac{\text{arc length } AB}{\text{radius } OA}$, where the circle has centre $O$. An angle in radians is a pure number, so 'an angle of size 1.3' means 1.3 radians.

- Conversions: $360° = 2\pi, \quad 180° = \pi, \quad 90° = \frac{\pi}{2}$.
- To convert from radians to degrees, multiply by $\dfrac{180°}{\pi}$. To convert from degrees to radians, multiply by $\dfrac{\pi}{180}$.
- $1 \text{ radian} = \dfrac{180°}{\pi} \doteq 57°18'$ and $1 \text{ degree} = \dfrac{\pi}{180} \doteq 0.0175$.
- Always decide whether the calculator should be in radians mode or degrees mode before using any trigonometric function. In calculus, use radians mode.

## EXACT VALUES AND ANGLES OF ANY SIZE (YEAR 11 ADVANCED 5A, 5D, 5E, 9G)

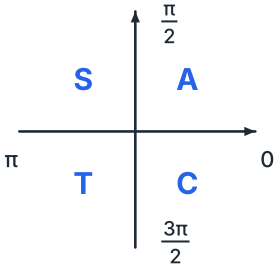
**Special angles** come from half a square and half an equilateral triangle: $\frac{\pi}{6} = 30°$, $\frac{\pi}{4} = 45°$ and $\frac{\pi}{3} = 60°$.




| $\theta$ | $\sin\theta$ | $\cos\theta$ | $\tan\theta$ |
|---|---|---|---|
| $\frac{\pi}{6}$ | $\frac{1}{2}$ | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{3}}$ |
| $\frac{\pi}{4}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{\sqrt{2}}$ | $1$ |
| $\frac{\pi}{3}$ | $\frac{\sqrt{3}}{2}$ | $\frac{1}{2}$ | $\sqrt{3}$ |

**The boundary angles** (read them off the graphs; * means undefined):

| $\theta$ | $0$ | $\frac{\pi}{2}$ | $\pi$ | $\frac{3\pi}{2}$ |
|---|---|---|---|---|
| $\sin\theta$ | 0 | 1 | 0 | $-1$ |
| $\cos\theta$ | 1 | 0 | $-1$ | 0 |
| $\tan\theta$ | 0 | * | 0 | * |




**Trigonometric functions of any angle.** 'All Stations To Central': in the four quadrants, A means all six functions are positive, S means only sine (and cosecant), T means only tangent (and cotangent), C means only cosine (and secant). Draw a quadrant diagram, place the ray in the correct quadrant and use the ASTC rule for the sign. Then find the related acute angle and evaluate the function at the related angle. The reciprocal functions are $\sec\theta = \dfrac{1}{\cos\theta}$, $\operatorname{cosec}\theta = \dfrac{1}{\sin\theta}$ and $\cot\theta = \dfrac{1}{\tan\theta}$.

**WORKED EXAMPLE** **Year 11 Advanced Example 15 (9G)**

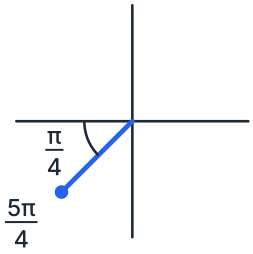
Use special angles to evaluate these trigonometric functions.

a $\sin \dfrac{5\pi}{4}$ b $\sec \dfrac{11\pi}{6}$

**SOLUTION**

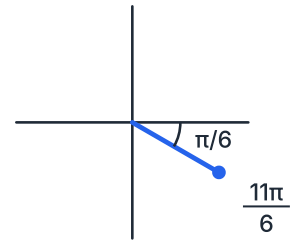
**a** Because $\frac{5\pi}{4}$ is in the third quadrant, with related angle $\frac{\pi}{4}$,

$$\begin{aligned}\sin \tfrac{5\pi}{4} &= -\sin \tfrac{\pi}{4} \\ &= -\sin 45^\circ \\ &= -\frac{1}{\sqrt{2}}.\end{aligned}$$




**b** Because $\frac{11\pi}{6}$ is in the fourth quadrant, with related angle $\frac{\pi}{6}$,

$$\begin{aligned}\sec \tfrac{11\pi}{6} &= +\sec \tfrac{\pi}{6} \\ &= \frac{1}{\cos \frac{\pi}{6}} \\ &= \frac{1}{\cos 30^\circ} \\ &= \frac{2}{\sqrt{3}}.\end{aligned}$$




**INTEGRATION BASICS FROM PART 1 (5B–5I)**

- A function $F(x)$ is a **primitive** (or **indefinite integral**) of $f(x)$ if $F'(x) = f(x)$. The constant of integration $C$ must always be included, except in definite integrals, where any primitive can be used.
- **Standard forms for integration.** Suppose that $n \neq -1$. Then, for some constant $C$,

  $$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \qquad \text{and} \qquad \int (ax+b)^n \, dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C.$$

  'Increase the index by 1 and divide by the new index.' The index $-1$ is excluded, because the rule would divide by zero: $\int x^{-1} \, dx$ is found in Lesson 6I.
- **The fundamental theorem of calculus.** Let $f(x)$ be continuous in a closed interval $[a, b]$. Then

  $$\int_a^b f(x) \, dx = \Big[F(x)\Big]_a^b = F(b) - F(a), \quad \text{where } F(x) \text{ is any primitive of } f(x).$$

  Substitute the upper limit, then the lower limit, and subtract. Whenever the primitive has two or more terms, use brackets for each substitution, and never omit a substitution of 0.

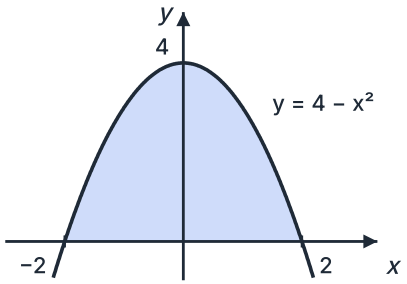
- **Signed areas.** $\int_a^b f(x)\,dx$ is the sum of the signed areas: $+$ for regions above the $x$-axis and $-$ for regions below it.
- **Properties.** Intervals of zero width, reversing the interval, and dissection of the interval:
  $$\int_a^a f(x)\,dx = 0 \qquad \int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$$
  $$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$
  The integral of a sum and the integral of a multiple:
  $$\int_a^b \big(f(x) + g(x)\big)\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$
  $$\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx$$
  Odd and even functions: if $f(x)$ is odd, then $\int_{-a}^a f(x)\,dx = 0$. If $f(x)$ is even, then $\int_{-a}^a f(x)\,dx = 2\int_0^a f(x)\,dx$.
- **Finding an area.** 1. Draw a sketch of the curves, showing relevant intercepts and intersections. 2. Create and evaluate the necessary definite integral or integrals. 3. Write a conclusion, giving the required area in square units. If $f(x) \le g(x)$ for $a \le x \le b$, then the area between the curves is $\int_a^b \big(g(x) - f(x)\big)\,dx$: the top curve minus the bottom curve.
- **The reverse chain rule.** Once a chain-rule differentiation has been performed, the result can be written down in reverse as an indefinite integral. As a formula,
  $$\int u^n \frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1} + C \qquad \text{OR} \qquad \int \big(f(x)\big)^n f'(x)\,dx = \frac{\big(f(x)\big)^{n+1}}{n+1} + C.$$
  The vital step is to identify $u$ or $f(x)$ and its derivative.

**WORKED EXAMPLE** **Textbook Example 22 (5F)**

Find the area of the region bounded by the curve $y = 4 - x^2$ and the $x$-axis.

**SOLUTION**

The curve meets the $x$-axis at $(2, 0)$ and $(-2, 0)$.
The region lies entirely above the $x$-axis and the relevant integral is




$$\begin{aligned}\int_{-2}^{2}(4-x^2)\,dx &= \left[4x - \frac{x^3}{3}\right]_{-2}^{2} \\ &= \left(8 - \tfrac{8}{3}\right) - \left(-8 + \tfrac{8}{3}\right) \\ &= 5\tfrac{1}{3} - \left(-5\tfrac{1}{3}\right) \\ &= 10\tfrac{2}{3},\end{aligned}$$

which is positive because the curve lies entirely above the $x$-axis.
Hence the required area is $10\frac{2}{3}$ square units.

**WORKED EXAMPLE** **Textbook Example 36 (5I)**

**a** Differentiate $(x^3 + 5)^4$ with full setting-out of the chain rule.
**b** Hence find a primitive of $12x^2(x^3 + 5)^3$.
**c** Hence find the primitive of $x^2(x^3 + 5)^3$.

**SOLUTION**

**a**

Let $y = (x^3 + 5)^4$.

Then $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$

$= 4(x^3 + 5)^3 \times 3x^2$

$= 12x^2(x^3 + 5)^3.$

Let $u = x^3 + 5$.
Then $y = u^4$.
Hence $\dfrac{du}{dx} = 3x^2$
and $\dfrac{dy}{du} = 4u^3$.

**b** By part **a**, $\dfrac{d}{dx}(x^3 + 5)^4 = 12x^2(x^3 + 5)^3$.

Reversing this, $\displaystyle\int 12x^2(x^3 + 5)^3\,dx = (x^3 + 5)^4$.

**c** Dividing by 12, $\displaystyle\int x^2(x^3 + 5)^3\,dx = \tfrac{1}{12}(x^3 + 5)^4 + C$, for some constant $C$.

**Note:** It is best not to add the arbitrary constant until the last line, because it would be pointless to divide $C$ by 12 as well.

## WARM-UP · answers are in the answers booklet

Each question is from the textbook exercise named, with some parts left out. Show the standard form or the chain-rule working in each part.

**1** Exercise 6B Q1

Use the standard form $\frac{d}{dx}\,e^{ax+b} = a\,e^{ax+b}$ to differentiate:

a $y = 4e^{3x}$

b $y = 6e^{\frac{1}{3}x}$

c $y = -\frac{1}{2}e^{-2x}$

d $y = e^{-3x+4}$

e $y = e^{-2x-7}$

**2** Exercise 6G Q8

Use the standard form $\frac{d}{dx}\log_e u = \frac{u'}{u}$ OR $\frac{d}{dx}\log_e f(x) = \frac{f'(x)}{f(x)}$ to differentiate:

a $\log_e(x^2+3x+2)$

b $\log_e(1+2x^3)$

c $\ln(e^x-2)$

**3** Exercise 6G Q5

Use the log laws to simplify each function, then differentiate it.

a $y = \ln x^3$

b $y = \ln x^{-3}$

c $y = \ln \sqrt{x}$

d $y = \ln \sqrt{x+1}$

**4** Exercise 7B Q2

Differentiate with respect to $x$.

a $\sin(2x - 1)$

b $2\cos(1 - x)$

c $7\sin(2 - 3x)$

d $6\sin\left(\frac{x+1}{2}\right)$

**5** Exercise 7B Q4

If $f(x) = \cos 2x$, find $f'(x)$ and then find:

a $f'(0)$

b $f'\left(\frac{\pi}{12}\right)$

c $f'\left(\frac{\pi}{6}\right)$

d $f'\left(\frac{\pi}{4}\right)$

**6** Exercise 5B Q4

Evaluate these definite integrals using the fundamental theorem. You will need to take care when finding powers of negative numbers.

a $\int_{-2}^{1} 3x^2\,dx$

b $\int_{-1}^{2} (4x^3 + 5)\,dx$

c $\int_{-2}^{2} (5x^4 + 6x^2)\,dx$

d $\int_{-2}^{-1} (4x^3 + 12x^2 - 3)\,dx$

LESSON 6D

# Integration of exponential functions

Finding primitives is the reverse of differentiation. Reversing the standard forms for differentiating $e^x$ and $e^{ax+b}$ gives the standard forms for integrating them: $\int e^x\,dx = e^x + C$ and $\int e^{ax+b}\,dx = \frac{1}{a}\,e^{ax+b} + C$. With these we evaluate definite integrals, find a function from its derivative and one known value, reverse a known derivative to find a new primitive, and use the reverse chain rule formula from Section 5I.

## CONCEPT 1 · standard forms for integration

### REVERSING THE STANDARD FORMS FOR DIFFERENTIATION

Reversing the standard forms for differentiating exponential functions gives the standard forms for integrating them.

Reversing $\dfrac{d}{dx}\,e^x = e^x$ gives $\displaystyle\int e^x\,dx = e^x + C,$ for some constant $C$.

Reversing $\dfrac{d}{dx}\,e^{ax+b} = a\,e^{ax+b}$ gives $\displaystyle\int a\,e^{ax+b}\,dx = e^{ax+b},$

and dividing through by $a$, $\displaystyle\int e^{ax+b}\,dx = \frac{1}{a}\,e^{ax+b} + C,$ for some constant $C$.

### WHERE THE CONSTANT GOES

The second line has no $C$. $C$ is added in the last line, after dividing by $a$, because $C$ divided by $a$ is just another constant.

### 4 STANDARD FORMS FOR INTEGRATION

- $\displaystyle\int e^x\,dx = e^x + C$, for some constant $C$
- $\displaystyle\int e^{ax+b}\,dx = \frac{1}{a}\,e^{ax+b} + C$, for some constant $C$

### THE REVERSE CHAIN RULE FOR EXPONENTIAL FUNCTIONS (FOR REFERENCE)

There is also an associated formula for the reverse chain rule, but it is not included in the course. For reference, this formula has the usual two forms:

$$\int e^u\,\frac{du}{dx}\,dx = e^u + C \qquad \text{OR} \qquad \int e^{f(x)}\,f'(x)\,dx = e^{f(x)} + C.$$

**WHEN THE INDEX IS NOT LINEAR**

The reference formula above is the reverse of the chain-rule result $\frac{d}{dx}\,e^{f(x)} = e^{f(x)}\,f'(x)$. Exercise 6D Q17 uses it.

The standard form $\int e^{ax+b}\,dx = \frac{1}{a}\,e^{ax+b} + C$ needs an index of the form $ax + b$, so that its derivative $a$ is a constant. Check every answer by differentiating it. For an index such as $x^2$, first differentiate $e^{x^2}$ with the chain rule and then reverse the result (Concept 4).

TUTOR DEMO **Indefinite integrals**

Find these indefinite integrals.

a $\int e^{5x-1}\,dx$

b $\int (2 + 3x - e^x)\,dx$

c $\int e^{4-2x}\,dx$

YOUR TURN **Same method**

Find these indefinite integrals.

a $\int e^{2x+7}\,dx$

b $\int (4 - x + e^x)\,dx$

c $\int e^{1-3x}\,dx$

WORKED EXAMPLE 11 **Keep for homework**

Find these indefinite integrals.

a $\int e^{3x+2}\,dx$

b $\int (1 - x + e^x)\,dx$

**SOLUTION**

**a** $\int e^{3x+2}\,dx = \frac{1}{3}e^{3x+2} + C$ ($a = 3$ and $b = 2$)

**b** $\int (1 - x + e^x)\,dx = x - \frac{1}{2}x^2 + e^x + C$ (integrating each term separately)

## CONCEPT 2 · definite integrals

**DEFINITE INTEGRALS**

Definite integrals are evaluated in the usual way by finding the primitive and substituting.

**SUBSTITUTING THE LIMITS**

Substitute the upper limit, then the lower limit, and subtract. Remember that $e^0 = 1$ and $e^{-1} = \frac{1}{e}$. Answers are usually left in exact form, and a negative index is often rewritten as a fraction.

TUTOR DEMO **Exact values**

Evaluate these definite integrals.

a $\int_0^3 e^x \, dx$

b $\int_2^3 e^{7-3x} \, dx$

YOUR TURN **Same method**

Evaluate these definite integrals.

a $\int_{-1}^1 e^x \, dx$

b $\int_0^1 e^{2-4x} \, dx$

**WORKED EXAMPLE 12** **Keep for homework**

Evaluate these definite integrals.

a $\int_0^2 e^x \, dx$

b $\int_2^3 e^{5-2x} \, dx$

**SOLUTION**

**a**

$$\begin{aligned}\int_0^2 e^x \, dx &= \left[e^x\right]_0^2 \\ &= e^2 - e^0 \\ &= e^2 - 1\end{aligned}$$

**b** $(a = -2 \text{ and } b = 5)$

$$\begin{aligned}\int_2^3 e^{5-2x} \, dx &= -\tfrac{1}{2}\left[e^{5-2x}\right]_2^3 \\ &= -\tfrac{1}{2}\left(e^{-1} - e\right) \\ &= -\frac{1}{2}\left(\frac{1}{e} - e\right) \\ &= \frac{e^2 - 1}{2e}\end{aligned}$$

**Note:** In part **b**, $5 - 2 \times 3 = -1$ and $5 - 2 \times 2 = 1$. The last line multiplies top and bottom by $e$:

$$-\frac{1}{2}\left(\frac{1}{e} - e\right) = \frac{1}{2}\left(e - \frac{1}{e}\right) = \frac{e^2 - 1}{2e}.$$

## CONCEPT 3 · given the derivative, find the function

**GIVEN THE DERIVATIVE, FIND THE FUNCTION**

As before, if the derivative of a function is known, and the value of the function at one point is also known, then the whole function can be determined.

**THE THREE STEPS**

- Take the primitive of the derivative, including the constant $C$.
- Substitute the known value to find $C$.
- Write the function with the value of $C$ in it, then use it.

TUTOR DEMO **Find the constant**

**1** It is known that $f'(x) = e^x$ and that $f(2) = 0$.
**a** Find the original function $f(x)$.
**b** Hence find $f(0)$.
**2 a** If $y' = 2 + 3e^{-x}$ and $y(0) = 1$, find $y$.
**b** Hence find $y(1)$.

YOUR TURN **Same method**

**1** It is known that $f'(x) = e^x$ and that $f(-1) = 0$.
**a** Find the original function $f(x)$.
**b** Hence find $f(0)$.
**2 a** If $y' = 3 - e^{-x}$ and $y(0) = 2$, find $y$.
**b** Hence find $y(1)$.

**WORKED EXAMPLE 13** **Keep for homework**

It is known that $f'(x) = e^x$ and that $f(1) = 0$.
**a** Find the original function $f(x)$.
**b** Hence find $f(0)$.

**SOLUTION**

**a**

It is given that $f'(x) = e^x$.
Taking the primitive, $f(x) = e^x + C$, for some constant $C$.
Substituting $f(1) = 0$, $0 = e^1 + C$
$C = -e$.
Hence $f(x) = e^x - e$.

**b**

Substituting $x = 0$, $f(0) = e^0 - e$
$= 1 - e$.

**WORKED EXAMPLE 14** **Keep for homework**

**a** If $y' = 1 + 2e^{-x}$ and $y(0) = 1$, find $y$.
**b** Hence find $y(1)$.

**SOLUTION**

**a**

It is given that $y' = 1 + 2e^{-x}$.
Taking the primitive, $y = x - 2e^{-x} + C$, for some constant $C$.
Substituting $y(0) = 1$, $1 = 0 - 2e^0 + C$
$1 = 0 - 2 + C$
$C = 3$.
Hence $y = x - 2e^{-x} + 3$.

**Note:** For $e^{-x}$, $a = -1$, so $\int 2e^{-x}\,dx = 2 \times \frac{1}{-1} e^{-x} = -2e^{-x}$.

**b**

Substituting $x = 1$, $y(1) = 1 - 2e^{-1} + 3$
$= 4 - 2e^{-1}$.

## CONCEPT 4 · given a derivative, find an integral

### GIVEN A DERIVATIVE, FIND AN INTEGRAL

The result of any differentiation can be reversed. This often allows a new primitive to be found.

### REVERSING A PRODUCT-RULE DERIVATIVE

In symbols: if $F'(x) = f(x)$, then $\int f(x)\,dx = F(x) + C$ and $\int_a^b f(x)\,dx = \big[F(x)\big]_a^b$.

Sometimes the derivative has two terms and only one of them is the integrand we want (Exercise 6D Q14). For example, by the product rule with $u = x$ and $v = e^{2x}$,

$$\frac{d}{dx}\left(xe^{2x}\right) = 1 \times e^{2x} + x \times 2e^{2x} = e^{2x} + 2xe^{2x}.$$

Reversing this, $\displaystyle\int_0^1 \left(e^{2x} + 2xe^{2x}\right) dx = \Big[xe^{2x}\Big]_0^1$. Split the left-hand side into two integrals and move the known one across:

$$\begin{aligned}\int_0^1 2xe^{2x}\,dx &= \Big[xe^{2x}\Big]_0^1 - \int_0^1 e^{2x}\,dx \\ &= \left(1 \times e^2 - 0 \times e^0\right) - \tfrac{1}{2}\Big[e^{2x}\Big]_0^1 \\ &= e^2 - \tfrac{1}{2}\left(e^2 - 1\right) \\ &= \tfrac{1}{2}\left(e^2 + 1\right),\end{aligned}$$

so, dividing by 2, $\displaystyle\int_0^1 xe^{2x}\,dx = \tfrac{1}{4}\left(e^2 + 1\right)$.

TUTOR DEMO **Reverse a chain-rule result**

**a** Use the chain rule to differentiate $e^{x^4}$.

**b** Hence find $\int_{-1}^{1} 4x^3 e^{x^4}\,dx$.

**c** Why is this result obvious without any calculation?

YOUR TURN **Same method**

**a** Use the chain rule to differentiate $e^{2x^2}$.

**b** Hence find $\int_{-1}^{1} 4xe^{2x^2}\,dx$.

**c** Why is this result obvious without any calculation?

**WORKED EXAMPLE 15** **Keep for homework**

**a** Use the chain rule to differentiate $e^{x^2}$.

**b** Hence find $\displaystyle\int_{-1}^{1} 2xe^{x^2}\,dx$.

**c** Why is this result obvious without any calculation?

**SOLUTION**

**a** Let $y = e^{x^2}$.

Applying the chain rule,

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= 2x\,e^{x^2}.\end{aligned}$$

Let $u = x^2$.

Then $y = e^u$.

Hence $\dfrac{du}{dx} = 2x$

and $\dfrac{dy}{du} = e^u$.

**b** Reversing part **a**, $\displaystyle\int 2xe^{x^2}\,dx = e^{x^2}$.

Hence

$$\begin{aligned}\int_{-1}^{1} 2xe^{x^2}\,dx &= \left[e^{x^2}\right]_{-1}^{1} \\ &= e^1 - e^1 \\ &= 0.\end{aligned}$$

**c** The function $2xe^{x^2}$ is odd, and the interval $[-1, 1]$ in the definite integral is symmetric in the $y$-axis, so the integral is zero.

**Note:** Here $(-1)^2 = 1$, so the lower limit also gives $e^1$. The function is odd because $2(-x)e^{(-x)^2} = -2xe^{x^2}$.

## CONCEPT 5 · using a formula for the reverse chain rule

**USING A FORMULA FOR THE REVERSE CHAIN RULE**

There are some situations where the reverse chain rule formula from Section 5I can be used.

**THE FORMULA FROM SECTION 5I**

$$\int u^n \frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1} + C \qquad \text{OR} \qquad \int \big(f(x)\big)^n f'(x)\,dx = \frac{\big(f(x)\big)^{n+1}}{n+1} + C,$$

where $n \neq -1$. Choose $u$ (or $f(x)$) to be the bracket that is raised to a power. Its derivative $u'$ must appear in the integrand, apart from a constant factor. Insert that constant inside the integral and balance it with its reciprocal outside.

TUTOR DEMO **A power of a bracket**

Use the reverse chain rule formula to find $\int \frac{e^{2x}}{(1+e^{2x})^3}\,dx$.

YOUR TURN **Same method**

Use the reverse chain rule formula to find $\int \frac{e^{3x}}{(1-e^{3x})^2}\,dx$.

**WORKED EXAMPLE 16** **Keep for homework**

Use the reverse chain rule formula to find $\int \frac{e^{2x}}{(1-e^{2x})^3}\,dx$.

**SOLUTION**

$$\begin{aligned}
&\int \frac{e^{2x}}{(1-e^{2x})^3}\,dx \\
&= -\tfrac{1}{2}\int \frac{-2e^{2x}}{(1-e^{2x})^3}\,dx \\
&= -\tfrac{1}{2} \times \left(-\tfrac{1}{2}\right) \times (1-e^{2x})^{-2} \\
&= \frac{1}{4(1-e^{2x})^2} + C,
\end{aligned}$$

for some constant $C$.

Let $u = 1 - e^{2x}$.
Then $u' = -2e^{2x}$.

$$\int u^{-3}\,\frac{du}{dx}\,dx = \frac{u^{-2}}{-2}$$

OR let $f(x) = 1 - e^{2x}$.
Then $f'(x) = -2e^{2x}$.

$$\int \big(f(x)\big)^{-3} f'(x)\,dx = \frac{\big(f(x)\big)^{-2}}{-2}$$

**Note:** The second line inserts the factor $-2$ to make $u' = -2e^{2x}$, and balances it with $-\frac{1}{2}$ outside. The third line uses $n = -3$, so $\dfrac{u^{n+1}}{n+1} = \dfrac{u^{-2}}{-2} = -\tfrac{1}{2}u^{-2}$.

## TEXTBOOK HOMEWORK · Exercise 6D

**Technology:** Some algebraic programs can display the primitive and evaluate the exact value of an integral. These can be used to check the questions in this exercise and also to investigate the effect of making small changes to the function or to the limits of integration.

### FOUNDATION

**1** Use the standard form $\int e^{ax+b}\,dx = \frac{1}{a}e^{ax+b} + C$ to find each indefinite integral.

a $\int e^{2x}\,dx$

b $\int e^{3x}\,dx$

c $\int e^{\frac{1}{3}x}\,dx$

d $\int e^{\frac{1}{2}x}\,dx$

e $\int 10e^{2x}\,dx$

f $\int 12e^{3x}\,dx$

g $\int e^{4x+5}\,dx$

h $\int e^{4x-2}\,dx$

i $\int 6e^{3x+2}\,dx$

j $\int 4e^{4x+3}\,dx$

k $\int e^{7-2x}\,dx$

l $\int \frac{1}{2}e^{1-3x}\,dx$

**2** Evaluate these definite integrals.

a $\int_0^1 e^x \, dx$ b $\int_1^2 e^x \, dx$ c $\int_{-1}^3 e^{-x} \, dx$ d $\int_{-2}^0 e^{-x} \, dx$

e $\int_0^2 e^{2x} \, dx$ f $\int_{-1}^2 20e^{-5x} \, dx$ g $\int_{-3}^1 8e^{-4x} \, dx$ h $\int_{-1}^3 9e^{6x} \, dx$

i $\int_{-1}^1 e^{2x+1} \, dx$ j $\int_{-2}^0 e^{4x-3} \, dx$ k $\int_{-2}^{-1} e^{3x+2} \, dx$ l $\int_{-\frac{1}{2}}^{\frac{1}{2}} e^{3-2x} \, dx$

m $\int_{-\frac{1}{3}}^{\frac{1}{3}} e^{2+3x} \, dx$ n $\int_1^2 6e^{3x+1} \, dx$ o $\int_2^3 12e^{4x-5} \, dx$ p $\int_1^2 12e^{8-3x} \, dx$

**3** Express each function using negative indices, and hence find its primitive.

a $\frac{1}{e^x}$ b $\frac{1}{e^{2x}}$ c $\frac{1}{e^{3x}}$

d $-\frac{3}{e^{3x}}$ e $\frac{6}{e^{2x}}$ f $\frac{8}{e^{-2x}}$

**4** a A function $f(x)$ has derivative $f'(x) = e^{2x}$. Find the equation of $f(x)$, which will involve an arbitrary constant.

b It is also known that $f(0) = -2$. Find the arbitrary constant and hence write down the equation of $f(x)$.

c Find $f(1)$ and $f(2)$.

**DEVELOPMENT**

**5** Find $f(x)$ and then find $f(1)$, given that:

a $f'(x) = 1 + 2e^x$ and $f(0) = 1$

b $f'(x) = 1 - 3e^x$ and $f(0) = -1$

c $f'(x) = 2 + e^{-x}$ and $f(0) = 0$

d $f'(x) = 4 - e^{-x}$ and $f(0) = 2$

e $f'(x) = e^{2x-1}$ and $f\left(\frac{1}{2}\right) = 3$

f $f'(x) = e^{1-3x}$ and $f\left(\frac{1}{3}\right) = \frac{2}{3}$

g $f'(x) = e^{\frac{1}{2}x+1}$ and $f(-2) = -4$

h $f'(x) = e^{\frac{1}{3}x+2}$ and $f(-6) = 2$

**6** Expand the brackets and then find primitives of:

a $e^x(e^x+1)$ b $e^x(e^x-1)$ c $e^{-x}(2e^{-x}-1)$

d $(e^x+1)^2$ e $(e^x-1)^2$ f $(e^x-2)^2$

g $(e^x+e^{-x})(e^x-e^{-x})$ h $(e^{5x}+e^{-5x})(e^{5x}-e^{-5x})$

**7** Use the standard form $\int e^{ax+b}\,dx = \frac{1}{a}\,e^{ax+b}+C$ to find these indefinite integrals.

a $\int e^{7x+q}\,dx$ b $\int e^{3x-k}\,dx$ c $\int e^{sx+1}\,dx$ d $\int e^{kx-1}\,dx$

e $\int pe^{px+q}\,dx$ f $\int me^{mx+k}\,dx$ g $\int Ae^{sx-t}\,dx$

**8** Express each function below as a power of $e$ or a multiple of a power of $e$, and hence find its primitive.

a $\frac{1}{e^{x-1}}$ b $\frac{1}{e^{3x-1}}$ c $\frac{1}{e^{2x+5}}$

d $\frac{4}{e^{2x-1}}$ e $\frac{10}{e^{2-5x}}$ f $\frac{12}{e^{3x-5}}$

**9** By writing each integrand as the sum of multiples of powers of $e$, find:

a $\displaystyle\int \frac{e^x+1}{e^x}\,dx$ b $\displaystyle\int \frac{e^{2x}+1}{e^x}\,dx$ c $\displaystyle\int \frac{e^x-1}{e^{2x}}\,dx$

d $\displaystyle\int \frac{e^x-3}{e^{3x}}\,dx$ e $\displaystyle\int \frac{2e^{2x}-3e^x}{e^{4x}}\,dx$ f $\displaystyle\int \frac{2e^x-e^{2x}}{e^{3x}}\,dx$

**10** a Find $y$ as a function of $x$ if $y' = e^{x-1}$, and $y = 1$ when $x = 1$. What is the $y$-intercept of this curve?

b The gradient of a curve is given by $y' = e^{2-x}$, and the curve passes through the point $(0, 1)$. What is the equation of this curve? What is its horizontal asymptote?

c It is known that $f'(x) = e^x + \dfrac{1}{e}$ and that $f(-1) = -1$. Find $f(0)$.

d Given that $f''(x) = e^x - e^{-x}$ and that $y = f(x)$ is horizontal as it passes through the origin, find $f(x)$.

**11** By first writing each integrand as a sum of powers of $e$, find:

a $\displaystyle\int_0^1 e^x(2e^x-1)\,dx$

b $\displaystyle\int_{-1}^1 (e^x+2)^2\,dx$

c $\displaystyle\int_0^1 (e^x-1)(e^{-x}+1)\,dx$

d $\displaystyle\int_{-1}^1 (e^{2x}+e^{-x})(e^{2x}-e^{-x})\,dx$

e $\displaystyle\int_0^1 \frac{e^{3x}+e^x}{e^{2x}}\,dx$

f $\displaystyle\int_{-1}^1 \frac{e^x-1}{e^{2x}}\,dx$

**12** a (i) Differentiate $e^{x^2+3}$. (ii) Hence find $\int 2xe^{x^2+3}\,dx$.

b (i) Differentiate $e^{x^2-2x+3}$. (ii) Hence find $\int (x-1)e^{x^2-2x+3}\,dx$.

c (i) Differentiate $e^{3x^2+4x+1}$. (ii) Hence find $\int (3x+2)e^{3x^2+4x+1}\,dx$.

d (i) Differentiate $y = e^{x^3}$. (ii) Hence find $\int_{-1}^{0} x^2e^{x^3}\,dx$.

**13** Write each integrand as a power of $e$, and hence find the indefinite integral.

a $\int \frac{1}{(e^x)^2}\,dx$ b $\int \frac{1}{(e^x)^3}\,dx$ c $\int \sqrt{e^x}\,dx$

d $\int \sqrt[3]{e^x}\,dx$ e $\int \frac{1}{\sqrt{e^x}}\,dx$ f $\int \frac{1}{\sqrt[3]{e^x}}\,dx$

**14** a Differentiate $y = xe^x$, and hence find $\int_0^2 xe^x \, dx$.

b Differentiate $y = xe^{-x}$, and hence find $\int_{-2}^0 xe^{-x} \, dx$.

**15** By first simplifying each integrand, find:

a $\int \frac{e^x - e^{-x}}{\sqrt{e^x}} \, dx$

b $\int \frac{e^x + e^{-x}}{\sqrt[3]{e^x}} \, dx$

**16** a Show that $f(x) = xe^{-x^2}$ is an odd function.

b Hence evaluate $\int_{-\sqrt{2}}^{\sqrt{2}} xe^{-x^2} \, dx$ without finding a primitive.

ENRICHMENT

**17** Under Box 4, we gave the reverse chain rule for integrals of exponential functions,

$$\int e^u \frac{du}{dx}\,dx = e^u + C \qquad \text{OR} \qquad \int e^{f(x)} f'(x)\,dx = e^{f(x)} + C.$$

Use one or other of the formulae to find primitives of these functions

a $xe^{x^2}$ b $4xe^{x^2-7}$ c $(3x+2)e^{3x^2+4x+1}$

d $(x^2-2x)e^{x^3-3x^2}$ e $x^{-2}\,e^{x^{-1}}$ f $-\sqrt{x}\,e^{-x\sqrt{x}}$

**18** Show that $\displaystyle\int \frac{e^x+1}{e^{\frac{1}{2}x}+e^{-\frac{1}{2}x}}\,dx = 2e^{\frac{1}{2}x} + C.$

**19** **A power series for $e^x$:** The intention of this question is to outline a reasonably rigorous proof of the famous result that $e^x$ can be written as the limit of the power series:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots,$$

where $n! = n \times (n-1) \times \cdots \times 2 \times 1$.

**a** For any positive number $R$, we know that $1 < e^t < e^R$ for $0 < t < R$, because $e^x$ is an increasing function. Integrate this inequality over the interval $t = 0$ to $t = x$, where $0 < x < R$, and hence show that $x < e^x - 1 < e^R x$.

**b** Change the variable to $t$, giving $t < e^t - 1 < e^R t$. Then integrate this new inequality from $t = 0$ to $t = x$, and hence show that $\frac{x^2}{2!} < e^x - 1 - x < \frac{e^R x^2}{2!}$.

**c** Do this process twice more, and prove that:

(i) $\frac{x^3}{3!} < e^x - 1 - x - \frac{x^2}{2!} < \frac{e^R x^3}{3!}$

(ii) $\frac{x^4}{4!} < e^x - 1 - x - \frac{x^2}{2!} - \frac{x^3}{3!} < \frac{e^R x^4}{4!}$

**d** Now use induction to prove that

$$\frac{x^{n+1}}{(n+1)!} < e^x - 1 - x - \frac{x^2}{2!} - \cdots - \frac{x^n}{n!} < \frac{e^R x^{n+1}}{(n+1)!}.$$

**e** Show that as $n \to \infty$, the left and right expressions converge to zero. Hence prove that the infinite power series converges to $e^x$ for $x > 0$. (Hint: Let $k$ be the smallest integer greater than $x$ and show that for $n > k$, each term in the sequence is less than the corresponding term of a geometric sequence with ratio $\frac{x}{k}$.)

**f** Prove that the power series also converges to $e^x$ for $x < 0$.

**20** a Use the power series in the previous question to show that

$$\frac{e^x + e^{-x}}{2} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^{2n}}{(2n)!} + \cdots.$$

b Find $\alpha$, the value of the right-hand side, correct to four decimal places when $x = 0.5$.

c Let $u = e^{0.5}$. Show that $u^2 - 2\alpha u + 1 = 0$.

d Solve this quadratic equation and hence estimate both $e^{0.5}$ and $e^{-0.5}$ correct to two decimal places. Compare your answers with the values obtained directly from the calculator.

## PAST-PAPER HOMEWORK · mini exam

**Suggested time: 10 minutes** **5 marks** Calculator allowed · show working · the hint is below the question if you get stuck

**PP 1** **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q11(b)** **5 marks**

(i) Find $\dfrac{dy}{dx}$ if $y = xe^{-x}$. 2

(ii) Hence find $\displaystyle\int_0^1 xe^{-x}\,dx$. 3

**Hint:** what does part (i) give you?

The school trial papers used for this pack (2020–2024) have only this one question that fits this lesson, so the mini exam is short. For more timed practice, do Questions 5(e), 10(c), 10(d), 11(f), 12(d) and 14(b) above in 25 minutes.

LESSON 6E

# Applications of integration

The normal methods of finding areas by integration can now be applied to functions involving $e^x$. We find the area between a curve and the $x$-axis, where a sketch is essential because regions below the axis give negative integrals, and the area between two curves, using the top curve minus the bottom curve.

## CONCEPT 1 · the area between a curve and the x-axis

### FINDING THE AREA BETWEEN A CURVE AND THE $x$-AXIS

A sketch is essential here, because the definite integral attaches a negative sign to the area of any region below the $x$-axis (provided that the integral does not run backwards).

### THE STEPS FOR FINDING AN AREA

- The integral does not run backwards when the lower limit is less than the upper limit.
- Sketch the curve, showing the intercepts (and any asymptote).
- Write down and evaluate the definite integral between the required $x$-values.
- If the region is below the $x$-axis, the integral is negative, and the area is its opposite. If the region crosses the $x$-axis, split the integral at the $x$-intercept and find each part separately.
- Write a conclusion, giving the area in square units.

### SKETCHING BY SHIFTING

The graph of $y = e^x - k$, where $k > 0$, is the graph of $y = e^x$ moved down $k$ units. Its $y$-intercept moves down from $1$ to $1 - k$, and its horizontal asymptote moves down from $y = 0$ to $y = -k$. Find the $x$-intercept by putting $y = 0$. For example, $e^x = e^2$ gives $x = 2$, because $e^x$ is an increasing function, so it takes the value $e^2$ only once.

TUTOR DEMO **A region below the $x$-axis**

**a** Use shifting to sketch $y = e^x - e^2$, showing the intercepts and asymptote.
**b** Find the area of the region between this curve, the $x$-axis and the $y$-axis.

YOUR TURN **Same method**

**a** Use shifting to sketch $y = e^x - e^3$, showing the intercepts and asymptote.
**b** Find the area of the region between this curve, the $x$-axis and the $y$-axis.

**WORKED EXAMPLE 17** **Keep for homework**

**a** Use shifting to sketch $y = e^x - e$, showing the intercepts and asymptote.
**b** Find the area of the region between this curve, the $x$-axis and the $y$-axis.

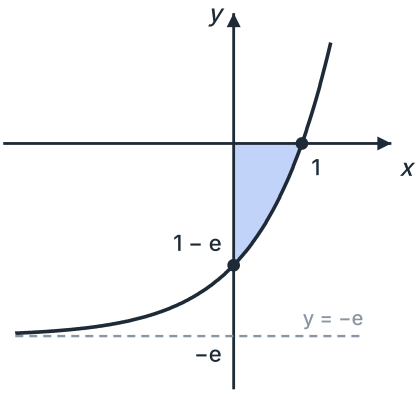
**SOLUTION**

**a** Move the graph of $y = e^x$ down $e$ units.

When $x = 0$, $y = e^0 - e$
$= 1 - e.$

When $y = 0$, $e^x = e$
$x = 1.$

The horizontal asymptote moves down to $y = -e$.




**b**

$$\int_0^1 (e^x - e)\,dx = \Big[e^x - ex\Big]_0^1 \qquad (\text{the number } e \text{ is a constant})$$
$$= (e^1 - e) - (e^0 - 0)$$
$$= -1.$$

This integral is negative because the region is below the $x$-axis.
Hence the required area is $1$ square unit.

**Note:** The primitive of the constant $e$ is $ex$, just as the primitive of $3$ is $3x$. In the second line, $e^1 - e = 0$ and $e^0 - 0 = 1$, so the value is $0 - 1 = -1$.

## CONCEPT 2 · areas between curves

**FINDING AREAS BETWEEN CURVES**

If a curve $y = f(x)$ is always above $y = g(x)$ in an interval $a \le x \le b$, then the area of the region between the curves is

$$\text{area between the curves} = \int_a^b \big(f(x) - g(x)\big)\,dx.$$

**TOP CURVE MINUS BOTTOM CURVE**

The formula is the top curve minus the bottom curve. It still works when part of the region lies below the $x$-axis, because only the difference $f(x) - g(x)$ matters, and it is positive throughout the interval.

TUTOR DEMO **Top curve minus bottom curve**

**a** Sketch the curves $y = e^x$ and $y = x^2$ in the interval $-3 \leq x \leq 3$.
**b** Find the area of the region between the curves, from $x = 0$ to $x = 3$.

YOUR TURN **Same method**

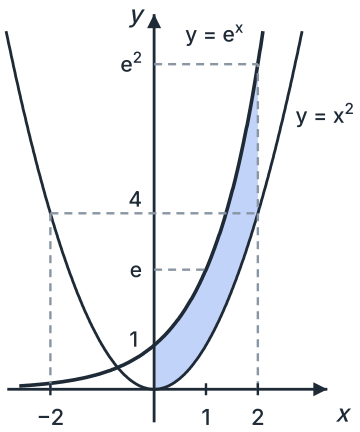
**a** Sketch the curves $y = e^x$ and $y = x^2$ in the interval $-1 \leq x \leq 1$.
**b** Find the area of the region between the curves, from $x = 0$ to $x = 1$.

**WORKED EXAMPLE 18** **Keep for homework**

**a** Sketch the curves $y = e^x$ and $y = x^2$ in the interval $-2 \leq x \leq 2$.
**b** Find the area of the region between the curves, from $x = 0$ to $x = 2$.

**SOLUTION**

**a** The graphs are drawn to the right.
Note that for $x > 0$, $y = e^x$ is always above $y = x^2$.



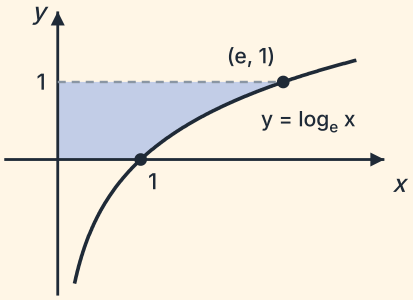

**b** Using the standard formula above,

$$\begin{aligned}\text{area} &= \int_0^2 \left(e^x - x^2\right) dx \\ &= \left[e^x - \tfrac{1}{3}x^3\right]_0^2 \\ &= \left(e^2 - \tfrac{8}{3}\right) - \left(e^0 - 0\right) \\ &= e^2 - 3\tfrac{2}{3} \text{ square units.}\end{aligned}$$

**Note:** The first line writes the formula with $f(x) = e^x$ on top and $g(x) = x^2$ below. In the third line, $\frac{1}{3} \times 2^3 = \frac{8}{3}$ and $e^0 = 1$, so the value is $e^2 - \frac{8}{3} - 1 = e^2 - 3\frac{2}{3}$.

**INTEGRATING WITH RESPECT TO $y$ (FROM SECTION 5F)**

A region bounded by a logarithmic curve is often easier to find by integrating with respect to $y$. Write $x$ as a function of $y$, then integrate between $y$-values, using the right-hand curve minus the left-hand curve.




For example, the region between $y = \log_e x$, the $y$-axis and the lines $y = 0$ and $y = 1$ is shown. Since $y = \log_e x$, we have $x = e^y$ by the definition of logarithms. So

$$\begin{aligned}\text{area} &= \int_0^1 e^y \, dy \\ &= \left[e^y\right]_0^1 \\ &= e^1 - e^0 \\ &= e - 1 \text{ square units.}\end{aligned}$$

## TEXTBOOK HOMEWORK · Exercise 6E

**Technology:** Graphing programs that can calculate the areas of specified regions may make the problems in this exercise clearer, particularly when no diagram has been given.

### FOUNDATION

**1** **a** Use the standard form $\int e^x\,dx = e^x + C$ to evaluate each definite integral. Then approximate it correct to two decimal places.

i $\int_0^1 e^x\,dx$ ii $\int_{-1}^0 e^x\,dx$ iii $\int_{-2}^0 e^x\,dx$ iv $\int_{-3}^0 e^x\,dx$

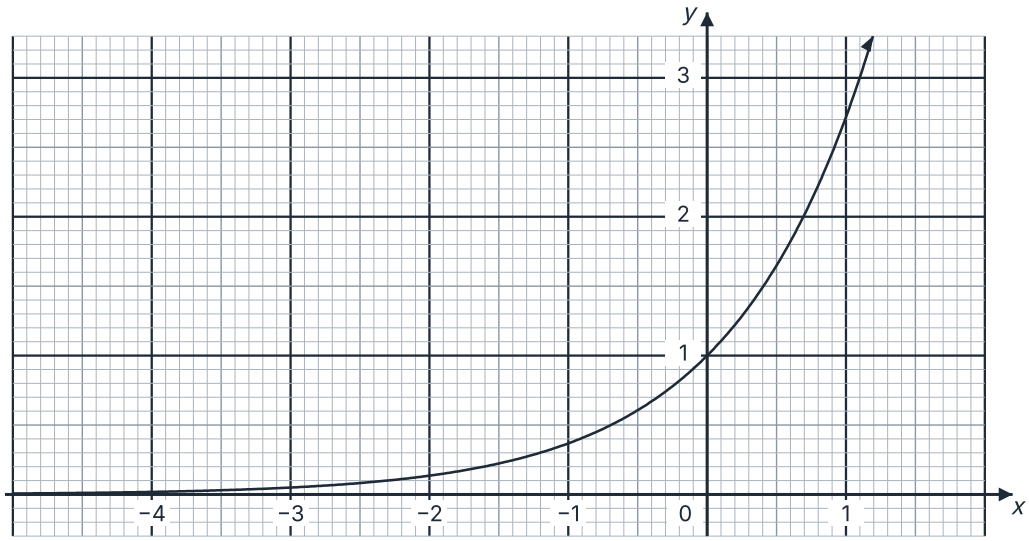
**b** The graph below shows $y = e^x$ from $x = -5$ to $x = 1$, with a scale of 10 divisions to 1 unit, so that 100 little squares equal 1 square unit. By counting squares under the curve from $x = 0$ to $x = 1$, find an approximation to $\int_0^1 e^x\,dx$, and compare it with the approximation obtained in part **a**.

**c** Count squares to the left of the $y$-axis to obtain approximations to:

i $\int_{-1}^0 e^x\,dx$, ii $\int_{-2}^0 e^x\,dx$, iii $\int_{-3}^0 e^x\,dx$,

and compare the results with the approximations obtained in part **a**.

**d** Continue counting squares to the left of $x = -3$, and estimate the total area under the curve to the left of the $y$-axis.



**2** Answer these questions first in exact form, then correct to four significant figures. In each case use the standard form $\int e^{ax+b}\,dx = \frac{1}{a}\,e^{ax+b} + C$.

a Find the area between the curve $y = e^{2x}$ and the $x$-axis from $x = 0$ to $x = 3$.

b Find the area between the curve $y = e^{-x}$ and the $x$-axis from $x = 0$ to $x = 1$.

c Find the area between the curve $y = e^{\frac{1}{3}x}$ and the $x$-axis from $x = -3$ to $x = 0$.

**3** In each case find the area between the $x$-axis and the given curve between the given $x$-values. Use the standard form $\int e^{ax+b}\,dx = \frac{1}{a}\,e^{ax+b} + C$.

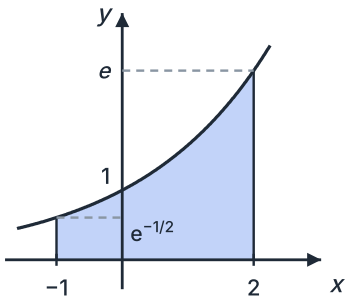
a $y = e^{x+3}$, for $-2 \leq x \leq 0$

b $y = e^{2x-1}$, for $0 \leq x \leq 1$

c $y = e^{-2x-1}$, for $-2 \leq x \leq -1$

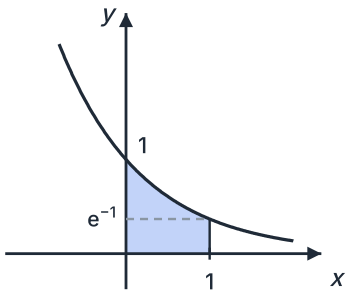
d $y = e^{\frac{1}{3}x+2}$, for $0 \leq x \leq 3$

**4** a




Find the area of the region bounded by the curve $y = e^{\frac{1}{2}x}$, the $x$-axis, and the lines $x = -1$ and $x = 2$.

b

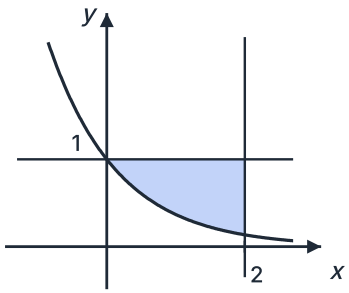



Find the area of the region bounded by the curve $y = e^{-x}$, the $x$-axis, the $y$-axis and the line $x = 1$.

**5** a Find the area between the curve $y = e^{x} + e^{-x}$ and the $x$-axis, from $x = -2$ to $x = 2$.

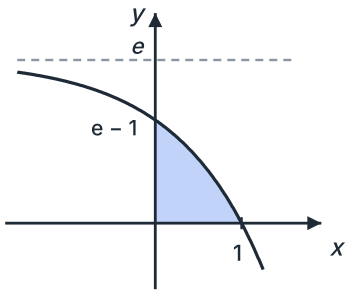
b Find the area between the curve $y = x^{2} + e^{x}$ and the $x$-axis, from $x = -3$ to $x = 3$.

**6** **a**



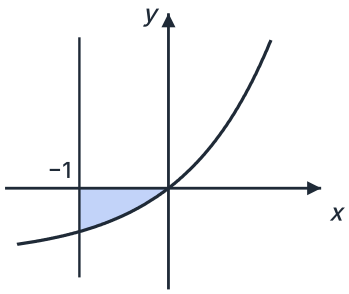

Find the area of the region bounded by the curve $y = e^{-x}$ and the lines $x = 2$ and $y = 1$.

**b**



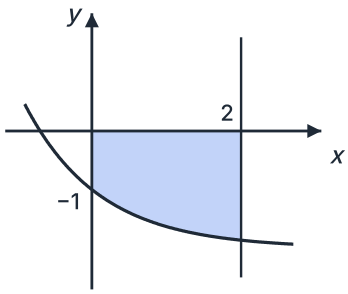

Find the area of the region in the first quadrant bounded by the coordinate axes and the curve $y = e - e^{x}$.

**c**



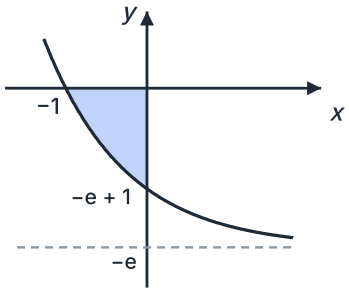

Find the area between the $x$-axis, the curve $y = e^{x} - 1$ and the line $x = -1$.

**d**



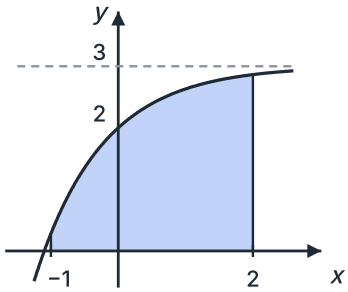

What is the area bounded by $x = 2$, $y = e^{-x} - 2$, the $x$-axis and the $y$-axis?

**e**




Find the area of the region bounded by the curve $y = e^{-x} - e$ and the coordinate axes.

**f**



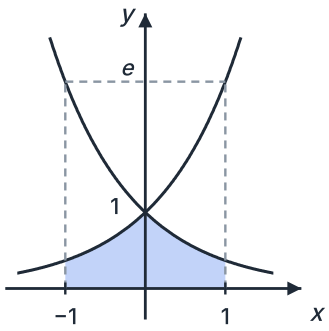

Find the area of the region bounded by the curve $y = 3 - e^{-x}$, the $x$-axis, and the lines $x = -1$ and $x = 2$.

DEVELOPMENT

**7** a Sketch the curves $y = e^x$ and $y = x + 1$, and shade the region between them, from $x = 0$ to $x = 1$. Then write down the area of this region as an integral and evaluate it.

b Sketch the curves $y = e^x$ and $y = 1 - x$, and shade the region between them, from $x = 0$ to $x = 1$. Then write down the area of this region as an integral and evaluate it.

**8** The diagram to the right shows the region above the $x$-axis, below both $y = e^x$ and $y = e^{-x}$, between $x = -1$ and $x = 1$.

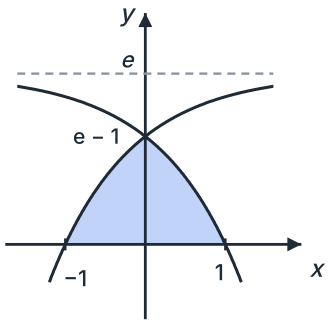



a Explain why the area of this region may be written as

$$\text{area} = 2\int_0^1 e^{-x}\,dx.$$

b Hence find the area of this region.

**9** The diagram to the right shows the region above the $x$-axis, below both $y = e - e^{-x}$ and $y = e - e^{x}$.

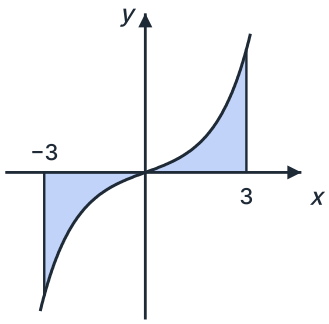



a Explain why the area of this region may be written as

$$\text{area} = 2\int_0^1 (e - e^x)\,dx.$$

b Hence find the area of this region.

**10** The diagram to the right shows the region between the curve $y = e^{x} - e^{-x}$, the $x$-axis and the lines $x = -3$ and $x = 3$.




a Show that $y = e^{x} - e^{-x}$ is an odd function.

b Hence write down the value of $\int_{-3}^{3} (e^{x} - e^{-x})\,dx$ without finding a primitive.

c Explain why the area of this region may be written as

$$\text{area} = 2\int_0^3 (e^x - e^{-x})\,dx.$$

d Hence find the area of this region.

**11** **a** Show that the curves $y = x^2$ and $y = e^{x+1}$ intersect at $x = -1$.

**b** Hence sketch the region in the second quadrant between these two curves and the $y$-axis.

**c** Find its area.

**12** Sketch the region between the graphs of $y = e^x$ and $y = x$, between the $y$-axis and $x = 2$, then find its area.

**13** In this question, give all approximations correct to four decimal places.

**a** Find the area between the curve $y = e^x$ and the $x$-axis, for $0 \leq x \leq 1$, by evaluating an appropriate integral. Then approximate the result.

**b** Estimate the area using the trapezoidal rule with two subintervals (that is, with three function values).

**c** Is the trapezoidal-rule approximation greater than or less than the exact value? Give a geometric explanation.

**14** a Differentiate $e^{-x^2}$ and hence write down a primitive of $xe^{-x^2}$.

b Hence find the area between the curve $y = xe^{-x^2}$ and the $x$-axis from $x = 0$ to $x = 2$, and from $x = -2$ to $x = 2$.

**15** a (i) Evaluate the integral $\int_N^0 e^x \, dx$. (ii) What is its limit as $N \to -\infty$?

b (i) Evaluate the integral $\int_0^N e^{-x} \, dx$. (ii) What is its limit as $N \to \infty$?

c Similarly, evaluate $\int_0^N 2xe^{-x^2} \, dx$, and find its limit as $N \to \infty$.

**ENRICHMENT**

**16** a Find $\int_\delta^1 \frac{e^{\sqrt{x}}}{\sqrt{x}} \, dx$.

b What happens to the integral as $\delta \to 0^+$?

**17**

a Differentiate $x\,e^{-x}$, and hence find $\displaystyle\int_0^N x\,e^{-x}\,dx$.

b Find the limit of this integral as $N \to \infty$ (use the dominance results).

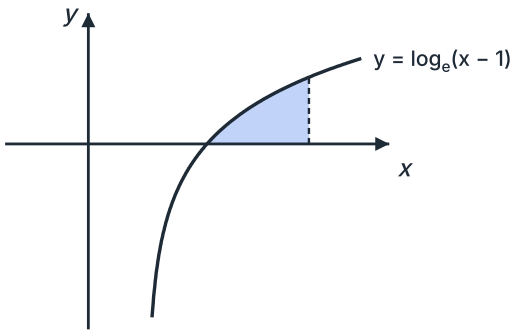
c Differentiate $x^2\,e^{-x}$, and hence find $\displaystyle\int_0^\infty x^2 e^{-x}\,dx$.

## PAST-PAPER HOMEWORK · mini exam

**Suggested time: 20 minutes** **14 marks** Calculator allowed · show working · each hint is below its question if you get stuck

**PP 1** **Pymble Ladies' College, 2023, Mathematics Extension 1 HSC Trial Examination Term 3, Q13(d)** 3 marks

Part of the graph of $y = \log_e(x - 1)$ is given.



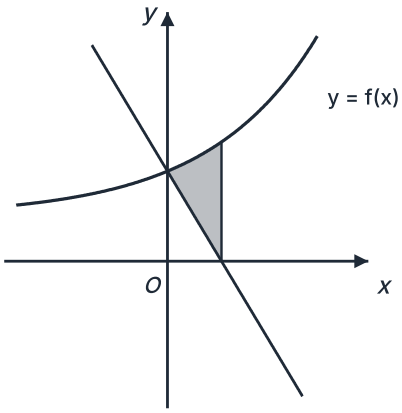

(i) Find the value of $y$ when $x = e + 1$. 1

(ii) Find the shaded area, bounded by the curve, the $x$-axis and the line $x = e + 1$. 2

**Hint:** what does the diagram tell you about the boundaries of the shaded region?

**PP 2** **Fort Street High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(c)** 4 marks

The graph of $f(x) = e^{\frac{x}{2}} + 1$ is shown. The normal to the graph of $f$ where it crosses the $y$-axis is also shown.




(i) Find the equation of the **normal** to the graph of $f$ where it crosses the $y$-axis. 2

(ii) Find the exact area of the shaded region. 2

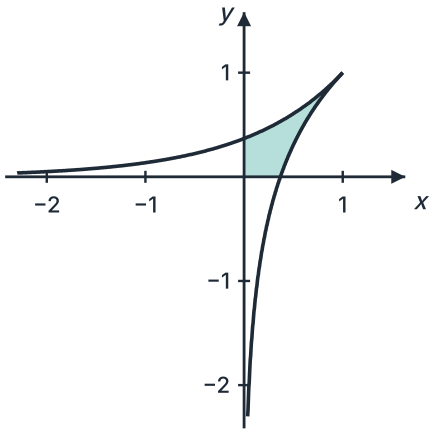
**Hint:** what does the diagram tell you about the boundaries of the shaded region?

**PP 3** **Caringbah High School, 2020, Higher School Certificate Trial Examination, Mathematics Extension 1, Q14(c)(i)(ii)** 4 marks

The function $f(x) = 1 + \ln x$ is defined in the domain $(0, 1]$.

(i) Show that $\dfrac{d}{dx}(x \ln x) = 1 + \ln x$ 1

The diagram shows the graphs of the function $y = f(x)$ and the inverse function $y = f^{-1}(x)$.



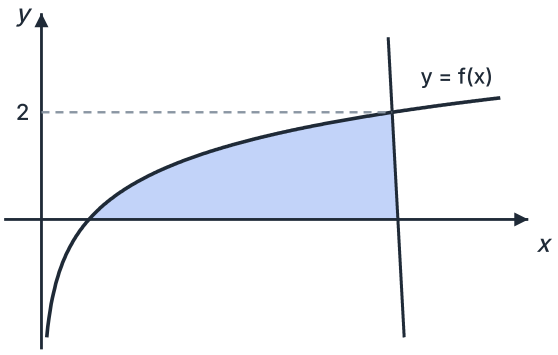

(ii) Find in simplest exact form the area of the shaded region in the first quadrant bounded by the curves $y = f(x)$, $y = f^{-1}(x)$ and the coordinate axes. 3

**Hint:** what does the diagram tell you about the boundaries of the shaded region?

**PP 4** **Saint Ignatius' College, Riverview, 2024, Mathematics Assessment Task, Year 12 Mathematics (Extension One) Task 4, Trial HSC Examination, Q13(d)(ii)** 3 marks

Let $f(x) = \ln\left(\frac{2x}{3}\right)$. Part (i), not included here, shows that the normal to the graph of $y = f(x)$ at the point where $x = \frac{3e^2}{2}$ has the equation $y = -\frac{3e^2x}{2} + \frac{9e^4}{4} + 2$.

(ii) Find the exact area enclosed by the graph of $y = f(x)$, the line $y = -\frac{3e^2x}{2} + \frac{9e^4}{4} + 2$ and the $x$ axis. 3




**Hint:** what does the diagram tell you about the boundaries of the shaded region?

The school trial papers used for this pack (2020–2024) have only these four questions that fit this lesson, so the mini exam is short. For more timed practice, do Questions 6(f), 9(b) and 11(c) above in 10 minutes.

LESSON 6I

# Integration of the reciprocal function

The usual rule for integrating powers of $x$ fails for $\frac{1}{x} = x^{-1}$, because it divides by zero. Reversing the standard form $\frac{d}{dx}\log_e x = \frac{1}{x}$ gives the missing primitive, $\int \frac{1}{x}\,dx = \log_e |x| + C$. We extend it to both sides of the origin, learn the three standard forms of Box 10, evaluate definite integrals without crossing an asymptote, find a function from its derivative, and find a primitive of $\log_e x$.

## CONCEPT 1 · the primitive of the reciprocal function

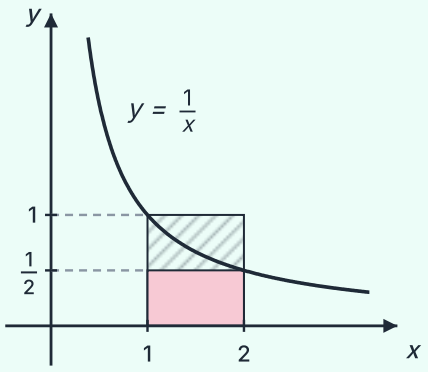
### WHY THE POWER RULE FAILS FOR $y = \frac{1}{x}$

The reciprocal function $y = \frac{1}{x}$ is an important function. We have seen that it is required whenever two quantities are inversely proportional to each other. So far, however, it has not been possible to integrate the reciprocal function, because the usual rule for integrating powers of $x$ gives nonsense:

$$\text{When } n = -1, \quad \int x^n\,dx = \frac{x^{n+1}}{n+1} \quad \text{gives} \quad \int x^{-1}\,dx = \frac{x^0}{0},$$

which is nonsense because of the division by zero.

Yet the graph of $y = \frac{1}{x}$ to the right shows that there should be no problem with definite integrals involving $\frac{1}{x}$, provided that the integral does not cross the discontinuity at $x = 0$. For example, the diagram shows the integral $\int_1^2 \frac{1}{x}\,dx$, which the little rectangles show has a value between $\frac{1}{2}$ and $1$.




### INTEGRATION OF THE RECIPROCAL FUNCTION

Reversing the standard form for differentiating $\log x$ will now give the necessary standard forms for integrating $\frac{1}{x}$.

Reversing $\frac{d}{dx}\log_e x = \frac{1}{x}$ gives $\int \frac{1}{x}\,dx = \log_e x + C.$

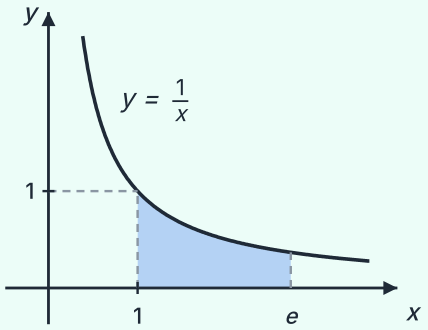
This is a new standard form for integrating the reciprocal function. The only qualification is that $x > 0$, otherwise $\log_e x$ is undefined, so we have

$$\int \frac{1}{x}\,dx = \log_e x + C, \quad \text{for some constant } C, \text{ provided that } x > 0.$$

### A CHARACTERISATION OF $e$

Integrating the reciprocal function from $1$ to $e$ gives an amazingly simple result:

$$\begin{aligned}\int_1^e \frac{1}{x}\,dx &= \Big[\log_e x\Big]_1^e \\ &= \log_e e - \log_e 1 \\ &= 1 - 0 \\ &= 1.\end{aligned}$$




The integral is sketched to the right. The example is very important because it characterises $e$ as the real number satisfying $\int_1^e \frac{1}{x}\,dx = 1$. In other expositions of the theory, this integral is taken as the definition of $e$.

TUTOR DEMO **Integrating $\frac{1}{x}$**

**a** Find the definite integral $\int_1^3 \frac{1}{x}\,dx$.

**b** Approximate the integral correct to three decimal places and verify that $\frac{2}{3} < \int_1^3 \frac{1}{x}\,dx < 2$.

**c** Find $\int_1^{e^3} \frac{1}{x}\,dx$.

**YOUR TURN** **Same method**

**a** Find the definite integral $\int_1^5 \frac{1}{x}\,dx$.

**b** Approximate the integral correct to three decimal places and verify that $\frac{4}{5} < \int_1^5 \frac{1}{x}\,dx < 4$.

**c** Find $\int_1^{e^4} \frac{1}{x}\,dx$.

**WORKED EXAMPLE 33** **Keep for homework**

**a** Find the definite integral $\int_1^2 \frac{1}{x}\,dx$ sketched above.

**b** Approximate the integral correct to three decimal places and verify that $\frac{1}{2} < \int_1^2 \frac{1}{x}\,dx < 1$.

**SOLUTION**

**a**

$$\begin{aligned}\int_1^2 \frac{1}{x}\,dx &= \Big[\log_e x\Big]_1^2 \\ &= \log_e 2 - \log_e 1 \\ &= \log_e 2, \quad \text{because } \log_e 1 = 0.\end{aligned}$$

**b** Hence $\int_1^2 \frac{1}{x}\,dx \doteq 0.693$, which is indeed between $\frac{1}{2}$ and $1$, as the diagram above indicated.

**READING THE RECTANGLES IN WORKED EXAMPLE 33**

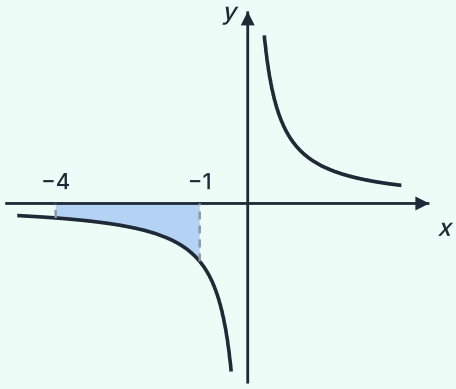
In the diagram, both rectangles have width $1$. The lower (shaded) rectangle has height $\frac{1}{2}$, the value of $\dfrac{1}{x}$ at $x = 2$, so its area is $\frac{1}{2}$. The upper (hatched) rectangle has height $1$, the value of $\dfrac{1}{x}$ at $x = 1$, so its area is $1$. Between $x = 1$ and $x = 2$ the curve lies between these two heights, so the area under it lies between $\frac{1}{2}$ and $1$. On the calculator, $\log_e 2 = 0.693\,147\ldots$

## CONCEPT 2 · the primitive on both sides of the origin

### THE PRIMITIVE OF $y = \dfrac{1}{x}$ ON BOTH SIDES OF THE ORIGIN

So far our primitive is restricted by the condition $x > 0$, meaning that we can only deal with definite integrals on the right-hand side of the origin. The full graph of the reciprocal function $y = \dfrac{1}{x}$, however, is a hyperbola, with two disconnected branches separated by the discontinuity at $x = 0$.

There is no reason why we should not integrate over a closed interval such as $-4 \leq x \leq -1$ on the left-hand side of the origin. We can take any definite integrals of $\dfrac{1}{x}$, provided only that we do not work across the asymptote at $x = 0$. If $x$ is negative, then $\log(-x)$ is well defined, and using our previous standard forms,




$$\frac{d}{dx}\log(-x) = -\left(\frac{1}{-x}\right) = \frac{1}{x}.$$

Reversing this, $\log(-x)$ is a primitive of $\dfrac{1}{x}$ when $x$ is negative,

$$\int \frac{1}{x}\,dx = \log_e(-x) + C, \quad \textbf{provided that } x < 0.$$

The absolute value function is designed for just these situations. We can combine the two results into one standard form for the whole reciprocal function,

$$\int \frac{1}{x}\,dx = \log_e|x| + C, \quad \textbf{provided that } x \neq 0.$$

Question 16 of Exercise 6G gives more detail about this standard form.

### THE DERIVATIVE OF $\log_e(-x)$ IN FULL

The line $\dfrac{d}{dx}\log(-x) = -\left(\dfrac{1}{-x}\right)$ uses the chain rule. Written out:

Let $y = \log_e(-x)$, where $x < 0$.

Let $u = -x$, so $y = \log_e u$.

Then $\dfrac{du}{dx} = -1$ and $\dfrac{dy}{du} = \dfrac{1}{u}$.

Hence $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx} = \dfrac{1}{-x} \times (-1) = \dfrac{1}{x}$.

### EACH BRANCH MAY HAVE ITS OWN CONSTANT OF INTEGRATION

Careful readers will realise that because $y = \dfrac{1}{x}$ has two disconnected branches, there can be different constants of integration in the two branches. So the general primitive of $\dfrac{1}{x}$ is

$$\int \frac{1}{x}\,dx = \begin{cases} \log_e x + A, & \text{for } x > 0, \\ \log(-x) + B, & \text{for } x < 0, \end{cases} \quad \textbf{where } A \textbf{ and } B \textbf{ are constants.}$$

If a boundary condition is given for one branch, this has no implication at all for the constant of integration in the other branch.

In any physical interpretation, however, the function would normally have meaning in only one of the two branches, so the complication discussed here is rarely needed, and the over-simplified forms in Box 10 below are standard and generally used. The qualification is understood and taken account of when necessary.

### THREE STANDARD FORMS

As always, reversing the other standard forms for differentiation gives two more standard forms.

**WHERE THE SECOND AND THIRD FORMS COME FROM**

Reversing each derivative, as for $\dfrac{1}{x}$:

Reversing $\dfrac{d}{dx}\log_e(ax+b) = \dfrac{a}{ax+b}$

gives $\displaystyle\int \frac{a}{ax+b}\,dx = \log_e|ax+b|,$

and dividing through by $a$, $\displaystyle\int \frac{1}{ax+b}\,dx = \frac{1}{a}\log_e|ax+b| + C.$

Reversing $\dfrac{d}{dx}\log_e u = \dfrac{u'}{u}$

gives $\displaystyle\int \frac{u'}{u}\,dx = \log_e|u| + C.$

**10 STANDARD FORMS FOR INTEGRATING RECIPROCAL FUNCTIONS**

- $\displaystyle\int \frac{1}{x}\,dx = \log_e|x| + C$
- $\displaystyle\int \frac{1}{ax+b}\,dx = \frac{1}{a}\log_e|ax+b| + C$
- $\displaystyle\int \frac{u'}{u}\,dx = \log_e|u| + C \qquad \text{OR} \qquad \int \frac{f'(x)}{f(x)}\,dx = \log_e|f(x)| + C$

No calculation involving these primitives may cross an asymptote.

**NEVER CROSS AN ASYMPTOTE**

The final warning always applies to the primitive of any function, but it is mentioned here because it is such an obvious issue.

**CHECKING FOR AN ASYMPTOTE**

Before substituting the limits, find where the denominator is zero. If that $x$-value lies between the limits, the definite integral is meaningless.

TUTOR DEMO **The first two standard forms**

Evaluate these definite integrals using the first two standard forms above.

a $\int_{e}^{e^3} \frac{4}{x}\,dx$ b $\int_{1}^{3} \frac{1}{1-3x}\,dx$ c $\int_{1}^{4} \frac{1}{x-3}\,dx$ d $\int_{-4}^{-1} \frac{1}{x}\,dx$

YOUR TURN **Same method**

Evaluate these definite integrals using the first two standard forms above.

a $\int_{e}^{e^4} \frac{2}{x}\,dx$ b $\int_{1}^{2} \frac{1}{1-2x}\,dx$ c $\int_{0}^{3} \frac{1}{x-1}\,dx$ d $\int_{-6}^{-2} \frac{1}{x}\,dx$

**WORKED EXAMPLE 34** **Keep for homework**

Evaluate these definite integrals using the first two standard forms above.

a $\displaystyle\int_e^{e^2} \frac{5}{x}\,dx$ b $\displaystyle\int_1^4 \frac{1}{1-2x}\,dx$ c $\displaystyle\int_1^5 \frac{1}{x-2}\,dx$

**SOLUTION**

**a**

$$\begin{aligned}\int_e^{e^2} \frac{5}{x}\,dx &= 5\Big[\log_e|x|\Big]_e^{e^2}\\ &= 5\big(\log_e e^2 - \log_e e\big)\\ &= 5(2-1)\\ &= 5\end{aligned}$$

**b** (here $a=-2$ and $b=1$)

$$\begin{aligned}\int_1^4 \frac{1}{1-2x}\,dx &= -\tfrac{1}{2}\Big[\log_e|1-2x|\Big]_1^4\\ &= -\tfrac{1}{2}\big(\log_e|-7| - \log_e|-1|\big)\\ &= -\tfrac{1}{2}\big(\log_e 7 - 0\big)\\ &= -\tfrac{1}{2}\log_e 7\end{aligned}$$

**c** This definite integral is meaningless because it crosses the asymptote at $x=2$.

**THE ARITHMETIC IN WORKED EXAMPLE 34**

In part **a**, $\log_e e^2 = 2$ and $\log_e e = 1$. In part **b**, $1-2\times 4 = -7$ and $1-2\times 1 = -1$, and $\log_e|-1| = \log_e 1 = 0$. In part **c**, the denominator $x-2$ is zero at $x=2$, which lies between the limits $1$ and $5$.

## CONCEPT 3 · using the third standard form

**USING THE THIRD STANDARD FORM**

The vital point in using the third standard form,

$$\int \frac{u'}{u}\,dx = \log_e|u| \qquad \text{OR} \qquad \int \frac{f'(x)}{f(x)}\,dx = \log_e|f(x)|,$$

is that the top must be the derivative of the bottom. Choose whichever form of the reverse chain rule you are most comfortable with.

**STEPS FOR THE THIRD STANDARD FORM**

- Let $u$ (or $f(x)$) be the bottom, and find its derivative $u'$.
- If the top is $u'$, the primitive is $\log_e|u|$.
- If the top is a constant multiple of $u'$, first make the top the derivative of the bottom: multiply the top by a constant, and multiply outside the integral by its reciprocal (Worked example 35b).

TUTOR DEMO **The top is the derivative of the bottom**

Evaluate these definite integrals using the third standard form above.

a $\int_0^2 \frac{2x}{x^2+5}\,dx$

b $\int_3^5 \frac{x}{1-x^2}\,dx$

c $\int_0^3 \frac{x^2}{8-x^3}\,dx$

YOUR TURN **Same method**

Evaluate these definite integrals using the third standard form above.

a $\int_0^1 \frac{2x}{x^2+3}\,dx$

b $\int_2^4 \frac{x}{1-x^2}\,dx$

c $\int_0^3 \frac{2x}{4-x^2}\,dx$

**WORKED EXAMPLE 35** **Keep for homework**

Evaluate these definite integrals using the third standard form above.

a $\displaystyle\int_0^1 \frac{2x}{x^2+2}\,dx$ b $\displaystyle\int_4^5 \frac{x}{9-x^2}\,dx$ c $\displaystyle\int_0^2 \frac{3x}{1-x^3}\,dx$

**SOLUTION**

**a**

Let $u = x^2+2$ OR $f(x) = x^2+2.$

Then $u' = 2x$ $f'(x) = 2x.$

Hence in the fraction $\dfrac{2x}{x^2+2}$, the top is the derivative of the bottom.

Thus, using $\displaystyle\int \frac{u'}{u}\,dx = \log_e|u|$ OR $\displaystyle\int \frac{f'(x)}{f(x)}\,dx = \log_e|f(x)|,$

$$\begin{aligned}\int_0^1 \frac{2x}{x^2+2}\,dx &= \Big[\log_e(x^2+2)\Big]_0^1 \\ &= \log_e 3 - \log_e 2.\end{aligned}$$

**Note:** The use of absolute value signs here is unnecessary (but is not wrong) because $x^2+2$ is never negative.

**b**

Let $u = 9-x^2$ OR $f(x) = 9-x^2.$

Then $u' = -2x$ $f'(x) = -2x.$

The first step is to make the top the derivative of the bottom,

$$\int_4^5 \frac{x}{9-x^2}\,dx = -\tfrac{1}{2}\int_4^5 \frac{-2x}{9-x^2}\,dx, \text{ of the form } \int \frac{u'}{u}\,dx \text{ OR } \int \frac{f'(x)}{f(x)}\,dx,$$

$$\begin{aligned}&= -\tfrac{1}{2}\Big[\log_e|9-x^2|\Big]_4^5 \\ &= -\tfrac{1}{2}\big(\log_e|-16| - \log_e|-7|\big) \\ &= -\tfrac{1}{2}\big(4\log_e 2 - \log_e 7\big) \\ &= -2\log_e 2 + \tfrac{1}{2}\log_e 7.\end{aligned}$$

**c** This definite integral is meaningless because it crosses the asymptote at $x = 1$.

**THE ARITHMETIC IN WORKED EXAMPLE 35**

In part **b**, $9-5^2 = -16$ and $9-4^2 = -7$, and $\log_e 16 = \log_e 2^4 = 4\log_e 2$. In part **c**, the denominator $1-x^3$ is zero at $x = 1$, which lies between the limits $0$ and $2$.

## CONCEPT 4 · given the derivative, find the function

**GIVEN THE DERIVATIVE, FIND THE FUNCTION**

Finding the function from the derivative involves a constant that can be found if the value of $y$ is known for some value of $x$.

**STEPS FOR FINDING THE FUNCTION**

- Take the primitive of the derivative, including the constant $C$.
- Substitute the known value to find $C$.
- Write the function with the value of $C$ in it, then use it. Check that the known point and the $x$-value you use are on the same side of any asymptote.

TUTOR DEMO **Find the constant**

**a** Find $f(x)$, if $f'(x) = \dfrac{3}{2-x}$ and the graph passes through the origin.

**b** Hence find $f(1)$.

YOUR TURN **Same method**

**a** Find $f(x)$, if $f'(x) = \dfrac{4}{5-x}$ and the graph passes through the origin.

**b** Hence find $f(4)$.

WORKED EXAMPLE 36 **Keep for homework**

**a** Find $f(x)$, if $f'(x) = \dfrac{2}{3-x}$ and the graph passes through the origin.

**b** Hence find $f(2)$.

**SOLUTION**

**a**

Here $f'(x) = \dfrac{2}{3-x}.$

Taking the primitive, $f(x) = -2\ln|3-x| + C,$ for some constant $C$.

Because $f(0) = 0$, $0 = -2\ln 3 + C$

$C = 2\ln 3.$

Hence $f(x) = 2\ln 3 - 2\ln|3-x|.$

**b**

Substituting $x = 2$ gives $f(2) = 2\ln 3 - 2\ln 1$

$= 2\ln 3.$

**Note:** As remarked above, this working is over-simplified, because each branch may have its own constant of integration. But there is no problem in this question because the asymptote is at $x = 3$, so that the given point $(0, 0)$ on the curve, and the value $x = 2$ in part **b**, are both on the same side of the asymptote.

**THE PRIMITIVE IN WORKED EXAMPLE 36**

In the second line, $\dfrac{2}{3-x} = 2 \times \dfrac{1}{-x+3}$, with $a = -1$ and $b = 3$, so the primitive is $2 \times \dfrac{1}{-1} \ln|3-x| = -2\ln|3-x|$. In part **b**, $|3-2| = 1$ and $\ln 1 = 0$.

## CONCEPT 5 · a primitive of log x

**A PRIMITIVE OF $\log_e x$**

The next worked example is more difficult, but it is important because it produces a primitive of $\log_e x$, which the theory has not yielded so far. There is no need to memorise the result.

**HOW THE METHOD WORKS**

The method reverses a derivative, as in Section 6D: differentiate a function whose derivative contains $\log_e x$, then reverse the result.

TUTOR DEMO **Reverse a derivative**

**a** Differentiate $x \log_e x - x$, using the product rule for the first term.
**b** Hence write down a primitive of $\log_e x$.
**c** Use this result to evaluate $\displaystyle\int_1^{e^2} \log_e x \, dx$.

YOUR TURN **Same method**

**a** Differentiate $x \log_e x - x$, using the product rule for the first term.

**b** Hence write down a primitive of $\log_e x$.

**c** Use this result to evaluate $\int_e^{e^2} \log_e x \, dx$.

**WORKED EXAMPLE 37** **Keep for homework**

**a** Differentiate $x \log_e x$ by the product rule.
**b** Show by differentiation that $x \log_e x - x$ is a primitive of $\log_e x$.
**c** Use this result to evaluate $\int_1^e \log_e x \, dx$.

**SOLUTION**

**a** Differentiating by the product rule,

$$\begin{aligned}\frac{d}{dx}(x \log_e x) &= vu' + uv' \\ &= \log_e x + x \times \frac{1}{x} \\ &= 1 + \log_e x.\end{aligned}$$

Let $u = x$
and $v = \log_e x$.
Then $u' = 1$
and $v' = \frac{1}{x}$.

**b**

Let $y = x \log_e x - x.$
Then $y' = (1 + \log_e x) - 1,$ using the result of part **a**,
$= \log_e x.$

Reversing this result gives the primitive of $\log_e x$,

$$\int \log_e x \, dx = x \log_e x - x + C.$$

**c** Part **b** can now be used to find the definite integral,

$$\begin{aligned}\int_1^e \log_e x \, dx &= \Big[x \log_e x - x\Big]_1^e \\ &= (e \log_e e - e) - (1 \log_e 1 - 1) \\ &= (e \log_e e - e) - (0 - 1) \\ &= (e - e) + 1 \\ &= 1.\end{aligned}$$

**THE ARITHMETIC IN WORKED EXAMPLE 37**

In part **c**, $\log_e e = 1$, so $e \log_e e - e = e - e = 0$, and $1 \log_e 1 = 1 \times 0 = 0$.

## TEXTBOOK HOMEWORK · Exercise 6I

**FOUNDATION**

**1** First rewrite each integral using the result $\int \frac{k}{x}\,dx = k\int \frac{1}{x}\,dx$, where $k$ is a constant. Then use the standard form $\int \frac{1}{x}\,dx = \log_e|x| + C$ to integrate it.

a $\int \frac{2}{x}\,dx$ b $\int \frac{1}{3x}\,dx$ c $\int \frac{4}{5x}\,dx$ d $\int \frac{3}{2x}\,dx$

**2** Use the standard form $\int \frac{1}{ax+b}\,dx = \frac{1}{a}\log_e|ax+b| + C$ to find these indefinite integrals.

a $\int \frac{1}{4x+1}\,dx$ b $\int \frac{1}{5x-3}\,dx$ c $\int \frac{6}{3x+2}\,dx$

d $\int \frac{15}{5x+1}\,dx$ e $\int \frac{4}{4x+3}\,dx$ f $\int \frac{dx}{3-x}$

g $\int \frac{dx}{7-2x}$ h $\int \frac{4\,dx}{5x-1}$ i $\int \frac{12\,dx}{1-3x}$

**3** Evaluate these definite integrals. Simplify your answers where possible.

a $\displaystyle\int_1^5 \frac{1}{x}\,dx$ b $\displaystyle\int_1^3 \frac{1}{x}\,dx$ c $\displaystyle\int_{-8}^{-2} \frac{1}{x}\,dx$

d $\displaystyle\int_{-3}^{9} \frac{1}{x}\,dx$ e $\displaystyle\int_1^4 \frac{dx}{2x}$ f $\displaystyle\int_{-15}^{-5} \frac{dx}{5x}$

**4** Evaluate these definite integrals, then use the function labelled $\boxed{\ln}$ on your calculator to approximate each integral correct to four significant figures.

a $\displaystyle\int_0^1 \frac{dx}{x+1}$ b $\displaystyle\int_{-7}^{-5} \frac{dx}{x+2}$ c $\displaystyle\int_{-5}^{-2} \frac{dx}{2x+3}$

d $\displaystyle\int_1^2 \frac{3}{5-2x}\,dx$ e $\displaystyle\int_{-1}^1 \frac{3}{7-3x}\,dx$ f $\displaystyle\int_0^{11} \frac{5}{2x-11}\,dx$

**5** Evaluate these definite integrals. Simplify your answers where possible.

a $\displaystyle\int_1^e \frac{dx}{x}$ b $\displaystyle\int_1^{e^2} \frac{dx}{x}$ c $\displaystyle\int_e^{e^4} \frac{dx}{x}$ d $\displaystyle\int_{\sqrt{e}}^e \frac{dx}{x}$

**6** Find primitives of these functions by first writing them as separate fractions.

a $\dfrac{x+1}{x}$ b $\dfrac{x+3}{5x}$ c $\dfrac{1-8x}{9x}$

d $\dfrac{3x^2-2x}{x^2}$ e $\dfrac{2x^2+x-4}{x}$ f $\dfrac{x^4-x+2}{x^2}$

**DEVELOPMENT**

**7** In each case show that the numerator is the derivative of the denominator. Then use the form $\displaystyle\int \frac{u'}{u}\,dx = \log_e|u| + C$ or $\displaystyle\int \frac{f'(x)}{f(x)}\,dx = \log_e|f(x)| + C$ to integrate the expression.

a $\dfrac{2x}{x^2-9}$ b $\dfrac{6x+1}{3x^2+x}$ c $\dfrac{2x+1}{x^2+x-3}$

d $\dfrac{5-6x}{2+5x-3x^2}$ g $\dfrac{e^x}{1+e^x}$ i $\dfrac{e^x-e^{-x}}{e^x+e^{-x}}$

Why is it unnecessary (but not wrong) to use absolute value signs in the answers to parts **g** and **i**?

**8** Use the standard form $\int \frac{1}{ax+b}\,dx = \frac{1}{a}\log_e|ax+b| + C$ to find these indefinite integrals.

a $\int \frac{1}{3x-k}\,dx$ b $\int \frac{1}{mx-2}\,dx$ c $\int \frac{p}{px+q}\,dx$ d $\int \frac{A}{sx-t}\,dx$

**9** Find $f(x)$, and then find $f(2)$, given that:

a $f'(x) = 1 + \frac{2}{x}$ and $f(1) = 1$

b $f'(x) = 2x + \frac{1}{3x}$ and $f(1) = 2$

c $f'(x) = 3 + \frac{5}{2x-1}$ and $f(1) = 0$

d $f'(x) = 6x^2 + \frac{15}{3x+2}$ and $f(1) = 5\ln 5$

**10**

a Given that the derivative of $f(x)$ is $\dfrac{x^2+x+1}{x}$ and $f(1)=1\frac{1}{2}$, find $f(x)$.

b Given that the derivative of $g(x)$ is $\dfrac{2x^3-3x-4}{x^2}$ and $g(2)=-3\ln 2$, find $g(x)$.

**11**

a Find $y$ as a function of $x$ if $y'=\dfrac{1}{4x}$ and $y=1$ when $x=e^2$. Where does this curve meet the $x$-axis on the right-hand side of the origin?

b The gradient of a curve is given by $y'=\dfrac{2}{x+1}$, and the curve passes through the point $(0,1)$. What is the equation of this curve?

c Find $y(x)$, given that $y'=\dfrac{2x+5}{x^2+5x+4}$ and $y=1$ when $x=1$. Hence evaluate $y(0)$.

d Write down the equation of the family of curves with the property $y'=\dfrac{2+x}{x}$. Hence find the curve that passes through $(1,1)$ and evaluate $y$ at $x=2$ for this curve.

e Given that $f''(x)=\dfrac{1}{x^2}$, $f'(1)=0$ and $f(1)=3$, find $f(x)$ and hence evaluate $f(e)$.

**12** Use one of the forms $\int \frac{u'}{u}\,dx = \log_e |u| + C$ OR $\int \frac{f'(x)}{f(x)}\,dx = \log_e |f(x)| + C$ to find:

a $\int \frac{3x^2}{x^3 - 5}\,dx$

b $\int \frac{4x^3 + 1}{x^4 + x - 5}\,dx$

c $\int \frac{x^3 - 3x}{x^4 - 6x^2}\,dx$

d $\int \frac{10x^3 - 7x}{5x^4 - 7x^2 + 8}\,dx$

e $\int_2^3 \frac{3x^2 - 1}{x^3 - x}\,dx$

f $\int_e^{2e} \frac{2x + 2}{x^2 + 2x}\,dx$

**13** a (i) Differentiate $y = x \log_e x - x$.

(ii) Hence find $\int \log_e x \, dx$ and $\int_{\sqrt{e}}^{e} \log_e x \, dx$.

b (i) Find the derivative of $y = 2x^2 \log_e x - x^2$.

(ii) Hence find $\int x \log_e x \, dx$ and $\int_{e}^{2} x \log_e x \, dx$.

c Differentiate $(\log_e x)^2$, and hence find $\int_{\sqrt{e}}^{e} \dfrac{\log_e x}{x} \, dx$.

d Differentiate $\ln(\ln x)$ and hence determine the family of primitives of $\dfrac{1}{x \ln x}$.

**14** Find the value of $a$ if:

a $\int_{1}^{a} \dfrac{1}{x} \, dx = 5$

b $\int_{a}^{e} \dfrac{1}{x} \, dx = 5$

c $\int_{a}^{-1} \dfrac{1}{x} \, dx = -2$

d $\int_{-e}^{a} \dfrac{1}{x} \, dx = -2$

**15** a Find $\displaystyle\int \frac{e^x}{e^x+1}\,dx$.

b Find $\displaystyle\int_1^e \left(x+\frac{1}{x^2}\right)^2 dx$.

c Show that $x\,e^x = e^{x+\log_e x}$, and hence differentiate $x\,e^x$ without using the product rule.

**16** Stella found the primitive of the function $\dfrac{1}{5x}$ by taking out a factor of $\frac{1}{5}$,

$$\int \frac{1}{5x}\,dx = \frac{1}{5}\int \frac{1}{x}\,dx = \tfrac{1}{5}\log_e|x| + C_1, \quad \textbf{for some constant } C_1.$$

Magar used the second standard form in Box 10 with $a=5$ and $b=0$,

$$\int \frac{1}{5x}\,dx = \tfrac{1}{5}\log_e|5x| + C_2, \quad \textbf{for some constant } C_2.$$

Explain what is going on. Will this affect their result when finding a definite integral?

**ENRICHMENT**

**17** A certain curve has gradient $y' = \dfrac{1}{x}$ and its two branches pass through the two points $(-1,2)$ and $(1,1)$. Find the equation of the curve.

**18 A power series for $\log_e(1+x)$:**

**a** Use the formula for the partial sum of a GP to prove that for $t \neq -1$,

$$1 - t + t^2 - t^3 + \cdots + t^{2n} = \frac{1}{1+t} + \frac{t^{2n+1}}{1+t}.$$

**b** Integrate both sides of this result from $t = 0$ to $t = x$ to show that for $x > -1$,

$$\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + \frac{x^{2n+1}}{2n+1} - \int_0^x \frac{t^{2n+1}}{1+t}\,dt.$$

**c** Explain why $\dfrac{t^{2n+1}}{1+t} \leq t^{2n+1}$, for $0 \leq t \leq 1$. Hence prove that for $0 \leq x \leq 1$, the integral

$$\int_0^x \frac{t^{2n+1}}{1+t}\,dt$$

converges to $0$ as $n \to \infty$. Hence show that

$$\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots, \quad \text{for } 0 \leq x \leq 1.$$

**d** (i) Use this series to approximate $\log_e \frac{3}{2}$ correct to two decimal places.
(ii) Write down the series converging to $\log_e 2$, called the *alternating harmonic series*.

**e** With a little more effort, it can be shown that the series in part **c** converges to the given limit for $-1 < x \leq 1$ (the proof is a reasonable challenge). Use this to write down the series converging to $\log_e(1-x)$ for $-1 \leq x < 1$, and hence approximate $\log_e \frac{1}{2}$ correct to two decimal places.

**f** Use both series to show that for $-1 < x < 1$,

$$\log_e\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right).$$

Use this result and an appropriate value of $x$ to find $\log_e 3$ correct to five significant figures.

## PAST-PAPER HOMEWORK · mini exam

**Suggested time: 5 minutes** **2 marks** Calculator allowed · show working · the hint is below the question if you get stuck

**PP 1** **North Sydney Boys High School, 2022, HSC Course Assessment Task 3-Trial Examination, Mathematics Extension 1, Q13(a)(i)** 2 marks

Find $\int \frac{1-\ln x}{x \ln x}\, dx$

**Hint:** what is being integrated here?

The school trial papers used for this pack (2020–2024) have only this one question that fits this lesson, so the mini exam is short. For more timed practice, do Questions 4(d), 7(d), 9(d), 11(c), 12(f) and 13(c) above in 25 minutes.

LESSON 6J

# Applications of integration of 1/x

The usual applications of integration can now be applied to the reciprocal function, whose primitive was previously unavailable. We find the area between a curve such as $y = \frac{2}{x} - 1$ and the $x$-axis, splitting the integral where the curve crosses the axis and never crossing the asymptote $x = 0$, and the area between a hyperbola and a line, using the top curve minus the bottom curve.

## CONCEPT 1 · areas between a curve and the x-axis

### APPLICATIONS OF INTEGRATION OF $1/x$

The usual applications of integration can now be applied to the reciprocal function, whose primitive was previously unavailable.

### THE STANDARD FORMS FROM LESSON 6I

- $\int \frac{1}{x}\,dx = \log_e |x| + C$
- $\int \frac{1}{ax+b}\,dx = \frac{1}{a}\log_e |ax+b| + C$
- $\int \frac{f'(x)}{f(x)}\,dx = \log_e |f(x)| + C$

Remember that $\log_e 1 = 0$ and $\log_e e = 1$. When $x > 0$, $|x| = x$, so $\log_e |x|$ may be written as $\log_e x$.

### THE STEPS FOR FINDING AN AREA

- Sketch the curve, showing the intercepts and the asymptotes. Curves such as $y = \frac{k}{x}$ have the vertical asymptote $x = 0$. A definite integral must never run across an asymptote.
- Write down and evaluate the definite integral between the required $x$-values.
- If the region is below the $x$-axis, the integral is negative, and the area is its opposite. If the region crosses the $x$-axis, split the integral at the $x$-intercept and find each part separately.
- Simplify with the log laws: $\log_e a + \log_e b = \log_e ab$, $\log_e a - \log_e b = \log_e \frac{a}{b}$ and $n \log_e a = \log_e a^n$.
- Write a conclusion, giving the area in square units.

### REGIONS TO THE LEFT OF THE $y$-AXIS

For $x < 0$, the primitive $\log_e |x|$ still works, because $|x|$ is positive. For example, the curve $y = \dfrac{1}{x}$ is below the $x$-axis for $x < 0$, and

$$\begin{aligned}\int_{-e}^{-1} \frac{1}{x}\,dx &= \Big[\log_e |x|\Big]_{-e}^{-1} \\ &= \log_e |-1| - \log_e |-e| \\ &= \log_e 1 - \log_e e \\ &= -1.\end{aligned}$$

The integral is negative because the region is below the $x$-axis, so the area between $y = \dfrac{1}{x}$ and the $x$-axis, from $x = -e$ to $x = -1$, is $1$ square unit.

TUTOR DEMO **A region that crosses the $x$-axis**

**a** Sketch $y = \dfrac{2}{x} - 1$, showing the $x$-intercept and both asymptotes.

**b** Find the area of the region bounded by the curve $y = \dfrac{2}{x} - 1$, the $x$-axis, $x = 1$ and $x = 4$.

YOUR TURN **Same method**

**a** Sketch $y = \dfrac{4}{x} - 1$, showing the $x$-intercept and both asymptotes.

**b** Find the area of the region bounded by the curve $y = \dfrac{4}{x} - 1$, the $x$-axis, $x = 2$ and $x = 8$.

## CONCEPT 2 · the area between a hyperbola and a line

### FINDING AREAS BY INTEGRATION

The next worked example involves finding the area between two given curves.

### TOP CURVE MINUS BOTTOM CURVE (FROM LESSON 6E)

If a curve $y = f(x)$ is always above $y = g(x)$ in an interval $a \leq x \leq b$, then the area of the region between the curves is $\displaystyle\int_a^b \big(f(x) - g(x)\big)\,dx$.

- Find where the curves meet by solving their equations simultaneously. These $x$-values are the limits of the integral.
- Decide which curve is on top by testing one $x$-value between the limits.

### THE HYPERBOLA $xy = k$

When $k > 0$, the hyperbola $xy = k$ is the curve $y = \dfrac{k}{x}$. Both axes are asymptotes, and its two branches lie in the first and third quadrants. In the first quadrant, it passes through $(1, k)$ and $(k, 1)$.

TUTOR DEMO **A hyperbola and a line**

**a** Find where the hyperbola $xy = 3$ meets the line $x + y = 4$.
**b** Sketch the situation.
**c** Find the area of the region between the two curves, in exact form, and then correct to three decimal places.

YOUR TURN **Same method**

**a** Find where the hyperbola $xy = 4$ meets the line $x + y = 5$.
**b** Sketch the situation.
**c** Find the area of the region between the two curves, in exact form, and then correct to three decimal places.

**WORKED EXAMPLE 38** **Keep for homework**

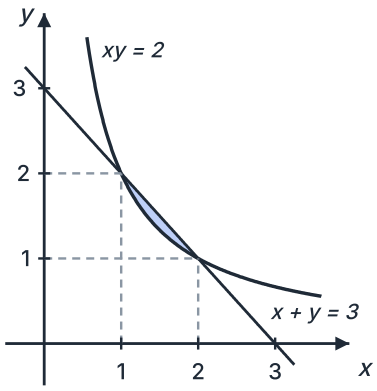
**a** Find where the hyperbola $xy = 2$ meets the line $x + y = 3$.
**b** Sketch the situation.
**c** Find the area of the region between the two curves, in exact form, and then correct to three decimal places.

**SOLUTION**

**a** Substituting the line $y = 3 - x$ into the hyperbola,

$$\begin{aligned} x(3-x) &= 2 \\ x^2 - 3x + 2 &= 0 \\ (x-1)(x-2) &= 0 \\ x &= 1 \text{ or } 2, \end{aligned}$$

so the curves meet at $(1, 2)$ and $(2, 1)$.




**b** The hyperbola $xy = 2$ has both axes as asymptotes.
The line $x + y = 3$ has $x$-intercept $(3, 0)$ and $y$-intercept $(0, 3)$.

**c**

$$\begin{aligned} \text{Area} &= \int_1^2 (\text{top curve} - \text{bottom curve})\, dx \\ &= \int_1^2 \left( (3-x) - \frac{2}{x} \right) dx \\ &= \left[ 3x - \tfrac{1}{2}x^2 - 2\log_e |x| \right]_1^2 \\ &= \left(6 - 2 - 2\log_e 2\right) - \left(3 - \tfrac{1}{2} - 2\log_e 1\right) \\ &= \left(4 - 2\log_e 2\right) - \left(2\tfrac{1}{2} - 0\right) \\ &= \left(1\tfrac{1}{2} - 2\log_e 2\right) \text{ square units} \\ &\doteq 0.114 \text{ square units.} \end{aligned}$$

**READING WORKED EXAMPLE 38, LINE BY LINE**

In part **a**, expanding $x(3 - x) = 2$ gives $3x - x^2 = 2$, and moving every term to the right gives $x^2 - 3x + 2 = 0$. In part **c**, the hyperbola is $y = \dfrac{2}{x}$, and the line $y = 3 - x$ is on top: at $x = 1\frac{1}{2}$, the line gives $1\frac{1}{2}$ and the hyperbola gives $1\frac{1}{3}$. In the fourth line, $3 \times 2 = 6$, $\frac{1}{2} \times 2^2 = 2$, $3 \times 1 = 3$ and $\frac{1}{2} \times 1^2 = \frac{1}{2}$. In the fifth line, $\log_e 1 = 0$. Finally, $1\frac{1}{2} - 2\log_e 2 \doteq 1.5 - 1.3863 = 0.1137$.

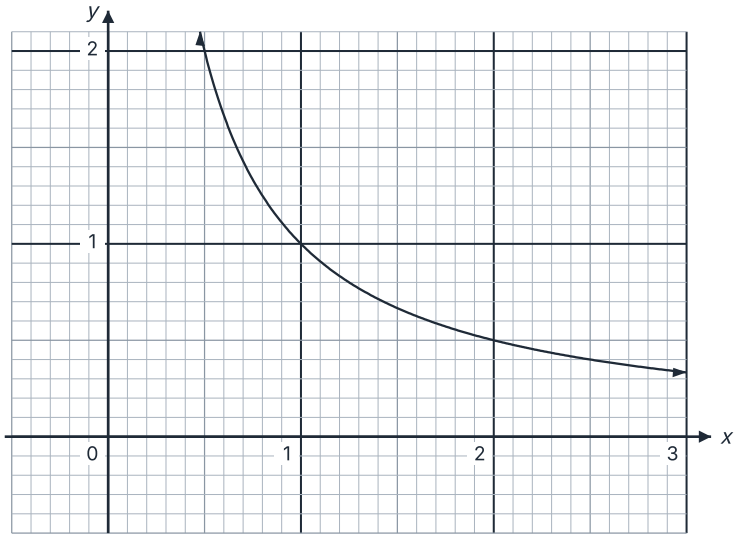
## TEXTBOOK HOMEWORK · Exercise 6J

**FOUNDATION**

**1** a Show that $\int_1^e \frac{1}{x}\,dx = 1$.

b This question uses the result in part **a** to estimate $e$ from a graph of $y = \frac{1}{x}$.

The diagram to the right shows the graph of $y = \frac{1}{x}$ from $x = 0$ to $x = 3$.




The graph has been drawn on graph paper with a scale of 10 little divisions to 1 unit, so that 100 of the little squares make 1 square unit.

Count the number of squares in the column from $x = 1.0$ to $1.1$, then the squares in the column from $x = 1.1$ to $1.2$, and so on.

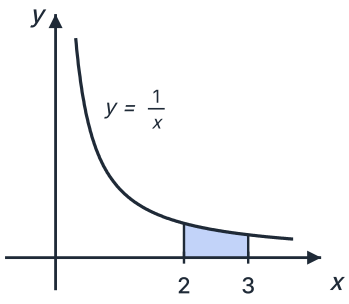
Continue until the number of squares equals 100: the $x$-value at this point will be an estimate of $e$.

**2** Find the area between the curve $y = \frac{1}{x}$ and the $x$-axis within the given interval. Answer first in exact form, and then correct to four significant figures.

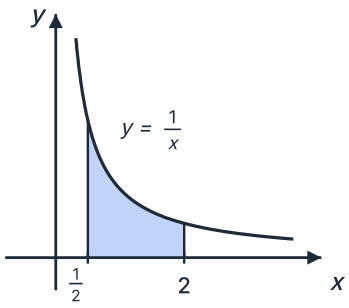
i $1 \leq x \leq 5$ ii $e \leq x \leq e^2$ iii $2 \leq x \leq 8$

**3** Find the area of each shaded region.

**a**




**b**




**4**

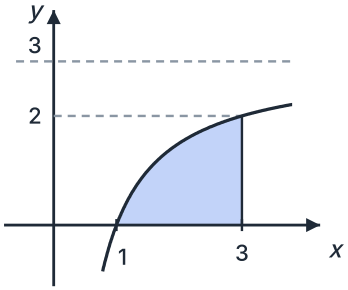
a Find the area between $y = \dfrac{1}{3x+2}$ and the $x$-axis for $0 \leq x \leq 1$.

b Find the area between $y = \dfrac{3}{x-1}$ and the $x$-axis for $2 \leq x \leq e^3 + 1$.

c Find the area between $y = \dfrac{1}{x} + x$ and the $x$-axis, from $x = \frac{1}{2}$ to $x = 2$.

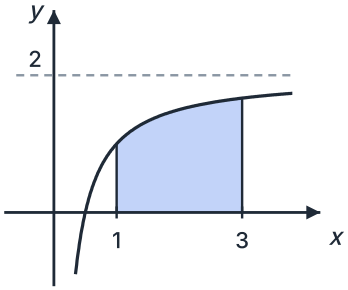
d Find the area between $y = \dfrac{1}{x} + x^2$ and the $x$-axis, from $x = 1$ to $x = 3$.

**5** a



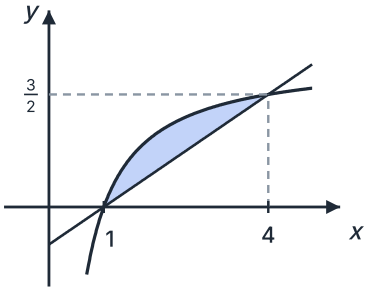

Find the area of the region bounded by $y = 3 - \dfrac{3}{x}$, the $x$-axis and $x = 3$.

b



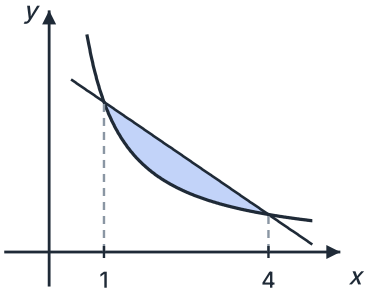

Find the area of the region bounded by $y = 2 - \dfrac{1}{x}$, the $x$-axis, $x = 1$ and $x = 3$.

**6** a



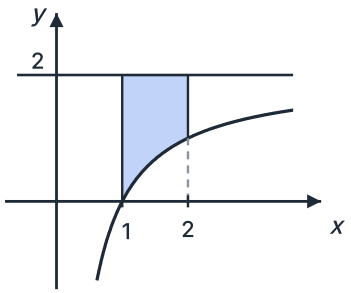

Find the area of the region bounded by $y = 2 - \dfrac{2}{x}$ and the line $y = \frac{1}{2}(x - 1)$.

b



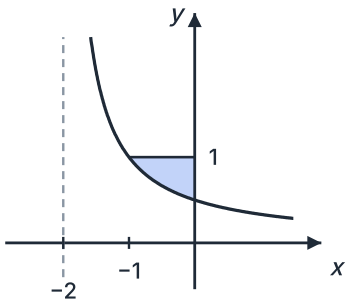

Find the area of the region between $y = \dfrac{2}{x}$ and the line $x + 2y - 5 = 0$.

**7** a



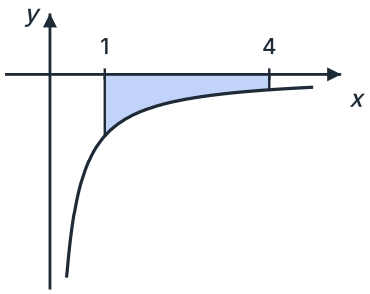

Find the area of the region in the first quadrant bounded by $y = 2 - \dfrac{2}{x}$ and $y = 2$, and lying between $x = 1$ and $x = 2$.

b



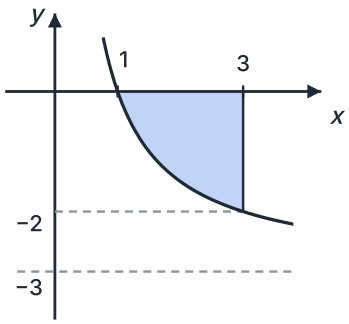

Find the area of the region bounded by the curve $y = \dfrac{1}{x+2}$, the $y$-axis and the horizontal line $y = 1$.

**8** a



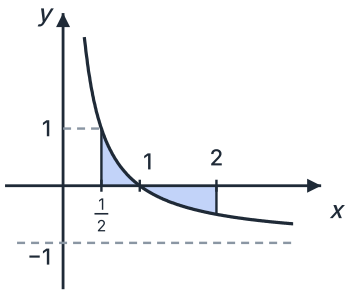

Find the area of the region bounded by $y = -\dfrac{1}{x}$, the $x$-axis, $x = 1$ and $x = 4$.

b



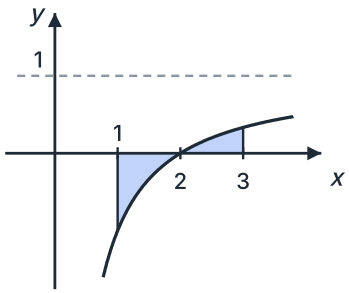

Find the area of the region bounded by $y = \dfrac{3}{x} - 3$, the $x$-axis and $x = 3$.

**9** a




Find the area of the region bounded by $y = \dfrac{1}{x} - 1$, the $x$-axis, $x = \frac{1}{2}$ and $x = 2$.

b




Find the area of the region bounded by $y = 1 - \dfrac{2}{x}$, the $x$-axis, $x = 1$ and $x = 3$.

**DEVELOPMENT**

**10** a Sketch the region bounded by $y = 1$, $x = 8$ and the curve $y = \dfrac{4}{x}$.

b Find the area of this region with the aid of an appropriate integral.

**11** a Find the two intersection points of the curve $y = \dfrac{1}{x}$ with the line $y = 4 - 3x$.

b Find the area between these two curves.

**12** a Differentiate $x^2 + 1$, and hence find the area under the graph $y = \dfrac{x}{x^2 + 1}$, between $x = 0$ and $x = 2$.

b Differentiate $x^2 + 2x + 3$, and hence find the area under the graph $y = \dfrac{x + 1}{x^2 + 2x + 3}$, between $x = 0$ and $x = 1$.

**13** a Sketch the region bounded by the $x$-axis, $y = x$, $y = \dfrac{1}{x}$ and $x = e$.

b Hence find the area of this region by using two appropriate integrals.

**14** a Use the trapezoidal rule with four subintervals to approximate the area between the curve $y = \ln x$ and the $x$-axis, between $x = 1$ and $x = 5$. Answer correct to four decimal places.

b Differentiate $y = x \log_e x$. Hence find the exact value of the area, and approximate it correct to four decimal places.

c Is the estimate greater than or less than the exact value? Explain how the graph could have predicted this.

**15** a Show that $4x = 2(2x + 1) - 2$.

b Hence evaluate the area under $y = \dfrac{4x}{2x + 1}$ between $x = 0$ and $x = 1$.

**16** **a** Show that the curves $y = \dfrac{6}{x}$ and $y = x^2 - 6x + 11$ intersect when $x = 1$, $2$ and $3$.

**b** Graph these two curves and shade the two areas enclosed by them.

**c** Find the total area enclosed by the two curves.

**17** **a** Use upper and lower rectangles to prove that $\frac{1}{2} < \displaystyle\int_{2^n}^{2^{n+1}} \frac{1}{x}\,dx < 1$, for $n \geq 0$.

**b** Hence prove that $\displaystyle\int_{1}^{2^n} \frac{1}{x}\,dx \to \infty$ as $n \to \infty$.

ENRICHMENT

**18** Consider the two curves $y = 6e^{-x}$ and $y = e^x - 1$.

a Let $u = e^x$. Show that the $x$-coordinate of the point of intersection of these two curves satisfies $u^2 - u - 6 = 0$.

b Hence find the coordinates of the point of intersection.

c Sketch the curves on the same number plane, and shade the region bounded by them and the $y$-axis.

d Find the area of the shaded region.

**19** The hyperbola $y = \dfrac{1}{x} + 1$ meets the $x$-axis at $(-1, 0)$. Find the area contained between the $x$-axis and the curve from:

a $x = -e$ to $x = -1$, b $x = -1$ to $x = -e^{-1}$, c $x = -e$ to $x = -e^{-1}$.

**20** Find a primitive of $\dfrac{1}{x + \sqrt{x}}$.

**21** a Show that:

$$\frac{d}{dx}\log_e\left(x+\sqrt{x^2+a^2}\right)=\frac{1}{\sqrt{x^2+a^2}} \quad \text{and}$$

$$\frac{d}{dx}\log_e\left|x+\sqrt{x^2-a^2}\right|=\frac{1}{\sqrt{x^2-a^2}}.$$

b Use these results to find:

i $\displaystyle\int_0^1 \frac{1}{\sqrt{x^2+1}}\,dx$ ii $\displaystyle\int_4^8 \frac{1}{\sqrt{x^2-16}}\,dx$

**22** Consider the area under $y=\dfrac{1}{x}$ between $x=n$ and $x=n+1$.

a Show that $\dfrac{1}{n+1}<\displaystyle\int_n^{n+1}\frac{1}{x}\,dx<\frac{1}{n}$.

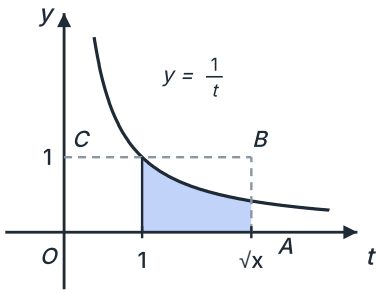
b Hence show that $\dfrac{n}{n+1}<\log_e\left(1+\frac{1}{n}\right)^n<1$.

c Take the limit as $n\to\infty$ to show that $\lim\limits_{n\to\infty}\left(1+\frac{1}{n}\right)^n=e$.

d Repeat the above steps, replacing $n+1$ with $n+t$ and show that $\lim\limits_{n\to\infty}\left(1+\frac{t}{n}\right)^n=e^t$.

**Proving the dominance limits:**
This question provides proofs of the basic dominance limits used in many of the curve-sketching questions in Exercises 6C and 6H.

**23** a The shaded region in the diagram to the right shows the definite integral $\int_1^{\sqrt{x}} \frac{1}{t}\,dt$.




i Explain why $0 < \int_1^{\sqrt{x}} \frac{1}{t}\,dt < \text{area } ABCO$.

ii Evaluate the integral, and prove that $0 < \frac{\log_e x}{x} < \frac{2}{\sqrt{x}}$.

iii Hence prove that $\lim_{x\to\infty} \frac{\log_e x}{x} = 0$.

b Substitute $x = \frac{1}{u}$ into the limit in part **a** to prove that $\lim_{x\to 0^+} x \log_e x = 0$.

c Substitute $x = e^u$ into the limit in part **b** to prove that $\lim_{x\to -\infty} xe^x = 0$.

d Substitute $x = e^u$ into the limit in part **a** to prove that $\lim_{x\to\infty} xe^{-x} = 0$.

## PAST-PAPER HOMEWORK · timed practice

The school trial papers used for this pack (2020–2024) have no question that first needs this lesson: each of their integration questions can be done with an earlier lesson's methods or needs later work. For timed practice, do Exercise 6J Questions 6(b), 9(b), 11 and 16 above in 30 minutes.

LESSON 6K

# Calculus with other bases

The change-of-base formula writes every logarithmic function as a constant multiple of $\log_e x$, and every exponential function $a^x$ as a power of $e$. With these we differentiate $\log_a x$, show that base $e$ is the only base whose logarithm has gradient 1 at its $x$-intercept, and differentiate and integrate $a^x$. This gives three new standard forms: the derivatives of $\log_a x$ and $a^x$, and the primitive of $a^x$.

## CONCEPT 1 · logarithmic functions to other bases

### CALCULUS WITH OTHER BASES

In applications of exponential functions where calculus is required, the base $e$ can generally be used. For example, the treatment of exponential growth in Chapter 11 of the Year 11 book was done entirely using base $e$.

The change-of-base formula, however, allows calculus to be applied to exponential and logarithmic functions of any base without conversion to base $e$. In this section, we develop three further standard forms that allow calculus to be applied straightforwardly to functions such as $y = 2^x$ and $y = 10^x$.

Throughout this section, the other base $a$ must be positive and not equal to 1.

### LOGARITHMIC FUNCTIONS TO OTHER BASES

Any logarithmic function can be expressed easily in terms of $\log_e x$ by using the change-of-base formula. For example,

$$\log_2 x = \frac{\log_e x}{\log_e 2}.$$

Thus every other logarithmic function is just a constant multiple of $\log_e x$. This allows any other logarithmic function to be differentiated easily.

### WHERE THE CHANGE-OF-BASE FORMULA COMES FROM

| | | |
|---|---|---|
| Let | $y = \log_a x.$ | |
| Then | $a^y = x,$ | by the definition of logarithms. |
| Take $\log_e$ of both sides: | $\log_e\left(a^y\right) = \log_e x$ | |
| | $y \log_e a = \log_e x,$ | by the log law $\log_e\left(a^n\right) = n \log_e a.$ |
| Divide by $\log_e a$: | $y = \dfrac{\log_e x}{\log_e a}.$ | ($\log_e a \neq 0$, because $a \neq 1$.) |

### 11 LOGARITHMIC FUNCTIONS WITH OTHER BASES

Every logarithmic function can be written as a multiple of a logarithmic function base $e$:

$$\log_a x = \frac{\log_e x}{\log_e a}, \qquad \text{that is} \qquad \log_a x = \frac{1}{\log_e a} \times \log_e x.$$

TUTOR DEMO **Change of base on the calculator**

**a** Express the function $y = \log_3 x$ in terms of the function $\log_e x$.
**b** Hence use the calculator function labelled $\boxed{\ln}$ to approximate, correct to four decimal places:

i $\log_3 40$ ii $\log_3 2$ iii $\log_3 0.05$

**c** Check the results of part **b** using the function labelled $\boxed{x^y}$.

YOUR TURN **Same method**

**a** Express the function $y = \log_7 x$ in terms of the function $\log_e x$.
**b** Hence use the calculator function labelled $\boxed{\ln}$ to approximate, correct to four decimal places:

i $\log_7 50$ ii $\log_7 3$ iii $\log_7 0.2$

**c** Check the results of part **b** using the function labelled $\boxed{x^y}$.

**WORKED EXAMPLE 39** **Keep for homework**

**a** Express the function $y = \log_5 x$ in terms of the function $\log_e x$.
**b** Hence use the calculator function labelled $\boxed{\ln}$ to approximate, correct to four decimal places:

i $\log_5 30$ ii $\log_5 2$ iii $\log_5 0.07$

**c** Check the results of part **b** using the function labelled $\boxed{x^y}$.

**SOLUTION**

**a** $\log_5 x = \dfrac{\log_e x}{\log_e 5}$

**b i** $\log_5 30 = \dfrac{\log_e 30}{\log_e 5} \doteq 2.1133$

**ii** $\log_5 2 = \dfrac{\log_e 2}{\log_e 5} \doteq 0.4307$

**iii** $\log_5 0.07 = \dfrac{\log_e 0.07}{\log_e 5} \doteq -1.6523$

**c** Checking these results using the function labelled $\boxed{x^y}$:

i $5^{2.1133} \doteq 30$ ii $5^{0.4307} \doteq 2$ iii $5^{-1.6523} \doteq 0.07$

**WHY PART C CHECKS PART B**

By the definition of logarithms, $\log_5 30 = y$ means $5^y = 30$. So if $\log_5 30 \doteq 2.1133$, then raising $5$ to the power $2.1133$ must give back about $30$. On the calculator, $5^{2.1133} = 30.0008\ldots$, which rounds to $30$.

## CONCEPT 2 · differentiating logarithmic functions with other bases

**DIFFERENTIATING LOGARITHMIC FUNCTIONS WITH OTHER BASES**

Once the function is expressed as a multiple of a logarithmic function base $e$, it can be differentiated using the previous standard forms.

Part **b** above gives the formula in the general case:

**12 DIFFERENTIATING LOGARITHMIC FUNCTIONS WITH OTHER BASES**

- Either use the change-of-base formula to convert to logarithms base $e$.
- Or use the standard form $\dfrac{d}{dx}\log_a x = \dfrac{1}{x \log_e a}$.

**THE CONSTANT FACTOR STAYS**

In the box above, "part **b** above" is part **b** of Worked example 40 below, which differentiates $y = \log_a x$.

After the change of base, $\log_a x = \dfrac{1}{\log_e a} \times \log_e x$. The factor $\dfrac{1}{\log_e a}$ is a constant, and the derivative of a constant times a function is the constant times the derivative of the function. Since $\dfrac{d}{dx}\log_e x = \dfrac{1}{x}$, the derivative is $\dfrac{1}{\log_e a} \times \dfrac{1}{x}$.

TUTOR DEMO **Both approaches**

**1** Use the change-of-base formula to differentiate:

a $y = \log_3 x$

b $y = 4\log_a x$

**2** Use the standard form to differentiate:

a $\log_5 x$

b $\log_{1.08} x$

YOUR TURN **Same method**

**1** Use the change-of-base formula to differentiate:

a $y = \log_7 x$

b $y = 2\log_a x$

**2** Use the standard form to differentiate:

a $\log_4 x$

b $\log_{1.1} x$

WORKED EXAMPLE 40 **Keep for homework**

Use the change-of-base formula to differentiate:

a $y = \log_2 x$

b $y = \log_a x$

**SOLUTION**

**a** Here $y = \log_2 x$.

Using the change-of-base formula,

$$y = \frac{\log_e x}{\log_e 2}.$$

Because $\log_e 2$ is a constant,

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{x} \times \frac{1}{\log_e 2} \\ &= \frac{1}{x \log_e 2}.\end{aligned}$$

**b** Here $y = \log_a x$.

Using the change-of-base formula,

$$y = \frac{\log_e x}{\log_e a}.$$

Because $\log_e a$ is a constant,

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{x} \times \frac{1}{\log_e a} \\ &= \frac{1}{x \log_e a}.\end{aligned}$$

WORKED EXAMPLE 41 **Keep for homework**

a Differentiate $\log_{10} x$.

b Differentiate $\log_{1.05} x$.

**SOLUTION**

a $$\frac{d}{dx}\log_{10} x = \frac{1}{x \log_e 10}$$

b $$\frac{d}{dx}\log_{1.05} x = \frac{1}{x \log_e 1.05}$$

## CONCEPT 3 · a characterisation of the logarithmic function

### A CHARACTERISATION OF THE LOGARITHMIC FUNCTION

We have already discussed in Section 6A that the tangent to $y = \log_e x$ at the $x$-intercept has gradient exactly 1.

The worked example below shows that this property distinguishes the logarithmic function base $e$ from all other logarithmic functions.

### 13 THE GRADIENT AT THE $x$-INTERCEPT

The function $y = \log_e x$ is the only logarithmic function whose gradient at the $x$-intercept is exactly 1.

### TWO FACTS FROM THE DEFINITION OF LOGARITHMS

$\log_a x = c$ means $x = a^c$. So $\log_a x = 0$ gives $x = a^0 = 1$. Thus every logarithmic function $y = \log_a x$ has $x$-intercept $(1, 0)$. In the same way, $\log_e a = 1$ gives $a = e^1 = e$.

TUTOR DEMO **The tangent at the $x$-intercept**

**a** Find the $x$-intercept of $y = \log_3 x$, and show that the tangent there has gradient $\dfrac{1}{\log_e 3}$.

**b** Find the equation of this tangent.

**c** Find the base $a$ for which the tangent to $y = \log_a x$ at the $x$-intercept has gradient $2$.

**YOUR TURN** **Same method**

**a** Find the $x$-intercept of $y = \log_5 x$, and show that the tangent there has gradient $\dfrac{1}{\log_e 5}$.

**b** Find the equation of this tangent.

**c** Find the base $a$ for which the tangent to $y = \log_a x$ at the $x$-intercept has gradient $\frac{1}{2}$.

**WORKED EXAMPLE 42** **Keep for homework**

**a** Show that the tangent to $y = \log_a x$ at the $x$-intercept has gradient $\dfrac{1}{\log_e a}$.

**b** Show that the function $y = \log_e x$ is the only logarithmic function whose gradient at the $x$-intercept is exactly 1.

**SOLUTION**

**a**

Here $y = \log_a x.$

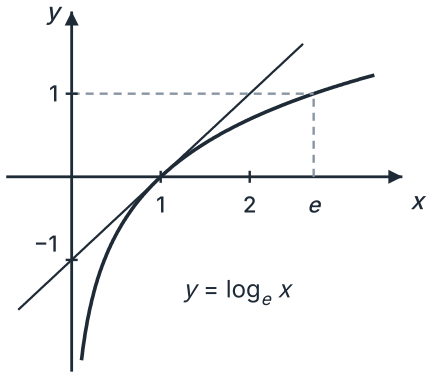
When $y = 0$, $\log_a x = 0$

$x = 1,$

so the $x$-intercept is $(1, 0)$.

Differentiating, $y' = \dfrac{1}{x \log_e a},$

so when $x = 1$, $y' = \dfrac{1}{\log_e a},$ as required.




**b** The gradient at the $x$-intercept is 1 if and only if

$\log_e a = 1$

$a = e^1$

$= e,$

that is, if and only if the original base $a$ is equal to $e$.

## CONCEPT 4 · exponential functions with other bases

### EXPONENTIAL FUNCTIONS WITH OTHER BASES

Before calculus can be applied to an exponential function $y = a^x$ with base $a$ different from $e$, it must be written as an exponential function with base $e$. The important identity used to do this is

$$e^{\log_e a} = a,$$

which simply expresses the fact that the functions $e^x$ and $\log_e x$ are inverse functions. Now $a^x$ can be written as

$$\begin{aligned} a^x &= \left(e^{\log_e a}\right)^x, && \text{replacing } a \text{ by } e^{\log_e a}, \\ &= e^{x \log_e a}, && \text{using the index law } \left(e^k\right)^x = e^{kx}. \end{aligned}$$

Thus $a^x$ has been expressed in the form $e^{kx}$, where $k = \log_e a$ is a constant.

### WHY $e^{\log_e a} = a$

By the definition of logarithms, $\log_e a$ is the index to which $e$ must be raised to give $a$. So raising $e$ to that index gives $a$ back. For example, $\log_e 2 \doteq 0.6931$, and $e^{0.6931} \doteq 2$. In the second line above, the index law is used with $k = \log_e a$, so $kx = x \log_e a$.

### 14 EXPONENTIAL FUNCTIONS WITH OTHER BASES

- Every positive real number can be written as a power of $e$:
  $a = e^{\log_e a}$
- Every exponential function can be written as an exponential function base $e$:
  $a^x = e^{x \log_e a}$

TUTOR DEMO **Powers of $e$**

Express these numbers and functions as powers of $e$.

a $3$ b $3^x$ c $7^{-x}$ d $10^{2x}$

**YOUR TURN** **Same method**

Express these numbers and functions as powers of $e$.

a $5$ b $5^x$ c $2^{-x}$ d $6^{3x}$

**WORKED EXAMPLE 43** **Keep for homework**

Express these numbers and functions as powers of $e$.

a $2$ b $2^x$ c $5^{-x}$

**SOLUTION**

**a** $2 = e^{\log_e 2}$

**b**

$$2^x = \left(e^{\log_e 2}\right)^x$$
$$= e^{x \log_e 2}$$

**c**

$$5^{-x} = \left(e^{\log_e 5}\right)^{-x}$$
$$= e^{-x \log_e 5}$$

## CONCEPT 5 · differentiating and integrating exponential functions with other bases

### DIFFERENTIATING AND INTEGRATING EXPONENTIAL FUNCTIONS WITH OTHER BASES

Write the function as a power of $e$. It can then be differentiated and integrated.

First, $a^x = e^{\log_e a^x}$
$= e^{x \log_e a}$.

Differentiating, $\frac{d}{dx} a^x = \frac{d}{dx} e^{x \log_e a}$
$= e^{x \log_e a} \times \log_e a$, because $\frac{d}{dx} e^{kx} = ke^{kx}$,
$= a^x \log_e a$, because $e^{x \log_e a} = a^x$.

### INTEGRATING EXPONENTIAL FUNCTIONS WITH OTHER BASES

Integrating, $\displaystyle\int a^x\,dx = \int e^{x\log_e a}\,dx$

$$= \frac{e^{x\log_e a}}{\log_e a}, \qquad \text{because } \int e^{kx}\,dx = \frac{1}{k}\,e^{kx},$$

$$= \frac{a^x}{\log_e a}, \qquad \text{because } e^{x\log_e a} = a^x.$$

This process can be carried through every time, or the results can be remembered as standard forms.

### 15 DIFFERENTIATION AND INTEGRATION WITH OTHER BASES

There are two approaches.

- Write all powers with base $e$ before differentiating or integrating.
- Alternatively, use the standard forms:

$$\frac{d}{dx}\,a^x = a^x\log_e a \qquad \text{and} \qquad \int a^x\,dx = \frac{a^x}{\log_e a} + C$$

### BASE $e$ IS THE SIMPLEST

**Note:** The formulae for differentiating and integrating $a^x$ both involve the constant $\log_e a$. This constant $\log_e a$ is 1 when $a = e$, so the formulae are simplest when the base is $e$. Again, this indicates that $e$ is the appropriate base to use for calculus with exponential functions.

### THE DERIVATION IN DETAIL

- The first line uses the identity $e^{\log_e b} = b$ with $b = a^x$. Then the log law $\log_e\left(a^x\right) = x\log_e a$ gives the second line.
- The standard forms for $e^{kx}$ are used with the constant $k = \log_e a$.
- The integration lines leave out the constant of integration. Box 15 adds it: $\displaystyle\int a^x\,dx = \frac{a^x}{\log_e a} + C.$

TUTOR DEMO **Both approaches for $3^x$**

**a** Differentiate $y = 3^x$. Hence find the gradient of $y = 3^x$ at the $y$-intercept, correct to three significant figures.

**b** Write $3^x$ as a power of $e$, and hence find $\int 3^x \, dx$.

YOUR TURN **Same method**

**a** Differentiate $y = 5^x$. Hence find the gradient of $y = 5^x$ at the $y$-intercept, correct to three significant figures.

**b** Write $5^x$ as a power of $e$, and hence find $\int 5^x \, dx$.

WORKED EXAMPLE 44 **Keep for homework**

Differentiate $y = 2^x$. Hence find the gradient of $y = 2^x$ at the $y$-intercept, correct to three significant figures.

**SOLUTION**

Here $y = 2^x$.

Using the standard form, $y' = 2^x \log_e 2$.

Hence when $x = 0$,

$$\begin{aligned} y' &= 2^0 \times \log_e 2 \\ &= \log_e 2 \\ &\doteq 0.693. \end{aligned}$$

**Note:** This result may be compared with the results of physically measuring this gradient in Question 1 of Exercise 11A in the Year 11 book.

**WHY $x = 0$**

The $y$-intercept is where $x = 0$. Since $2^0 = 1$, the $y$-intercept of $y = 2^x$ is $(0, 1)$, and the gradient there is $1 \times \log_e 2 = \log_e 2$.

## CONCEPT 6 · areas involving other bases

**THE AREA BETWEEN A LINE AND AN EXPONENTIAL CURVE (AS IN LESSON 6E)**

- Find where the two graphs meet. These $x$-values are the limits of the integral.
- Decide which graph is on top by substituting one $x$-value between the limits into both.
- Area $= \int (\text{upper curve} - \text{lower curve})\, dx$, using $\int a^x\, dx = \dfrac{a^x}{\log_e a} + C$ for the exponential term.

TUTOR DEMO **The region between $y = 3^x$ and a line**

**a** Show that the line $y = 2x + 1$ meets the curve $y = 3^x$ at $A(0, 1)$ and $B(1, 3)$.
**b** Sketch the two curves and shade the region contained between them.
**c** Find the area of this shaded region, correct to four significant figures.

YOUR TURN **Same method**

**a** Show that the line $y = 3x + 1$ meets the curve $y = 4^x$ at $A(0, 1)$ and $B(1, 4)$.
**b** Sketch the two curves and shade the region contained between them.
**c** Find the area of this shaded region, correct to four significant figures.

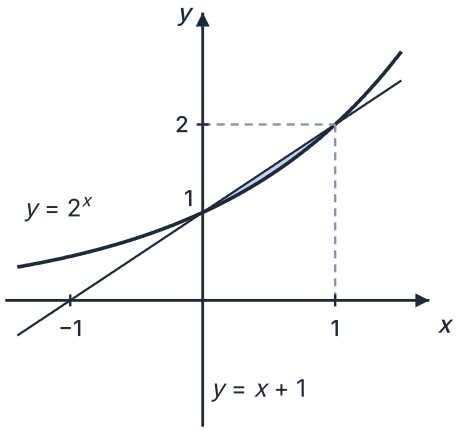
WORKED EXAMPLE 45 **Keep for homework**

**a** Show that the line $y = x + 1$ meets the curve $y = 2^x$ at $A(0, 1)$ and $B(1, 2)$.
**b** Sketch the two curves and shade the region contained between them.
**c** Find the area of this shaded region, correct to four significant figures.

**SOLUTION**

**a** Simple substitution of $x = 0$ and $x = 1$ into both functions verifies the result.

**b** The graph is drawn to the right.




**c**

$$\begin{aligned}
\text{Area} &= \int_0^1 (\text{upper curve} - \text{lower curve})\, dx \\
&= \int_0^1 \left(x + 1 - 2^x\right) dx \\
&= \left[\tfrac{1}{2}x^2 + x - \frac{2^x}{\log_e 2}\right]_0^1 \\
&= \left(\tfrac{1}{2} + 1 - \frac{2}{\log_e 2}\right) - \left(0 + 0 - \frac{1}{\log_e 2}\right) \\
&= 1\tfrac{1}{2} - \frac{1}{\log_e 2} \text{ square units} \\
&\doteq 0.05730 \text{ square units.}
\end{aligned}$$

**THE LINES THAT WORKED EXAMPLE 45 LEAVES OUT**

**a** When $x = 0$: $x + 1 = 1$ and $2^0 = 1$, so both pass through $A(0, 1)$. When $x = 1$: $x + 1 = 2$ and $2^1 = 2$, so both pass through $B(1, 2)$.

**c** The line is the upper curve: at $x = \frac{1}{2}$, $x + 1 = 1.5$ while $2^{\frac{1}{2}} = \sqrt{2} \doteq 1.414$. Substituting the limits uses $2^1 = 2$ and $2^0 = 1$. Then

$$\left(\tfrac{1}{2} + 1 - \frac{2}{\log_e 2}\right) - \left(0 + 0 - \frac{1}{\log_e 2}\right) = 1\tfrac{1}{2} - \frac{2}{\log_e 2} + \frac{1}{\log_e 2} = 1\tfrac{1}{2} - \frac{1}{\log_e 2},$$

and $1.5 - 1.442\,695\ldots = 0.057\,304\ldots \doteq 0.05730$.

## TEXTBOOK HOMEWORK · Exercise 6K

**FOUNDATION**

**1** Use the change-of-base formula $\log_a x = \dfrac{\log_e x}{\log_e a}$ and the function labelled $\boxed{\ln}$ on your calculator to evaluate each expression correct to three significant figures. Then check your answers using the function labelled $\boxed{x^y}$.

a $\log_2 3$ b $\log_2 10$ c $\log_5 26$ d $\log_3 0.0047$

**2** Use the change-of-base formula to express these with base $e$, then differentiate them.

a $y = \log_2 x$ b $y = \log_{10} x$ c $y = 3\log_5 x$

**3** Use the standard form $\dfrac{d}{dx}\log_a x = \dfrac{1}{x \log_e a}$ to differentiate:

a $y = \log_3 x$ b $y = \log_7 x$ c $y = 5\log_6 x$

**4** Express these functions as powers of $e$, then differentiate them.

a $y = 3^x$ b $y = 4^x$ c $y = 2^x$

**5** Use the standard form $\dfrac{d}{dx}\, a^x = a^x \log_e a$ to differentiate:

a $y = 10^x$ b $y = 8^x$ c $y = 3 \times 5^x$

**6** Convert each integrand to a power of $e$ and then integrate.

a $\displaystyle\int 2^x \, dx$ b $\displaystyle\int 6^x \, dx$ c $\displaystyle\int 7^x \, dx$ d $\displaystyle\int 3^x \, dx$

**7** Use the result $\displaystyle\int a^x \, dx = \frac{a^x}{\log_e a} + C$ to find each primitive, then evaluate the definite integral correct to four significant figures.

a $\displaystyle\int_0^1 2^x \, dx$ b $\displaystyle\int_0^1 3^x \, dx$ c $\displaystyle\int_{-1}^1 5^x \, dx$ d $\displaystyle\int_0^2 4^x \, dx$

**8** **a** Complete the table of values to the right, giving your answers correct to two decimal places where necessary.

**b** Use this table of values to sketch the three curves $y = \log_2 x$, $y = \log_e x$ and $y = \log_4 x$ on the same set of axes.

| $x$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $1$ | $2$ | $4$ |
|---|---|---|---|---|---|
| $\log_2 x$ | | | | | |
| $\log_e x$ | | | | | |
| $\log_4 x$ | | | | | |

**DEVELOPMENT**

**9** **a** Differentiate $y = \log_2 x$. Hence find the gradient of the tangent to the curve at $x = 1$.

**b** Hence find the equation of the tangent there.

**c** Do likewise for:

i $y = \log_3 x$,

ii $y = \log_5 x$.

**10** Give the exact value of each integral, then evaluate it correct to four decimal places.

a $\displaystyle\int_1^3 2^x \, dx$ b $\displaystyle\int_{-1}^1 \left(3^x + 1\right) dx$ c $\displaystyle\int_0^2 \left(10^x - 10x\right) dx$

**11** Use the change-of-base formula to express $y = \log_{10} x$ with base $e$, and hence find $y'$.

a Find the gradient of the tangent to this curve at the point $(10, 1)$.

b Thus find the equation of this tangent in general form.

c At what value of $x$ will the tangent have gradient 1?

**12** a Find the equations of the tangents to each of $y = \log_2 x$, $y = \log_e x$ and $y = \log_4 x$ at the points where $x = 3$.

b Show that the three tangents all meet at the same point on the $x$-axis.

**13** **a** Show that the curves $y = 2^x$ and $y = 1 + 2x - x^2$ intersect at $A(0, 1)$ and $B(1, 2)$.

**b** Sketch the curves and find the area between them.

**14** Find the intercepts of the curve $y = 8 - 2^x$, and hence find the area of the region bounded by this curve and the coordinate axes.

**15** **a** Sketch the curve $y = 3 - 3^x$, showing the intercepts and asymptote.

**b** Find the area contained between the curve and the axes.

**16** **a** Show that the curves $y = x + 1$ and $y = 4^x$ intersect at the $y$-intercept and at $\left(-\frac{1}{2}, \frac{1}{2}\right)$.

**b** Write the area of the region enclosed between these two curves as an integral.

**c** Evaluate the integral found in part **b**.

**17** **a** Show that the tangent to $y = \log_3 x$ at $x = e$ passes through the origin.

**b** Show that the tangent to $y = \log_5 x$ at $x = e$ passes through the origin.

**c** Show that the same is true for $y = \log_a x$, for any base $a$.

**18** **a** Differentiate $x \log_e x - x$, and hence find $\int \log_e x \, dx$.

**b** Use the change-of-base formula and the integral in part **a** to evaluate $\int_1^{10} \log_{10} x \, dx$.

ENRICHMENT

**19** As always, the three standard forms in this section have linear extensions. The pronumeral $m$ is used here instead of the usual $a$ because $a$ is being used for the base.

a Use the standard form $\dfrac{d}{dx}\log_a(mx+b) = \dfrac{m}{(mx+b)\log_e a}$ to differentiate:

i $y = \log_3 x$ ii $y = \log_7(2x+3)$ iii $y = 5\log_6(4-9x)$

b Use the standard form $\dfrac{d}{dx}\,a^{mx+b} = ma^{mx+b}\log_e a$ to differentiate:

i $y = 10^x$ ii $y = 8^{4x-3}$ iii $y = 3 \times 5^{2-7x}$

c Use the standard form $\displaystyle\int a^{mx+b}\,dx = \frac{a^{mx+b}}{m\log_e a} + C$ to find:

i $\displaystyle\int 3^{5x}\,dx$ ii $\displaystyle\int 6^{2x+7}\,dx$ iii $\displaystyle\int 5 \times 7^{4-9x}\,dx$

**20** If the positive base $a$ of $y = a^x$ and $y = \log_a x$ is small enough, then the two curves will intersect. What base must be chosen so that the two are tangent at the point of contact? Proceed as follows:

**a** Rewrite both equations with base $e$, and let $k = \log_e a$.

**b** Explain why the gradient of the tangent at the point of contact must be 1.

**c** Use the last part to obtain two equations for the gradient.

**d** Solve these simultaneously to find $k$, and hence write down the base $a$.

## PAST-PAPER HOMEWORK · timed practice

The school trial papers used for this pack (2020–2024) have no integration question that first needs this lesson: each of their integration questions can be done with an earlier lesson's methods or needs later work. For timed practice, do Questions 10(c), 12, 14 and 16 above in 30 minutes.

LESSON 7D

# Integrating the trigonometric functions

Reversing the standard derivatives of $\sin x$, $\cos x$ and $\tan x$ gives three new standard integrals, and replacing $x$ by $ax + b$ extends them. With these we find areas and definite integrals, find the primitives of $\tan x$ and $\cot x$, find a function from its derivative and one known value, and reverse chain-rule derivatives to find new primitives.

## CONCEPT 1 · the standard forms for integrating the trigonometric functions

### THE STANDARD FORMS FOR INTEGRATING THE TRIGONOMETRIC FUNCTIONS

As always, the standard forms for differentiation can be reversed to give standard forms for integration.

When the standard forms for differentiating $\sin x$, $\cos x$ and $\tan x$ are reversed, they give three new standard integrals.

First, $\dfrac{d}{dx}\sin x = \cos x,$ and reversing this, $\displaystyle\int \cos x\,dx = \sin x.$

Secondly, $\dfrac{d}{dx}\cos x = -\sin x,$ and reversing this, $\displaystyle\int (-\sin x)\,dx = \cos x$

$\boxed{\times(-1)}$ $\displaystyle\int \sin x\,dx = -\cos x.$

Thirdly, $\dfrac{d}{dx}\tan x = \sec^2 x,$ and reversing this, $\displaystyle\int \sec^2 x\,dx = \tan x.$

This gives three new standard integrals. These three standard forms should be carefully memorised — pay attention to the signs in the first two standard forms.

### 8 STANDARD TRIGONOMETRIC INTEGRALS

- $\displaystyle\int \cos x\,dx = \sin x + C$, for some constant $C$
- $\displaystyle\int \sin x\,dx = -\cos x + C$, for some constant $C$
- $\displaystyle\int \sec^2 x\,dx = \tan x + C$, for some constant $C$

No calculation involving a primitive may cross an asymptote.

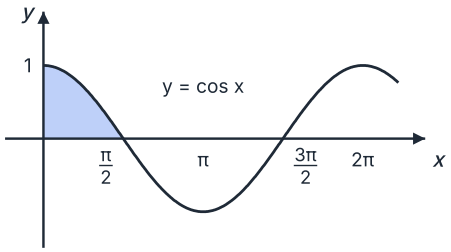
**WHERE THE MINUS SIGN COMES FROM**

The box $\times(-1)$ means: multiply both sides of $\int(-\sin x)\,dx = \cos x$ by $-1$. This gives $\int \sin x\,dx = -\cos x$. Check it by differentiating: $\frac{d}{dx}(-\cos x) = -(-\sin x) = \sin x$.

The asymptotes in this lesson belong to $\tan x$ and to $\sec^2 x$, the reciprocal of $\cos^2 x$. They are the lines $x = \frac{\pi}{2}$, $x = \frac{3\pi}{2}$, $x = -\frac{\pi}{2}$, and so on, where $\cos x = 0$.

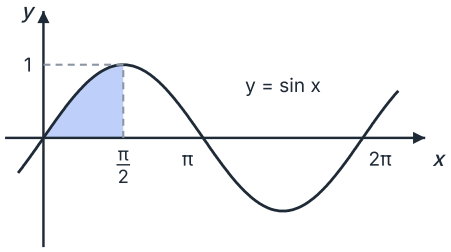
TUTOR DEMO **An area under $y = \cos x$**

The curve $y = \cos x$ is sketched to the right. Show that the region shaded in the diagram, from $x = 0$ to $x = \dfrac{\pi}{2}$, has area $1$ square unit.




YOUR TURN **Same method**

The curve $y = \sin x$ is sketched to the right. Show that the region shaded in the diagram, from $x = 0$ to $x = \dfrac{\pi}{2}$, has area $1$ square unit.

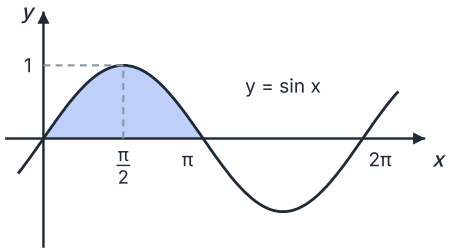


**WORKED EXAMPLE 16** **Keep for homework**

The curve $y = \sin x$ is sketched below. Show that the first arch of the curve, as shaded in the diagram, has area 2 square units.

**SOLUTION**

Because the region is entirely above the $x$-axis,




$$\begin{aligned}\text{area} &= \int_0^{\pi} \sin x \, dx \\ &= \Big[ -\cos x \Big]_0^{\pi} \\ &= -\cos \pi + \cos 0 \\ &= -(-1) + 1\end{aligned}$$

(the graph of $y = \cos x$ shows that $\cos \pi = -1$)

$= 2$ square units.

**THE SUBSTITUTION IN WORKED EXAMPLE 16**

Substituting the upper limit and then the lower limit into $-\cos x$, and subtracting,

$$\Big[ -\cos x \Big]_0^{\pi} = (-\cos \pi) - (-\cos 0) = -\cos \pi + \cos 0.$$

Then $\cos \pi = -1$ and $\cos 0 = 1$, so the area is $-(-1) + 1 = 1 + 1 = 2$ square units.

**NOTE**

This simple answer confirms again that radians are the right units to use for calculus with trigonometric functions. Similar simple results were obtained earlier when $e$ was used as the base for powers. For example, Question 39c of the Chapter 6 Review gathered together three remarkably simple results:

$$\int_1^e \frac{1}{x} \, dx = \int_1^e \log_e x \, dx = \int_0^1 x e^x \, dx = 1$$

TUTOR DEMO **Definite integrals**

Evaluate these definite integrals.

a $\int_0^{2\pi} \cos x \, dx$

b $\int_0^{\frac{\pi}{4}} \sec^2 x \, dx$

c $\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \sec^2 x \, dx$

YOUR TURN **Same method**

Evaluate these definite integrals.

a $\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \cos x \, dx$

b $\int_0^{\frac{\pi}{6}} \sec^2 x \, dx$

c $\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \sec^2 x \, dx$

**WORKED EXAMPLE 17** **Keep for homework**

Evaluate these definite integrals.

a $\int_0^{\pi} \cos x \, dx$ b $\int_0^{\frac{\pi}{3}} \sec^2 x \, dx$ c $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \sec^2 x \, dx$

**SOLUTION**

**a**

$$\begin{aligned}\int_0^{\pi} \cos x \, dx &= \Big[\sin x\Big]_0^{\pi} \\ &= \sin \pi - \sin 0 \\ &= 0\end{aligned}$$

(Use the graph of $y = \sin x$ to see that $\sin \pi = 0$ and $\sin 0 = 0$.)

**b**

$$\begin{aligned}\int_0^{\frac{\pi}{3}} \sec^2 x \, dx &= \Big[\tan x\Big]_0^{\frac{\pi}{3}} \\ &= \tan \tfrac{\pi}{3} - \tan 0 \\ &= \sqrt{3}\end{aligned}$$

(Here $\tan \frac{\pi}{3} = \sqrt{3}$ and $\tan 0 = 0$.)

**c** This integral is meaningless because it crosses the asymptote at $x = \frac{\pi}{2}$.

**WHY PART C IS MEANINGLESS**

The integrand $\sec^2 x$ is the reciprocal of $\cos^2 x$, so it is undefined at $x = \frac{\pi}{2}$, because $\cos \frac{\pi}{2} = 0$, and $\frac{\pi}{2}$ lies between the limits $\frac{\pi}{4}$ and $\frac{3\pi}{4}$. The primitive $\tan x$ has an asymptote there. If the asymptote were ignored, the calculation would give $\tan \frac{3\pi}{4} - \tan \frac{\pi}{4} = -1 - 1 = -2$, a negative number from a function $\sec^2 x$ that is never negative. This shows that such a calculation cannot be trusted.

## CONCEPT 2 · replacing x by ax + b

### REPLACING $x$ BY $ax + b$

Reversing the standard forms for derivatives in Section 7B gives a further set of standard forms. Again, the constants of integration have been ignored until the boxed statement of the standard forms.

First, $\dfrac{d}{dx}\sin(ax+b) = a\cos(ax+b),$

so $\displaystyle\int a\cos(ax+b)\,dx = \sin(ax+b)$

and dividing by $a$, $\displaystyle\int \cos(ax+b)\,dx = \frac{1}{a}\sin(ax+b).$

Secondly, $\dfrac{d}{dx}\cos(ax+b) = -a\sin(ax+b),$

so $\displaystyle\int -a\sin(ax+b)\,dx = \cos(ax+b)$

and dividing by $-a$, $\displaystyle\int \sin(ax+b)\,dx = -\frac{1}{a}\cos(ax+b).$

Thirdly, $\dfrac{d}{dx}\tan(ax+b) = a\sec^2(ax+b),$

so $\displaystyle\int a\sec^2(ax+b)\,dx = \tan(ax+b).$

and dividing by $a$, $\displaystyle\int \sec^2(ax+b)\,dx = \frac{1}{a}\tan(ax+b).$

The result is extended forms of the three standard integrals. These extended standard forms should also be carefully memorised.

### 9 STANDARD INTEGRALS FOR FUNCTIONS OF $ax + b$

- $\displaystyle\int \cos(ax+b)\,dx = \frac{1}{a}\sin(ax+b) + C$, for some constant $C$
- $\displaystyle\int \sin(ax+b)\,dx = -\frac{1}{a}\cos(ax+b) + C$, for some constant $C$
- $\displaystyle\int \sec^2(ax+b)\,dx = \frac{1}{a}\tan(ax+b) + C$, for some constant $C$

### READING OFF $a$ AND $b$

Compare the angle with $ax + b$. For $\sin\frac{1}{4}x$, the angle is $\frac{1}{4}x + 0$, so $a = \frac{1}{4}$, $b = 0$ and $-\frac{1}{a} = -4$. For $\sec^2(2x + \pi)$, $a = 2$ and $b = \pi$, so $\frac{1}{a} = \frac{1}{2}$. For $\cos(7 - 5x)$, the angle is $-5x + 7$, so $a = -5$. The $\frac{1}{a}$ can be checked by differentiating the answer.

## TUTOR DEMO Functions of $ax + b$

Evaluate these definite integrals.

a $\int_0^{\frac{\pi}{4}} \cos 2x \, dx$

b $\int_0^{\pi} \sin \frac{1}{3}x \, dx$

c $\int_0^{\frac{\pi}{6}} \sec^2(2x + \pi) \, dx$

## YOUR TURN Same method

Evaluate these definite integrals.

a $\int_0^{\frac{\pi}{8}} \cos 4x \, dx$

b $\int_0^{\frac{3\pi}{2}} \sin \frac{1}{3}x \, dx$

c $\int_0^{\frac{\pi}{12}} \sec^2(3x + \pi) \, dx$

**WORKED EXAMPLE 18** **Keep for homework**

Evaluate these definite integrals.

a $\displaystyle\int_{\pi}^{2\pi} \sin \tfrac{1}{4}x \, dx$ b $\displaystyle\int_{0}^{\frac{\pi}{8}} \sec^2(2x+\pi) \, dx$

**SOLUTION**

**a**

$$\begin{aligned}\int_{\pi}^{2\pi} \sin \tfrac{1}{4}x \, dx &= -4\Big[\cos \tfrac{1}{4}x\Big]_{\pi}^{2\pi} \qquad \text{(because the reciprocal of } \tfrac{1}{4} \text{ is 4)} \\ &= -4\cos \tfrac{\pi}{2} + 4\cos \tfrac{\pi}{4} \\ &= 0 + 4 \times \frac{\sqrt{2}}{2} \\ &= 2\sqrt{2}\end{aligned}$$

**b**

$$\begin{aligned}\int_{0}^{\frac{\pi}{8}} \sec^2(2x+\pi) \, dx &= \tfrac{1}{2}\Big[\tan(2x+\pi)\Big]_{0}^{\frac{\pi}{8}} \\ &= \tfrac{1}{2}\Big(\tan \tfrac{5\pi}{4} - \tan \pi\Big) \\ &= \tfrac{1}{2}(1-0) \qquad \Big(\tfrac{5\pi}{4} \text{ is in quadrant 3 with related angle } \tfrac{\pi}{4}\Big) \\ &= \tfrac{1}{2}\end{aligned}$$

**THE VALUES IN WORKED EXAMPLE 18**

In part **a**, $-4\Big[\cos \tfrac{1}{4}x\Big]_{\pi}^{2\pi} = -4\Big(\cos \tfrac{2\pi}{4} - \cos \tfrac{\pi}{4}\Big) = -4\cos \tfrac{\pi}{2} + 4\cos \tfrac{\pi}{4}$. Then $\cos \tfrac{\pi}{2} = 0$ and $\cos \tfrac{\pi}{4} = \tfrac{\sqrt{2}}{2}$, and $4 \times \tfrac{\sqrt{2}}{2} = 2\sqrt{2}$.

In part **b**, the upper limit gives $2 \times \tfrac{\pi}{8} + \pi = \tfrac{\pi}{4} + \pi = \tfrac{5\pi}{4}$, and the lower limit gives $2 \times 0 + \pi = \pi$. The tangent is positive in quadrant 3, so $\tan \tfrac{5\pi}{4} = \tan \tfrac{\pi}{4} = 1$, and $\tan \pi = 0$.

## CONCEPT 3 · the primitives of tan x and cot x

### THE PRIMITIVES OF $\tan x$ AND $\cot x$

The primitives of $\tan x$ and $\cot x$ can be found by using the ratio formulae $\tan x = \dfrac{\sin x}{\cos x}$ and $\cot x = \dfrac{\cos x}{\sin x}$ and then applying the standard form from the previous chapter, in either one of its two versions,

$$\int \frac{u'}{u}\,dx = \log_e |u| + C \qquad \text{or} \qquad \int \frac{f'(x)}{f(x)}\,dx = \log_e |f(x)| + C.$$

TUTOR DEMO **The angle is $2x$**

Find primitives of these functions.

a $\cot 2x$

b $\tan 2x$

YOUR TURN **Same method**

Find primitives of these functions.

a $\cot 3x$

b $\tan 3x$

**WORKED EXAMPLE 19** **Keep for homework**

Find primitives of these functions.

a $\cot x$ b $\tan x$

**SOLUTION**

**a**

$$\int \cot x \, dx$$

$$= \int \frac{\cos x}{\sin x} \, dx$$

$$= \log_e |\sin x| + C.$$

Let $u = \sin x$.
Then $u' = \cos x$.

$$\int \frac{u'}{u} \, dx = \log_e |u|$$

OR

Let $f(x) = \sin x$.
Then $f'(x) = \cos x$.

$$\int \frac{f'(x)}{f(x)} \, dx = \log_e |f(x)|$$

**b**

$$\int \tan x \, dx$$

$$= \int \frac{\sin x}{\cos x} \, dx$$

$$= -\int \frac{-\sin x}{\cos x} \, dx$$

$$= -\log_e |\cos x| + C.$$

Let $u = \cos x$.
Then $u' = -\sin x$.

$$\int \frac{u'}{u} \, dx = \log_e |u|$$

OR

Let $f(x) = \cos x$.
Then $f'(x) = -\sin x$.

$$\int \frac{f'(x)}{f(x)} \, dx = \log_e |f(x)|$$

(This can also be written as $\log_e |\sec x|$.)

**NOTE**

Do not run across a zero of $\sin x$ when using part **a**, or a zero of $\cos x$ when using part **b**.

**THE STEPS IN WORKED EXAMPLE 19**

In part **b**, the top $\sin x$ is not yet the derivative of the bottom $\cos x$, which is $-\sin x$. Multiplying the top by $-1$ and the outside by $-1$ balances it:

$$\frac{\sin x}{\cos x} = -\frac{-\sin x}{\cos x}.$$

The last form uses a log law:

$$-\log_e |\cos x| = \log_e |\cos x|^{-1} = \log_e \left| \frac{1}{\cos x} \right| = \log_e |\sec x|.$$

When the angle is $ax$, balance the constant in the same way. For example, the derivative of $\sin 2x$ is $2\cos 2x$, so

$$\cot 2x = \frac{\cos 2x}{\sin 2x} = \frac{1}{2} \times \frac{2\cos 2x}{\sin 2x}.$$

## CONCEPT 4 · finding a function whose derivative is known

**FINDING A FUNCTION WHOSE DERIVATIVE IS KNOWN**

If the derivative of a function is known, and the value of the function at one point is also known, then the whole function can be found.

### TUTOR DEMO Find the constant

**1** The derivative of a certain function is $y' = \sin x$, and the graph of the function has $y$-intercept $(0, 2)$. Find the original function $f(x)$ and then find $f\left(\frac{\pi}{2}\right)$.

**2** Given that $f'(x) = \cos 2x$ and $f\left(\frac{\pi}{4}\right) = 1$:

**a** find the function $f(x)$,

**b** find $f\left(\frac{\pi}{12}\right)$.

### YOUR TURN Same method

**1** The derivative of a certain function is $y' = -\sin x$, and the graph of the function has $y$-intercept $(0, 4)$. Find the original function $f(x)$ and then find $f(\pi)$.

**2** Given that $f'(x) = \cos 2x$ and $f\left(\frac{\pi}{2}\right) = 2$:

**a** find the function $f(x)$,

**b** find $f\left(\frac{\pi}{4}\right)$.

**WORKED EXAMPLE 20** **Keep for homework**

The derivative of a certain function is $y' = \cos x$, and the graph of the function has $y$-intercept $(0, 3)$. Find the original function $f(x)$ and then find $f\left(\frac{\pi}{2}\right)$.

**SOLUTION**

Here $y' = \cos x,$

and taking the primitive, $y = \sin x + C,$ for some constant $C$.

When $x = 0$, $y = 3$, so substituting $x = 0$,

$$3 = \sin 0 + C$$
$$C = 3.$$

Hence $y = \sin x + 3.$

When $x = \frac{\pi}{2}$, $y = \sin \frac{\pi}{2} + 3$

$= 4,$ because $\sin \frac{\pi}{2} = 1.$

**WORKED EXAMPLE 21** **Keep for homework**

Given that $f'(x) = \sin 2x$ and $f(\pi) = 1$:
**a** find the function $f(x)$,
**b** find $f\left(\frac{\pi}{4}\right)$.

**SOLUTION**

**a**

Here $f'(x) = \sin 2x,$

and taking the primitive, $f(x) = -\frac{1}{2}\cos 2x + C,$ for some constant $C$.

It is known that $f(\pi) = 1$, so substituting $x = \pi$,

$$1 = -\frac{1}{2}\cos 2\pi + C$$
$$1 = -\frac{1}{2} \times 1 + C$$
$$C = 1\frac{1}{2}.$$

Hence $f(x) = -\frac{1}{2}\cos 2x + 1\frac{1}{2}.$

**b**

Substituting $x = \frac{\pi}{4}$, $f\left(\frac{\pi}{4}\right) = -\frac{1}{2} \times \cos \frac{\pi}{2} + 1\frac{1}{2}$

$= 1\frac{1}{2},$ because $\cos \frac{\pi}{2} = 0.$

**THE VALUES IN WORKED EXAMPLE 21**

In part **a**, $\int \sin 2x\,dx = -\frac{1}{2}\cos 2x$, because $a = 2$. Substituting $x = \pi$ makes the angle $2\pi$, and $\cos 2\pi = 1$. Then $1 = -\frac{1}{2} + C$ gives $C = 1 + \frac{1}{2} = 1\frac{1}{2}$.

In part **b**, substituting $x = \frac{\pi}{4}$ makes the angle $2 \times \frac{\pi}{4} = \frac{\pi}{2}$, and $\cos \frac{\pi}{2} = 0$, so $f\left(\frac{\pi}{4}\right) = 0 + 1\frac{1}{2} = 1\frac{1}{2}$.

## CONCEPT 5 · given a chain-rule derivative, find an integral

**GIVEN A CHAIN-RULE DERIVATIVE, FIND AN INTEGRAL**

As always, the results of a chain-rule differentiation can be reversed to give a primitive.

TUTOR DEMO **Reverse a chain-rule result**

**a** Use the chain rule to differentiate $\cos^7 x$.

**b** Hence find $\int_0^{\pi} \sin x \cos^6 x\,dx$.

YOUR TURN **Same method**

**a** Use the chain rule to differentiate $\cos^3 x$.

**b** Hence find $\int_0^{\frac{\pi}{3}} \sin x \cos^2 x \, dx$.

**WORKED EXAMPLE 22** **Keep for homework**

**a** Use the chain rule to differentiate $\cos^5 x$.

**b** Hence find $\int_0^{\pi} \sin x \cos^4 x \, dx$.

**SOLUTION**

**a**

Let $y = \cos^5 x$.

By the chain rule, $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$

$= -5 \sin x \cos^4 x.$

Let $u = \cos x$.
Then $y = u^5$.
Hence $\dfrac{du}{dx} = -\sin x$
and $\dfrac{dy}{du} = 5u^4$.

**b**

From part **a**, $\dfrac{d}{dx}(\cos^5 x) = -5 \sin x \cos^4 x.$

Reversing this, $\int (-5 \sin x \cos^4 x) \, dx = \cos^5 x.$

$\boxed{\div(-5)}$ $\int \sin x \cos^4 x \, dx = -\frac{1}{5}\cos^5 x.$

Hence

$$\int_0^{\pi} \sin x \cos^4 x \, dx = -\tfrac{1}{5}\Big[\cos^5 x\Big]_0^{\pi}$$
$$= -\tfrac{1}{5}(-1-1)$$
$$= \tfrac{2}{5}$$

**THE STEPS IN WORKED EXAMPLE 22**

In part **a**, $\frac{dy}{du} \times \frac{du}{dx} = 5u^4 \times (-\sin x) = -5 \sin x \cos^4 x$, after putting back $u = \cos x$.
In part **b**, the box $\div(-5)$ means: divide both sides by $-5$. For the limits, $\cos \pi = -1$, so $\cos^5 \pi = (-1)^5 = -1$, and $\cos 0 = 1$, so $\cos^5 0 = 1^5 = 1$. Hence $-\frac{1}{5}\Big[\cos^5 x\Big]_0^{\pi} = -\frac{1}{5}(-1-1) = -\frac{1}{5} \times (-2) = \frac{2}{5}$.

## CONCEPT 6 · using a formula for the reverse chain rule

**USING A FORMULA FOR THE REVERSE CHAIN RULE**

The integral in the worked example above could have been done using the reverse chain rule for powers of $u$ or $f(x)$.

**THE FORMULA FROM SECTION 5I**

The reverse chain rule for powers has two forms:

$$\int u^n \frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1} + C \qquad \text{OR} \qquad \int \big(f(x)\big)^n f'(x)\,dx = \frac{\big(f(x)\big)^{n+1}}{n+1} + C,$$

where $n \neq -1$. Let $u$ be the function that is raised to the power, such as $u = \cos x$ in $\sin x \cos^4 x$. Its derivative $u'$ must appear in the integrand, apart from a constant factor. Insert that constant inside the integral and balance it with its reciprocal outside.

**WORKED EXAMPLE 22 BY THE FORMULA**

$$\begin{aligned}
&\int_0^{\pi} \sin x \cos^4 x\,dx \\
&= -\int_0^{\pi} (-\sin x)\cos^4 x\,dx \\
&= -\left[\tfrac{1}{5}\cos^5 x\right]_0^{\pi} \\
&= -\tfrac{1}{5}\left(\cos^5 \pi - \cos^5 0\right) \\
&= -\tfrac{1}{5}(-1-1) \\
&= \tfrac{2}{5}
\end{aligned}$$

Let $u = \cos x$.
Then $u' = -\sin x$.

$$\int u^4 \frac{du}{dx}\,dx = \frac{u^5}{5}$$

The second line inserts the factor $-1$ to make $u' = -\sin x$, and balances it with $-1$ outside.

TUTOR DEMO **A power of $\sin x$**

Use the reverse chain rule to find $\int \cos x \sin^6 x\,dx$.

YOUR TURN **Same method**

Use the reverse chain rule to find $\int \sin x \cos^6 x\,dx$.

## TEXTBOOK HOMEWORK · Exercise 7D

**FOUNDATION**

**1** Find the following indefinite integrals.

a $\int \sec^2 x \, dx$

b $\int \cos x \, dx$

c $\int \sin x \, dx$

d $\int -\sin x \, dx$

e $\int 2\cos x \, dx$

f $\int \cos 2x \, dx$

g $\int \frac{1}{2}\cos x \, dx$

h $\int \cos \frac{1}{2}x \, dx$

i $\int \sin 2x \, dx$

j $\int \sec^2 5x \, dx$

k $\int \cos 3x \, dx$

l $\int \sec^2 \frac{1}{3}x \, dx$

m $\int \sin \frac{x}{2} \, dx$

n $\int -\cos \frac{1}{5}x \, dx$

o $\int -4\sin 2x \, dx$

p $\int \frac{1}{4}\sin \frac{1}{4}x \, dx$

q $\int 12\sec^2 \frac{1}{3}x \, dx$

r $\int 2\cos \frac{x}{3} \, dx$

**2** Find the value of:

a $\int_0^{\frac{\pi}{2}} \cos x \, dx$

b $\int_0^{\frac{\pi}{6}} \cos x \, dx$

c $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin x \, dx$

d $\int_0^{\frac{\pi}{3}} \sec^2 x \, dx$

e $\int_0^{\frac{\pi}{4}} 2\cos 2x \, dx$

f $\int_0^{\frac{\pi}{3}} \sin 2x \, dx$

g $\int_0^{\frac{\pi}{2}} \sec^2 \left(\frac{1}{2}x\right) dx$

h $\int_{\frac{\pi}{3}}^{\pi} \cos \left(\frac{1}{2}x\right) dx$

i $\int_0^{\pi} (2\sin x - \sin 2x) \, dx$

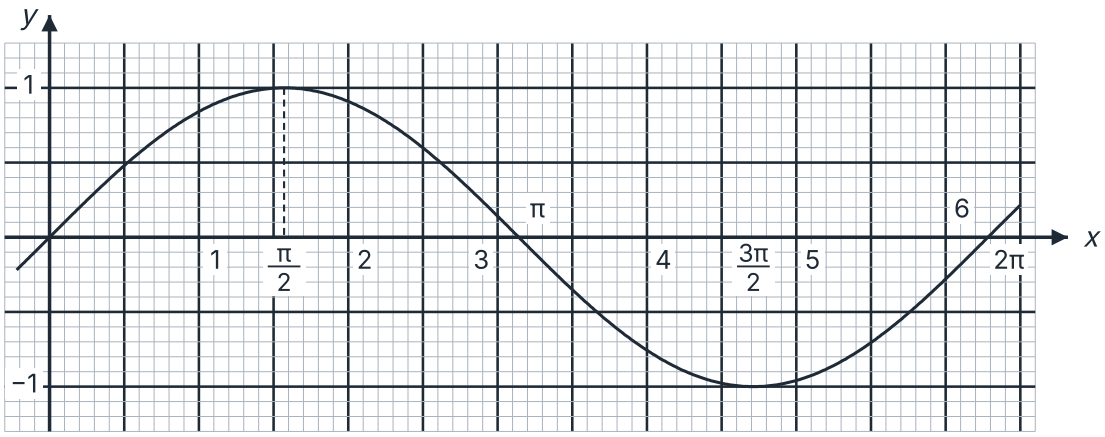
**3**

a The gradient function of a certain curve is given by $\frac{dy}{dx} = \sin x$. If the curve passes through the origin, find its equation.

b Another curve passing through the origin has gradient function $y' = \cos x - 2\sin 2x$. Find its equation.

c If $\frac{dy}{dx} = \sin x + \cos x$, and $y = -2$ when $x = \pi$, find $y$ as a function of $x$.

**4**




The graph of $y = \sin x$ is sketched above.

a The first worked exercise in the notes for this section proved that $\int_0^{\pi} \sin x \, dx = 2$.

Count squares on the graph of $y = \sin x$ above to confirm this result.

b On the same graph of $y = \sin x$, count squares and use symmetry to find:

i $\int_0^{\frac{\pi}{4}} \sin x \, dx$ ii $\int_0^{\frac{\pi}{2}} \sin x \, dx$ iii $\int_0^{\frac{3\pi}{4}} \sin x \, dx$

iv $\int_0^{\frac{5\pi}{4}} \sin x \, dx$ v $\int_0^{\frac{3\pi}{2}} \sin x \, dx$ vi $\int_0^{\frac{7\pi}{4}} \sin x \, dx$

c Evaluate these integrals using the fact that $-\cos x$ is a primitive of $\sin x$, and confirm the results of part **b**.

**5** [Technology]

Programs that sketch the graph and then approximate definite integrals would help reinforce the previous very important investigation. The investigation could then be continued past $x = \pi$, after which the definite integral decreases again.

Similar investigation with the graphs of $\cos x$ and $\sec^2 x$ would also be helpful, comparing the results of computer integration with the exact results obtained by integration using the standard primitives.

**6** Find the following indefinite integrals.

a $\int \cos(x+2)\,dx$ b $\int \cos(2x+1)\,dx$ c $\int \sin(x+2)\,dx$

d $\int \sin(2x+1)\,dx$ e $\int \cos(3x-2)\,dx$ f $\int \sin(7-5x)\,dx$

g $\int \sec^2(4-x)\,dx$ h $\int \sec^2\left(\frac{1-x}{3}\right)dx$ i $\int \sin\left(\frac{1-x}{3}\right)dx$

**7**

a Find $\int \left(6\cos 3x - 4\sin \tfrac{1}{2}x\right)dx$.

b Find $\int \left(8\sec^2 2x - 10\cos \tfrac{1}{4}x + 12\sin \tfrac{1}{3}x\right)dx$.

**8** a If $f'(x) = \pi \cos \pi x$ and $f(0) = 0$, find $f(x)$ and $f\left(\frac{1}{3}\right)$.

b If $f'(x) = \cos \pi x$ and $f(0) = \frac{1}{2\pi}$, find $f(x)$ and $f\left(\frac{1}{6}\right)$.

c If $f''(x) = 18 \cos 3x$ and $f'(0) = f\left(\frac{\pi}{2}\right) = 1$, find $f(x)$.

## DEVELOPMENT

**9** Find the following indefinite integrals, where $a$, $b$, $u$ and $v$ are constants.

a $\int a \sin(ax + b)\, dx$

b $\int \pi^2 \cos \pi x\, dx$

c $\int \frac{1}{u} \sec^2(v + ux)\, dx$

d $\int \frac{a}{\cos^2 ax}\, dx$

**10** a Copy and complete $1 + \tan^2 x = \ldots$, and hence find $\int \tan^2 x\, dx$.

b Simplify $1 - \sin^2 x$, and hence find the value of $\int_0^{\frac{\pi}{3}} \frac{2}{1 - \sin^2 x}\, dx$.

**11** a Copy and complete $\displaystyle\int \frac{f'(x)}{f(x)}\,dx = \ldots.$

b Hence show that $\displaystyle\int_0^{\frac{\pi}{6}} \frac{\cos x}{1+\sin x}\,dx \doteqdot 0.4.$

**12** a Use the fact that $\tan x = \dfrac{\sin x}{\cos x}$ to show that $\displaystyle\int_0^{\frac{\pi}{4}} \tan x\,dx = \tfrac{1}{2}\ln 2.$

b Use the fact that $\cot x = \dfrac{\cos x}{\sin x}$ to find $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot x\,dx.$

**13** a Find $\dfrac{d}{dx}(\sin^5 x)$, and hence find $\displaystyle\int \sin^4 x \cos x\,dx.$

b Find $\dfrac{d}{dx}(\tan^3 x)$, and hence find $\displaystyle\int \tan^2 x \sec^2 x\,dx.$

**14**

a Differentiate $e^{\sin x}$, and hence find the value of $\int_0^{\frac{\pi}{2}} \cos x\, e^{\sin x}\, dx$.

b Differentiate $e^{\tan x}$, and hence find the value of $\int_0^{\frac{\pi}{4}} \sec^2 x\, e^{\tan x}\, dx$.

**15**

a Show that $\frac{d}{dx}(\sin x - x\cos x) = x\sin x$, and hence find $\int_0^{\frac{\pi}{2}} x\sin x\, dx$.

b Show that $\frac{d}{dx}\left(\frac{1}{3}\cos^3 x - \cos x\right) = \sin^3 x$, and hence find $\int_0^{\frac{\pi}{3}} \sin^3 x\, dx$.

**16** Use the reverse chain rule $\int f'(x)\big(f(x)\big)^n\,dx = \dfrac{\big(f(x)\big)^{n+1}}{n+1}$, to evaluate:

a $\displaystyle\int_0^{\pi} \sin x \cos^8 x\,dx$ b $\displaystyle\int_0^{\frac{\pi}{2}} \sin x \cos^n x\,dx$ c $\displaystyle\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \sin^7 x\,dx$

d $\displaystyle\int_0^{\frac{\pi}{6}} \cos x \sin^n x\,dx$ e $\displaystyle\int_0^{\frac{\pi}{3}} \sec^2 x \tan^7 x\,dx$ f $\displaystyle\int_0^{\frac{\pi}{4}} \sec^2 x \tan^n x\,dx$

**17** a Show, by finding the integral in two different ways, that for constants $C$ and $D$,

$$\int \sin x \cos x\,dx = \tfrac{1}{2}\sin^2 x + C = -\tfrac{1}{4}\cos 2x + D.$$

b How may the two answers be reconciled?

**18** Find $\dfrac{d}{dx}\left(x \sin 2x\right)$, and hence find $\displaystyle\int_0^{\frac{\pi}{4}} x \cos 2x \, dx$.

**19** a Show that $\dfrac{d}{dx}(\tan^3 x) = 3(\sec^4 x - \sec^2 x)$.

b Hence find $\displaystyle\int_0^{\frac{\pi}{4}} \sec^4 x \, dx$.

ENRICHMENT

**20** a Show that $\sin(A+B)+\sin(A-B)=2\sin A\cos B$.

b Hence find:

i $\displaystyle\int_0^{\frac{\pi}{2}} 2\sin 3x\cos 2x\,dx$

ii $\displaystyle\int_0^{\pi} \sin 3x\cos 4x\,dx$

c Show that $\displaystyle\int_{-\pi}^{\pi} \sin mx\cos nx\,dx=0$ for positive integers $m$ and $n$:

i using the primitive,

ii using symmetry arguments.

**21** a Find the values of $A$ and $B$ in the identity

$A(2\sin x+\cos x)+B(2\cos x-\sin x)=7\sin x+11\cos x.$

b Hence show that $\displaystyle\int_0^{\frac{\pi}{2}} \frac{7\sin x+11\cos x}{2\sin x+\cos x}\,dx=\frac{1}{2}(5\pi+6\ln 2).$

**22** [The power series for $\sin x$ and $\cos x$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \qquad \text{and}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots .]$$

a We know that $\cos t \leq 1$, for $t$ positive. Integrate this inequality over the interval $0 \leq t \leq x$, where $x$ is positive, and hence show that $\sin x \leq x$.

b Change the variable to $t$, integrate the inequality $\sin t \leq t$ over $0 \leq t \leq x$, and hence show that $\cos x \geq 1 - \frac{x^2}{2!}$.

c Do it twice more, and show that:

i $\sin x \geq x - \frac{x^3}{3!}$ ii $\cos x \leq 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$

d Now use induction (informally) to show that for all positive integers $n$,

$$\sin x \leq x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + \frac{x^{4n+1}}{(4n+1)!} \leq \sin x + \frac{x^{4n+3}}{(4n+3)!},$$

and use this inequality to conclude that $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ converges, with limit $\sin x$.

e Proceeding similarly, prove that $1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$ converges, with limit $\cos x$.

f Use evenness and oddness to extend the results of (d) and (e) to negative values of $x$.

## PAST-PAPER HOMEWORK · mini exam

**Suggested time: 25 minutes** **16 marks** Calculator allowed · show working · each hint is below its question if you get stuck

**PP 1** **Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q7** **1 mark**

Which of the following is a correct primitive of $\tan x$?

(A) $\sec^2 x + c$ (B) $\ln|A\sin x|$ (C) $\dfrac{1}{1+x^2} + c$ (D) $-\ln|A\cos x|$

**Hint:** what is being integrated here?

**PP 2** **Caringbah High School, 2021, Year 12 Mathematics Extension 1 HSC Course Assessment Task 4, Q6** **1 mark**

Find $\displaystyle\int \sec^2\theta\tan^2\theta\,d\theta$

(A) $\sec^2\theta + \dfrac{1}{2}\tan^2\theta + C$ (B) $\dfrac{1}{3}\tan^3\theta + C$

(C) $\dfrac{1}{3}\tan^3\theta + \dfrac{1}{5}\tan^5\theta + C$ (D) $\tan^4\theta - \ln\left|\cos^4\theta\right| + C$

**Hint:** what is being integrated here?

**PP 3** **Cherrybrook Technology High School, 2020, Year 12 AP4, Mathematics Extension 1, Q5** **1 mark**

Find the primitive function of $\dfrac{dy}{dx} = \sin x\cos^3 x$.

A. $y = \dfrac{1}{4}\cos^4 x + C$ B. $y = -\dfrac{1}{4}\cos^4 x + C$

C. $y = \dfrac{1}{8}\sin^2 x\cos^4 x + C$ D. $y = -\dfrac{1}{8}\sin^2 x\cos^4 x + C$

**Hint:** what is being integrated here?

**PP 4** **Caringbah High School, 2020, Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(b)** 2 marks

Find $\int_0^{\frac{\pi}{6}} \sin x \cos x \, dx$

**Hint:** what is being integrated here?

**PP 5** **Fort Street High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q12(e)** 2 marks

The curve with equation $y = f(x)$ passes through the point $\left(\frac{\pi}{8}, 2\right)$ and has a gradient of $-1$ at this point. Find the exact gradient of the curve at $x = \frac{\pi}{12}$ given that $y'' = -\sec^2(2x)$.

**Hint:** what does the information at the point tell you?

**PP 6** **Hornsby Girls High School, 2022, Mathematics Extension 1, Year 12 Higher School Certificate Trial Examination Term 3, Q14(a)** 3 marks

If $f'(x) = \cot x + x$ and $f\left(\frac{\pi}{2}\right) = 0$, find an expression for $f(x)$.

**Hint:** what information are you given about $f$?

**PP 7** **Abbotsleigh, 2023, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q11(d)(ii)** 3 marks

Part (i), not included here, shows that $2\sqrt{3}\sin x + 2\cos x = 4\sin\left(x + \frac{\pi}{6}\right)$.

(ii) Hence find the value of $k$, where $0 \leq k \leq 2\pi$, for which 3

$$\int_0^k \left(2\sqrt{3}\cos x - 2\sin x\right) dx = 2.$$

**Hint:** what does the result of part (i) give you?

**PP 8** **Killara High School, 2020, Mathematics Extension 1 Trial HSC Examination, Q12(d)(ii)** 3 marks

Part (i), not included here, uses the substitution $t = \tan \dfrac{x}{2}$ to show that $\operatorname{cosec} x + \cot x = \cot \dfrac{x}{2}$.

(ii) Hence, evaluate $\displaystyle\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\operatorname{cosec} x + \cot x)\, dx$. 3

(Answer in simplest form)

**Hint:** what does part (i) give you?

LESSON 7E

# Applications of integration

The trigonometric integrals from Lesson 7D can now be used to find areas in the usual way. We find the area between a curve such as $y = \cos \frac{1}{2}x$ and the $x$-axis, where the integral is split at every $x$-intercept, and the area between two curves, such as $y = \sin x$ and $y = \sin 2x$, using the top curve minus the bottom curve.

## CONCEPT 1 · the area between a curve and the x-axis

### APPLICATIONS OF INTEGRATION

The trigonometric integrals can now be used to find areas in the usual way.

### FINDING AREAS BY INTEGRATION

As always, *a sketch is essential*, because areas below the $x$-axis are represented as a negative number by the definite integral.

It is best to evaluate the separate integrals first and then make a conclusion about areas.

### THE STANDARD FORMS FROM LESSON 7D

- $\displaystyle\int \cos(ax+b)\,dx = \frac{1}{a}\sin(ax+b) + C$
- $\displaystyle\int \sin(ax+b)\,dx = -\frac{1}{a}\cos(ax+b) + C$
- $\displaystyle\int \sec^2(ax+b)\,dx = \frac{1}{a}\tan(ax+b) + C$

For example, $\displaystyle\int \cos \tfrac{1}{2}x\,dx = \frac{1}{\frac{1}{2}}\sin \tfrac{1}{2}x + C = 2\sin \tfrac{1}{2}x + C$, because the reciprocal of $\frac{1}{2}$ is $2$.

### SKETCHING $y = \cos nx$ AND $y = \sin nx$ (YEAR 11)

The curves $y = \cos nx$ and $y = \sin nx$, where $n > 0$, have amplitude $1$ and period $2\pi \div n$. The curve $y = \cos nx$ crosses the $x$-axis where $nx = \dfrac{\pi}{2}, \dfrac{3\pi}{2}, \dfrac{5\pi}{2}, \ldots$, and the curve $y = \sin nx$ crosses it where $nx = 0,\ \pi,\ 2\pi,\ \ldots$. The curve is above the $x$-axis and below it in turn between these intercepts.

**THE STEPS FOR FINDING AN AREA**

- Sketch the curve in the given interval, and find every $x$-intercept inside it.
- Evaluate a separate integral over each piece between the intercepts.
- A piece below the $x$-axis gives a negative integral. Its area is the opposite of that integral.
- Add the areas of the pieces, and write the conclusion in square units.

TUTOR DEMO **Three pieces, one below the axis**

**a** Sketch $y = \cos \frac{1}{3}x$ in the interval $0 \leq x \leq 6\pi$, marking both $x$-intercepts.
**b** Hence find the area between the curve and the $x$-axis, for $0 \leq x \leq 6\pi$.

YOUR TURN **Same method**

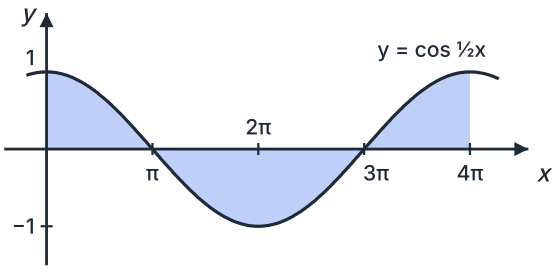
**a** Sketch $y = \cos 2x$ in the interval $0 \leq x \leq \pi$, marking both $x$-intercepts.
**b** Hence find the area between the curve and the $x$-axis, for $0 \leq x \leq \pi$.

**WORKED EXAMPLE 24** **Keep for homework**

**a** Sketch $y = \cos \frac{1}{2}x$ in the interval $0 \leq x \leq 4\pi$, marking both $x$-intercepts.
**b** Hence find the area between the curve and the $x$-axis, for $0 \leq x \leq 4\pi$.

**SOLUTION**

**a** The curve $y = \cos \frac{1}{2}x$ has amplitude 1, and the period is $2\pi \div \frac{1}{2} = 4\pi$.
The two $x$-intercepts in the interval are $x = \pi$ and $x = 3\pi$.




**b** We must integrate separately over the three intervals $[0, \pi]$ and $[\pi, 3\pi]$ and $[3\pi, 4\pi]$.

First,

$$\begin{aligned}\int_0^{\pi} \cos \tfrac{1}{2}x \, dx &= \left[2 \sin \tfrac{1}{2}x\right]_0^{\pi} \\ &= 2 \sin \tfrac{\pi}{2} - 2 \sin 0 \\ &= 2 - 0 \\ &= 2,\end{aligned}$$

which is positive, because the curve is above the $x$-axis for $0 \leq x \leq \pi$.

Secondly,

$$\begin{aligned}\int_{\pi}^{3\pi} \cos \tfrac{1}{2}x \, dx &= \left[2 \sin \tfrac{1}{2}x\right]_{\pi}^{3\pi} \\ &= 2 \sin \tfrac{3\pi}{2} - 2 \sin \tfrac{\pi}{2} \\ &= -2 - 2 \\ &= -4,\end{aligned}$$

which is negative, because the curve is below the $x$-axis for $\pi \leq x \leq 3\pi$.

Thirdly,

$$\begin{aligned}\int_{3\pi}^{4\pi} \cos \tfrac{1}{2}x \, dx &= \left[2 \sin \tfrac{1}{2}x\right]_{3\pi}^{4\pi} \\ &= 2 \sin 2\pi - 2 \sin \tfrac{3\pi}{2} \\ &= 0 - (-2) \\ &= 2,\end{aligned}$$

which is positive, because the curve is above the $x$-axis for $3\pi \leq x \leq 4\pi$.

Hence

$$\begin{aligned}\text{total area} &= 2 + 4 + 2 \\ &= 8 \text{ square units.}\end{aligned}$$

**READING WORKED EXAMPLE 24, LINE BY LINE**

In part **a**, the curve crosses the $x$-axis where $\frac{1}{2}x = \frac{\pi}{2}$ or $\frac{1}{2}x = \frac{3\pi}{2}$, that is, where $x = \pi$ or $x = 3\pi$. In part **b**, the primitive $2\sin\frac{1}{2}x$ comes from the first standard form with $a = \frac{1}{2}$. At the limits, $\frac{1}{2} \times \pi = \frac{\pi}{2}$, $\frac{1}{2} \times 3\pi = \frac{3\pi}{2}$ and $\frac{1}{2} \times 4\pi = 2\pi$. The values are $\sin 0 = 0$, $\sin\frac{\pi}{2} = 1$, $\sin\frac{3\pi}{2} = -1$ and $\sin 2\pi = 0$. The middle integral is $-4$, so the middle piece has area $4$, and the total area is $2 + 4 + 2 = 8$.

## CONCEPT 2 · areas between curves

**FINDING AREAS BETWEEN CURVES**

The next worked example uses the principle that if $y = f(x)$ is above $y = g(x)$ throughout some interval $a \le x \le b$, then the area between the curves is given by the formula

$$\text{area between the curves} = \int_a^b \big(f(x) - g(x)\big)\,dx.$$

**TOP CURVE MINUS BOTTOM CURVE**

- Find where the curves meet. These $x$-values are usually the limits of the integral.
- Decide which curve is on top by testing one $x$-value between the limits.
- Integrate the top curve minus the bottom curve. The formula still works when part of the region lies below the $x$-axis, because $f(x) - g(x)$ is positive throughout the interval.

To show that two curves meet at a given $x$-value, substitute it into both equations and show that the $y$-values are equal.

TUTOR DEMO **The other region between the curves**

**a** Show that the curves $y = \sin x$ and $y = \sin 2x$ intersect when $x = \dfrac{\pi}{3}$.

**b** Sketch these curves in the interval $[0, \pi]$.

**c** Find the area contained between the curves in the interval $\left[\dfrac{\pi}{3}, \pi\right]$.

YOUR TURN **Same method**

**a** Show that the curves $y = \cos x$ and $y = \sin 2x$ intersect when $x = \frac{\pi}{6}$.

**b** Sketch these curves in the interval $\left[0, \frac{\pi}{2}\right]$.

**c** Find the area contained between the curves in the interval $\left[0, \frac{\pi}{6}\right]$.

**WORKED EXAMPLE 25** **Keep for homework**

**a** Show that the curves $y = \sin x$ and $y = \sin 2x$ intersect when $x = \dfrac{\pi}{3}$.

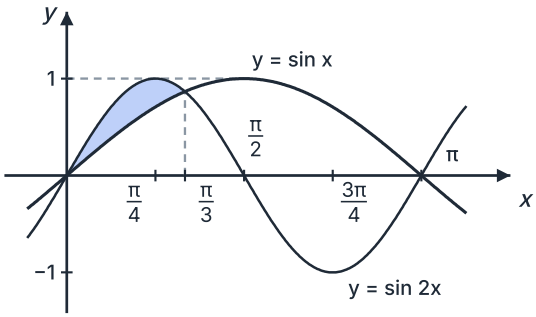
**b** Sketch these curves in the interval $[0, \pi]$.

**c** Find the area contained between the curves in the interval $\left[0, \dfrac{\pi}{3}\right]$.

**SOLUTION**

**a** The curves intersect at $x = \dfrac{\pi}{3}$ because $\sin\dfrac{\pi}{3} = \sin\dfrac{2\pi}{3} = \frac{1}{2}\sqrt{3}$.

**b** The curves are sketched to the right.




**c** In the interval $0 \le x \le \dfrac{\pi}{3}$, the curve $y = \sin 2x$ is always above $y = \sin x$,

$$\text{so area between} = \int_0^{\frac{\pi}{3}} (\sin 2x - \sin x)\, dx$$

$$= \left[ -\tfrac{1}{2}\cos 2x + \cos x \right]_0^{\frac{\pi}{3}}$$

$$= \left( -\tfrac{1}{2}\cos\tfrac{2\pi}{3} + \cos\tfrac{\pi}{3} \right) - \left( -\tfrac{1}{2}\cos 0 + \cos 0 \right).$$

Because $\cos 0 = 1$ and $\cos\dfrac{\pi}{3} = \frac{1}{2}$ and $\cos\dfrac{2\pi}{3} = -\frac{1}{2}$,

$$\text{area} = \left(\tfrac{1}{4} + \tfrac{1}{2}\right) - \left(-\tfrac{1}{2} + 1\right)$$

$$= \tfrac{1}{4} \text{ square units.}$$

**READING WORKED EXAMPLE 25, LINE BY LINE**

In part **a**, $\sin 2x$ at $x = \dfrac{\pi}{3}$ is $\sin\dfrac{2\pi}{3}$. The angle $\dfrac{2\pi}{3}$ is in quadrant 2 with related angle $\dfrac{\pi}{3}$, and sine is positive there, so $\sin\dfrac{2\pi}{3} = \sin\dfrac{\pi}{3} = \frac{1}{2}\sqrt{3}$. In part **c**, a test value shows which curve is on top: at $x = \dfrac{\pi}{4}$, $\sin 2x = \sin\dfrac{\pi}{2} = 1$ and $\sin x = \frac{1}{2}\sqrt{2}$, which is less than $1$. The primitive of $\sin 2x$ is $-\frac{1}{2}\cos 2x$, and the primitive of $-\sin x$ is $\cos x$. In the last lines, $-\frac{1}{2} \times \left(-\frac{1}{2}\right) = \frac{1}{4}$ and $-\frac{1}{2} \times 1 + 1 = \frac{1}{2}$, so the area is $\frac{3}{4} - \frac{1}{2} = \frac{1}{4}$.

## CONCEPT 3 · two curves that cross twice

**WHERE $y = \sin x$ AND $y = \cos x$ MEET**

To solve $\sin x = \cos x$, divide both sides by $\cos x$. This is allowed, because $\cos x \neq 0$ at any solution: if $\cos x = 0$, then $\sin x = \pm 1$, and the two sides are not equal. Dividing gives $\tan x = 1$. In each interval of length $2\pi$ there are two solutions, $\pi$ apart, one in quadrant 1 and one in quadrant 3.

The curves then cross twice, and the region between the crossings lies partly above and partly below the $x$-axis. Top curve minus bottom curve still gives its area in one integral.

TUTOR DEMO **A region across the $x$-axis**

**a** Show that in the interval $0 \leq x \leq 2\pi$, the curves $y = \sin x$ and $y = -\cos x$ intersect when $x = \dfrac{3\pi}{4}$ and when $x = \dfrac{7\pi}{4}$.

**b** Sketch the curves in this interval and find the area contained between them.

YOUR TURN **Same method**

**a** Show that in the interval $-\pi \leq x \leq \pi$, the curves $y = \sin x$ and $y = \cos x$ intersect when $x = -\dfrac{3\pi}{4}$ and when $x = \dfrac{\pi}{4}$.

**b** Sketch the curves in this interval and find the area contained between them.

## TEXTBOOK HOMEWORK · Exercise 7E

**Technology:** Some graphing programs can perform numerical integration on specified regions. Such programs would help to confirm the integrals in this exercise and to investigate quickly further integrals associated with these curves.

### FOUNDATION

**1** Find the exact area between the curve $y = \cos x$ and the $x$-axis:

a from $x = 0$ to $x = \dfrac{\pi}{2}$,

b from $x = 0$ to $x = \dfrac{\pi}{6}$.

**2** Find the exact area between the curve $y = \sec^2 x$ and the $x$-axis:

a from $x = 0$ to $x = \dfrac{\pi}{4}$,

b from $x = 0$ to $x = \dfrac{\pi}{3}$.

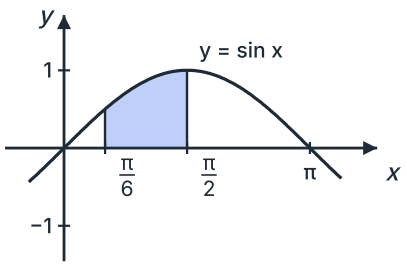
**3** Find the exact area between the curve $y = \sin x$ and the $x$-axis:

a from $x = 0$ to $x = \dfrac{\pi}{4}$,

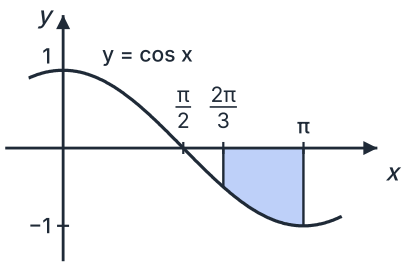
b from $x = 0$ to $x = \dfrac{\pi}{6}$.

**4** Calculate the area of the shaded region in each diagram below (and then observe that the two regions have equal area).

a




b




**5** Find the area enclosed between each curve and the $x$-axis over the specified domain.

a $y = \sin x$, from $x = \dfrac{\pi}{3}$ to $x = \dfrac{\pi}{2}$

b $y = \sin 2x$, from $x = \dfrac{\pi}{4}$ to $x = \dfrac{\pi}{2}$

c $y = \cos x$, from $x = \dfrac{\pi}{3}$ to $x = \dfrac{\pi}{2}$

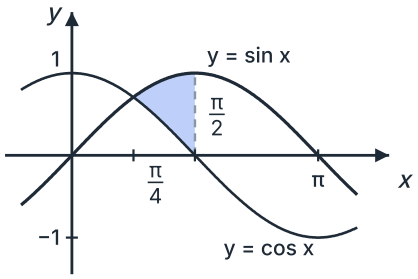
d $y = \cos 3x$, from $x = \dfrac{\pi}{12}$ to $x = \dfrac{\pi}{6}$

e $y = \sec^2 x$, from $x = \dfrac{\pi}{6}$ to $x = \dfrac{\pi}{3}$

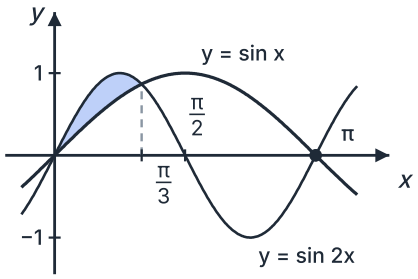
f $y = \sec^2 \frac{1}{2}x$, from $x = -\dfrac{\pi}{2}$ to $x = \dfrac{\pi}{2}$

**6** Calculate the area of the shaded region in each diagram below.

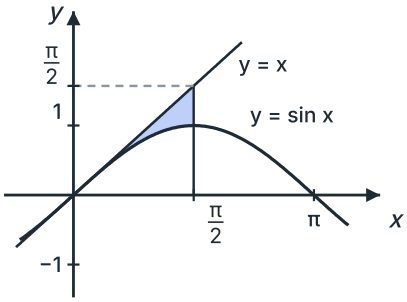
**a**



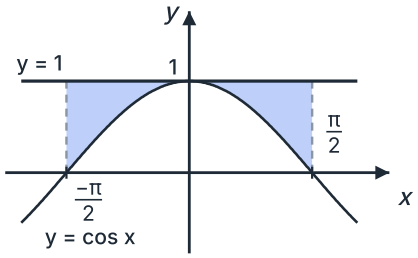

**b**




**c**

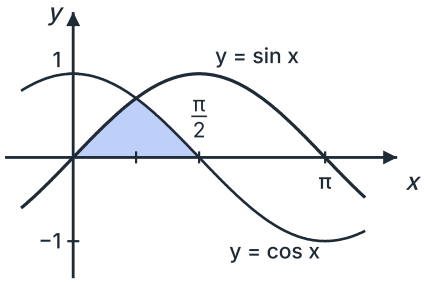



**d**

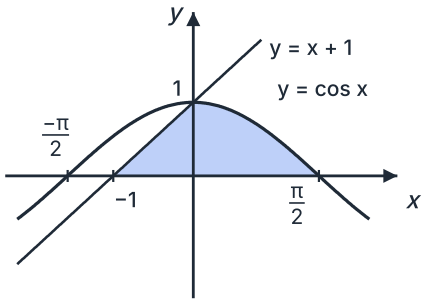


**7** Calculate the area of the shaded region in each diagram below.

**a**




**b**




**DEVELOPMENT**

**8** Find, using a diagram, the area bounded by one arch of each curve and the $x$-axis.

a $y = \sin x$

b $y = \cos 2x$

**9** Sketch the area enclosed between each curve and the $x$-axis over the specified domain, and then find the exact value of the area. (Make use of symmetry wherever possible.)

a $y = \cos x$, from $x = 0$ to $x = \pi$

b $y = \sin x$, from $x = \dfrac{\pi}{4}$ to $x = \dfrac{3\pi}{4}$

c $y = \cos 2x$, from $x = 0$ to $x = \pi$

d $y = \sin 2x$, from $x = \dfrac{\pi}{3}$ to $x = \dfrac{2\pi}{3}$

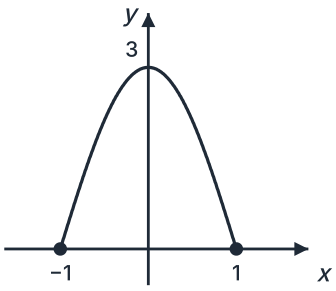
e $y = \sin x$, from $x = -\dfrac{5\pi}{6}$ to $x = \dfrac{7\pi}{6}$

f $y = \cos 3x$, from $x = \dfrac{\pi}{6}$ to $x = \dfrac{2\pi}{3}$

**10** a Sketch the curve $y = 2\cos \pi x$, for $-1 \leq x \leq 1$, clearly marking the two $x$-intercepts.

b Find the exact area bounded by the curve $y = 2\cos \pi x$ and the $x$-axis, between the two $x$-intercepts.

**11** An arch window 3 metres high and 2 metres wide is made in the shape of the curve $y = 3\cos\left(\frac{\pi}{2}x\right)$, as shown to the right. Find the area of the window in square metres, correct to one decimal place.




**12** The graphs of $y = x - \sin x$ and $y = x$ are sketched together in a worked exercise in Section 7C. Find the total area enclosed between these graphs, from $x = 0$ to $x = 2\pi$.

**13** The region $R$ is bounded by the curve $y = \tan x$, the $x$-axis and the vertical line $x = \frac{\pi}{3}$. Show that $R$ has area $\ln 2$ square units.

**14** a Sketch the region bounded by the graphs of $y = \sin x$ and $y = \cos x$, and by the vertical lines $x = -\dfrac{\pi}{2}$ and $x = \dfrac{\pi}{6}$.

b Find the area of the region in part **a**.

**15** a Show by substitution that $y = \sin x$ and $y = \cos 2x$ meet at $x = -\dfrac{\pi}{2}$ and $x = \dfrac{\pi}{6}$.

b On the same number plane, sketch $y = \sin x$ and $y = \cos 2x$, for $-\dfrac{\pi}{2} \leq x \leq \dfrac{\pi}{6}$.

c Hence find the area of the region bounded by the two curves.

**16** a Show that $\sqrt{2}\sin\left(x+\frac{\pi}{4}\right)=\sin x+\cos x$.

b Hence, or otherwise, find the exact area under one arch of the curve $y=\sin x+\cos x$.

**17** a Show that for all positive integers $n$:

i $\int_0^{2\pi}\sin nx\,dx=0$ ii $\int_0^{2\pi}\cos nx\,dx=0$

b Sketch each of the following graphs, and then find the area between the curve and the $x$-axis, from $x=0$ to $x=2\pi$.

i $y=\sin x$ ii $y=\sin 2x$ iii $y=\sin 3x$ iv $y=\sin nx$ v $y=\cos nx$

**18** a Show that $\int_0^n (1+\sin 2\pi x)\,dx = n$, for all positive integers $n$.

b Sketch $y = 1+\sin 2\pi x$, and interpret the result geometrically.

**19** Sketch $y = |\sin x|$ for $0 \le x \le 6\pi$, and hence evaluate $\int_0^{6\pi} |\sin x|\,dx$.

**20** a Using the fact that $\sin x < x < \tan x$ for $0 < x < \dfrac{\pi}{2}$, explain why $x^2 \sin x < x^3 < x^2 \tan x$ for $0 < x < \dfrac{\pi}{2}$.

b Hence show that $\int_0^{\frac{\pi}{4}} x^2 \sin x\,dx < \dfrac{\pi^4}{4^5} < \int_0^{\frac{\pi}{4}} x^2 \tan x\,dx$.

**21** a Given that $y = \dfrac{1}{1+\sin x}$, show that $y' = -\dfrac{\cos x}{(1+\sin x)^2}$.

b Hence explain why the function $y = \dfrac{1}{1+\sin x}$ is decreasing for $0 < x < \dfrac{\pi}{2}$.

c Sketch the curve for $0 \leq x \leq \dfrac{\pi}{2}$, and hence show that $\dfrac{\pi}{4} < \displaystyle\int_0^{\frac{\pi}{2}} \frac{1}{1+\sin x}\,dx < \frac{\pi}{2}$.

**22** Use symmetry arguments to help evaluate:

a $\displaystyle\int_{-4\pi}^{4\pi} \sin 3x\,dx$

b $\displaystyle\int_{-2\pi}^{2\pi} \cos^2 x \sin^3 x\,dx$

c $\displaystyle\int_{-\frac{5\pi}{2}}^{\frac{5\pi}{2}} \cos x\,dx$

d $\displaystyle\int_{-\pi}^{\pi} \sec^2 \tfrac{1}{3}x\,dx$

e $\displaystyle\int_{-\pi}^{\pi} (3+2x+\sin x)\,dx$

f $\displaystyle\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin 2x + \cos 3x + 3x^2)\,dx$

ENRICHMENT

**23** a Show that $\dfrac{d}{dx}\left(-\frac{1}{2}e^{-x}(\sin x+\cos x)\right)=e^{-x}\sin x$.

b Find $\displaystyle\int_0^N e^{-x}\sin x\,dx$, and show that $\displaystyle\int_0^\infty e^{-x}\sin x\,dx$ converges to $\dfrac{1}{2}$.

c Find $\displaystyle\int_0^\pi e^{-x}\sin x\,dx$, $\displaystyle\int_{2\pi}^{3\pi} e^{-x}\sin x\,dx$, ..., and show that the areas of the arches above the $x$-axis form a GP with limiting sum $\dfrac{e^\pi}{2(e^\pi-1)}$.

d Show that the areas of the arches below the $x$-axis also form a GP, and hence show that the total area contained between the curve and the $x$-axis, to the right of the $y$-axis, is $\dfrac{e^\pi+1}{2(e^\pi-1)}$. Also confirm by subtraction the result of part **b**.

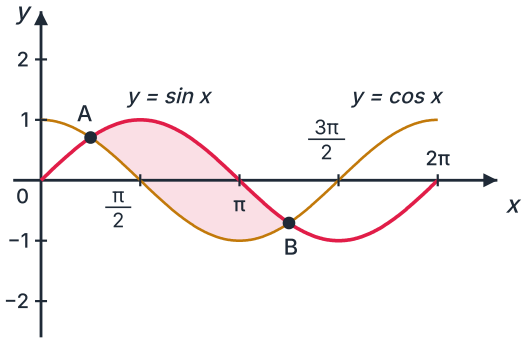
## PAST-PAPER HOMEWORK · mini exam

**Suggested time: 25 minutes** **17 marks** Calculator allowed · show working · each hint is below its question if you get stuck

**PP 1** **Fort Street High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q10** **1 mark**

The area enclosed by the curves $y = \sin x$ and $y = \cos x$ is shaded in the diagram.

Which expression could be used to calculate this area?




A. $\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \left(\cos(x) - \sin(x)\right) dx$

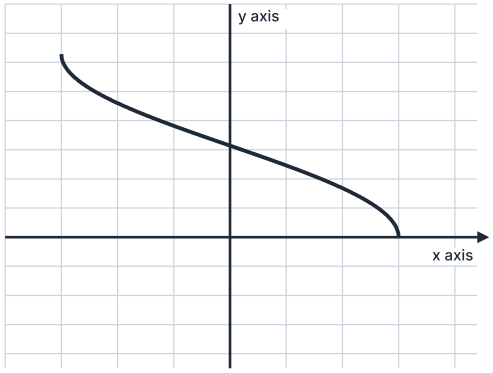
B. $\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \left(\sin(x) - \cos(x)\right) dx$

C. $\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \left(\cos(x) + \sin(x)\right) dx$

D. $\int_{\frac{\pi}{3}}^{\frac{4\pi}{3}} \left(\cos(x) - \sin(x)\right) dx$

**Hint:** what does the diagram tell you about the boundaries of the shaded region?

**PP 2** **Caringbah High School, 2021, Year 12 Mathematics Extension 1 HSC Course Assessment Task 4, Q12(c)** **5 marks**




The graph above is the function $f(x) = 2\cos^{-1}\dfrac{x}{3}$.

(i) Find the value of the $y$ intercept. 1

(ii) Find the domain and range of $f(x) = 2\cos^{-1}\dfrac{x}{3}$. 2

(iii) Calculate the area of the region under the function $f(x) = 2\cos^{-1}\dfrac{x}{3}$ between $x = 0$ and $x = 3$. 2

**Hint:** what does the graph tell you about the region?

**PP 3** **Northern Beaches Secondary College Manly Selective Campus, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q11(c)** 3 marks

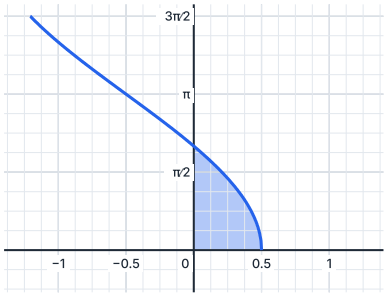
Find the area enclosed by the curve $y = \tan^{-1} x$, the $y$-axis and the lines $y = \pm\frac{\pi}{4}$.

**(Express your answer in simplest exact form).**

**Hint:** what does the region look like?

**PP 4** **Hornsby Girls High School, 2023, Year 12 Higher School Certificate Trial Examination Term 3, Mathematics Extension 1, Q14(b)** 3 marks

The graph of $y = 2\cos^{-1}\left(x + \frac{1}{2}\right)$ is shown below.




Calculate the shaded area.

**Hint:** what are the boundaries of the shaded region?

**PP 5** **Pymble Ladies' College, 2024, Mathematics Extension 1 HSC Trial Examination Term 3, Q13(d)** **5 marks**

(i) Sketch the curve $y = -\tan^{-1} x$ and label the point where $x = 1$. 2

(ii) Find the area bounded by the curve, the $x$-axis and the line $x = 1$. 3

**Hint:** what does your sketch show about the region?

REVIEW

# Chapter review exercise

The integration questions from the Chapter 6 review (exponential and logarithmic functions) and the Chapter 7 review (trigonometric functions), for mixed practice across this pack. The question numbers are the textbook's.

## CHAPTER 6 REVIEW · integration questions

**16** Find:

a $\int e^{5x}\,dx$ b $\int 10e^{2-5x}\,dx$ c $\int e^{\frac{1}{5}x}\,dx$ d $\int 3e^{5x-4}\,dx$

**17** Find the exact value of:

a $\int_0^2 e^{x}\,dx$ b $\int_0^1 e^{2x}\,dx$ c $\int_{-1}^0 e^{-x}\,dx$

d $\int_{-\frac{2}{3}}^0 e^{3x+2}\,dx$ e $\int_0^{\frac{1}{2}} e^{3-2x}\,dx$ f $\int_0^2 2e^{\frac{1}{2}x}\,dx$

**18** Find the primitive of:

a $\dfrac{1}{e^{5x}}$ b $e^{3x} \times e^{x}$ c $\dfrac{6}{e^{3x}}$ d $(e^{3x})^2$

e $\dfrac{e^{3x}}{e^{5x}}$ f $\dfrac{e^{3x}+1}{e^{2x}}$ g $e^{2x}(e^{x}+e^{-x})$ h $(1+e^{-x})^2$

**19** Find the exact value of:

a $\displaystyle\int_0^1 (1+e^{-x})\,dx$ b $\displaystyle\int_0^2 (e^{2x}+x)\,dx$ c $\displaystyle\int_0^1 \frac{2}{e^{x}}\,dx$

d $\displaystyle\int_0^{\frac{1}{3}} e^{3x}(1-e^{-3x})\,dx$ e $\displaystyle\int_0^1 \frac{e^{2x}+1}{e^{x}}\,dx$ f $\displaystyle\int_0^1 (e^{x}+1)^2\,dx$

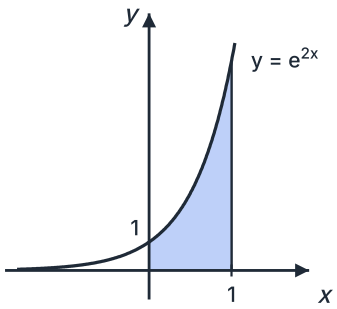
**20** If $f'(x) = e^{x} - e^{-x} - 1$ and $f(0) = 3$, find $f(x)$ and then find $f(1)$.

**21** a Differentiate $e^{x^3}$.

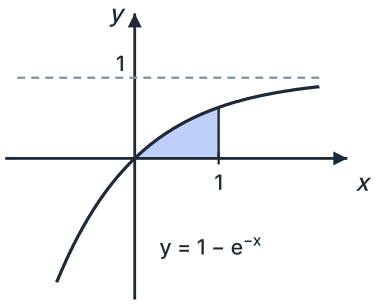
b Hence find $\int_0^1 x^2 e^{x^3}\,dx$.

**22** Find the area of each region correct to three significant figures.

a

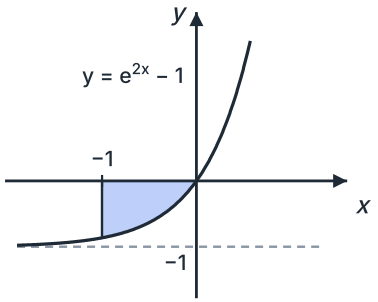



b

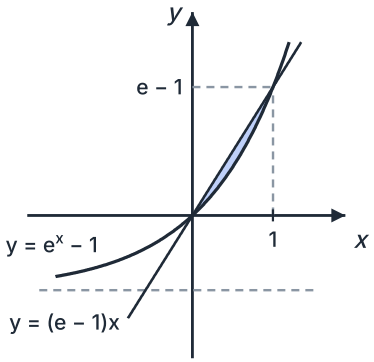


**23** Find the exact area of the shaded region.

**a**




**b**




**32** Find these indefinite integrals.

a $\int \frac{1}{x}\,dx$ b $\int \frac{3}{x}\,dx$ c $\int \frac{1}{5x}\,dx$ d $\int \frac{1}{x+7}\,dx$

e $\int \frac{1}{2x-1}\,dx$ f $\int \frac{1}{2-3x}\,dx$ g $\int \frac{2}{2x+9}\,dx$ h $\int \frac{8}{1-4x}\,dx$

**33** Evaluate these definite integrals.

a $\int_0^1 \frac{1}{x+2}\,dx$ b $\int_1^4 \frac{1}{4x-3}\,dx$ c $\int_1^e \frac{1}{x}\,dx$ d $\int_{e^2}^{e^3} \frac{1}{x}\,dx$

**34** Use the standard form $\int \frac{u'}{u}\,dx = \log_e |u| + C$ OR $\int \frac{f'(x)}{f(x)}\,dx = \log_e |f(x)| + C$ to find:

a $\int \frac{2x}{x^2+4}\,dx$ b $\int \frac{3x^2-5}{x^3-5x+7}\,dx$ c $\int \frac{x}{x^2-3}\,dx$ d $\int \frac{x^3-1}{x^4-4x}\,dx$

**35** Find the area of the region bounded by the curve $y = \frac{1}{x}$, the $x$-axis and the lines $x = 2$ and $x = 4$.

**36** **a** By solving the equations simultaneously, show that the curve $y = \dfrac{5}{x}$ and the line $y = 6 - x$ intersect at the points $(1, 5)$ and $(5, 1)$.

**b** By sketching both graphs on the same number plane, find the area of the region enclosed between them.

**38** Find these indefinite integrals.

**a** $\int e^x \, dx$ **b** $\int 2^x \, dx$ **c** $\int 3^x \, dx$ **d** $\int 5^x \, dx$

**39**

a Differentiate $x \log_e x$, and hence find $\int \log_e x \, dx$.

b Differentiate $xe^x$, and hence find $\int xe^x \, dx$.

c Hence prove that $\int_1^e \frac{1}{x} \, dx = \int_1^e \log_e x \, dx = \int_0^1 xe^x \, dx = 1$.

**41**

a Find $\int_0^3 2^x \, dx$ and $\int_{-3}^0 2^x \, dx$.

b Explain geometrically why the first is 8 times the second.

## CHAPTER 7 REVIEW · integration questions

**5** Find:

a $\int 4\cos x\,dx$ b $\int \sin 4x\,dx$ c $\int \sec^2 \frac{1}{4}x\,dx$

**6** Find the value of:

a $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sec^2 x\,dx$ b $\int_0^{\frac{\pi}{4}} \cos 2x\,dx$ c $\int_0^{\frac{1}{3}} \pi \sin \pi x\,dx$

**7** Find the value of $\int_0^{\frac{1}{4}} \sin 3x\,dx$, correct to three decimal places.

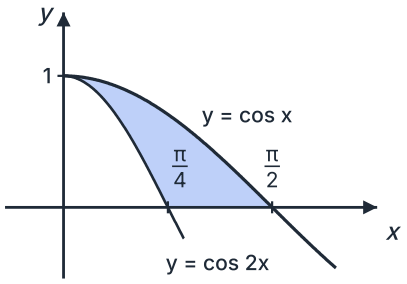
**8** A curve has gradient function $y' = \cos \frac{1}{2}x$ and passes through the point $(\pi, 1)$. Find its equation.

**9** a Sketch the curve $y = 2\sin 2x$, for $0 \leq x \leq \pi$, and then shade the area between the curve and the $x$-axis from $x = \dfrac{\pi}{4}$ to $x = \dfrac{3\pi}{4}$.

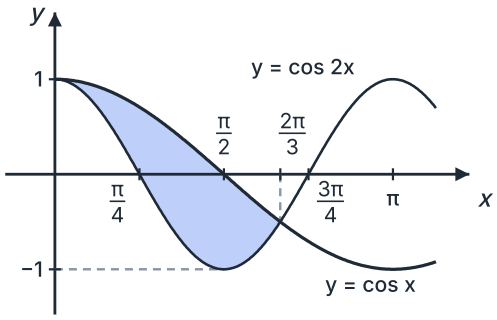
b Calculate the shaded area in part **a**.

**10** Find the area of the shaded region in each diagram below.

a

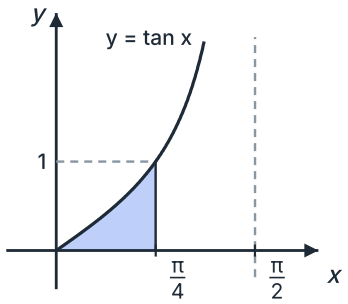



b



**11** a Write $\tan x$ in terms of $\sin x$ and $\cos x$.

b Hence find the exact area of the shaded region in the diagram to the right.




**The following questions are more difficult:**

**16** Find:

a $\int e^{2x} \cos e^{2x}\, dx$

b $\int \frac{\sin e^{-2x}}{e^{2x}}\, dx$

c $\int \frac{\sec^2 x}{3\tan x + 1}\, dx$

d $\int \frac{3\sin x}{4 + 5\cos x}\, dx$

e $\int \frac{1 - \cos^3 x}{1 - \sin^2 x}\, dx$

f $\int_0^{\pi} \sin x \cos^2 x\, dx$

**17** a Show that $\tan^3 x = \tan x \sec^2 x - \tan x$.

b Hence find:

i $\int \tan^3 x \, dx$

ii $\int \tan^5 x \, dx$

**18** a Sketch $y = 1 - \tan x$, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, and shade the region $R$ bounded by the curve and the coordinate axes.

b Find the area of $R$.

**19** a Show that $\cos(A+B)+\cos(A-B)=2\cos A\cos B$.

b Hence find:

i $\displaystyle\int_0^{\frac{\pi}{2}} 2\cos 3x\cos 2x\,dx$

ii $\displaystyle\int_0^{\pi} \cos 3x\cos 4x\,dx$

c Find $\displaystyle\int \cos mx\cos nx\,dx$, where $m$ and $n$ are:

i distinct positive integers,

ii equal positive integers.